



Boards and Beyond: Cardiology Slides

Color slides for USMLE Step 1 preparation
from the Boards and Beyond Website

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2020 Edition

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Cardiac Anatomy

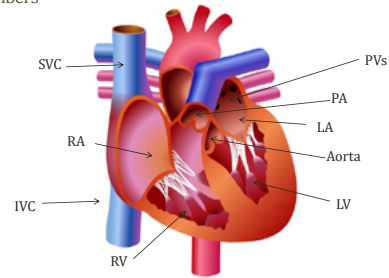
Cardiac Anatomy

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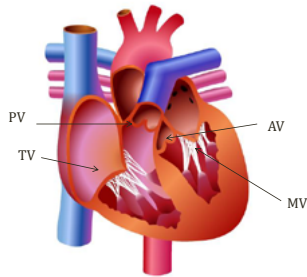
The Heart

Chambers

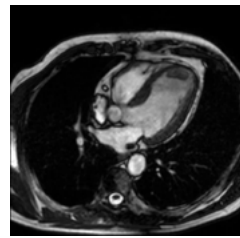


The Heart

Valves



Anterior-Posterior Structures



Right Ventricle → Anterior

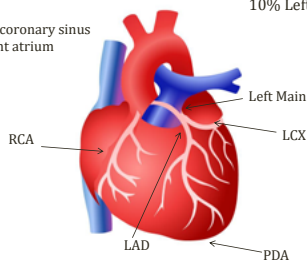
Left atrium → Posterior



Coronary Arteries

Heart drains to coronary sinus
CS → Right atrium

90% Right dominant
10% Left dominant

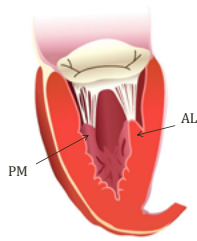


Coronary Artery Territories

- Anterior wall, anterior septum, apex → LAD
- Lateral wall → LCX
- Inferior wall, inferior septum → PDA
 - RCA 90% of the time
 - 10% of people "left dominant" - LCX supplies PDA
- Occlusion occurs LAD > RCA > LCX



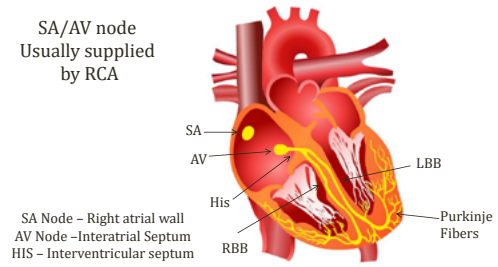
Mitral Valve



- Two papillary muscles
 - Anterolateral (AL)
 - Posteromedial (PM)
- AL has dual blood supply
 - LAD/LCX
- PM → single blood supply
 - RCA (or LCX)
- Inferior infarction can lead to rupture of PM pap
 - Severe mitral regurgitation
 - Acute heart failure

Cardiac Electrical System

SA/AV node
Usually supplied
by RCA



SA Node - Right atrial wall
AV Node - Interatrial Septum
HIS - Interventricular septum

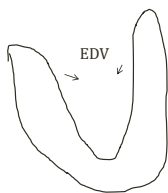
Cardiac Physiology

Cardiac Physiology

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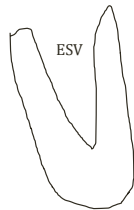


Heart Volumes



End diastolic Volume

Filling completed
Contraction beginning



End systolic Volume

Emptying completed
Relaxation beginning



Important Terms

- Stroke Volume (SV) = EDV - ESV
- Ejection Fraction (EF) = SV / EDV
- Cardiac Output (CO) = SV * HR



Important Terms

- Venous Return (VR)
 - Blood returned to left ventricle
 - Should be equal to the cardiac output
- Total peripheral resistance
 - Resistance to blood flow from peripheral structures
 - Vasoconstriction → ↑ TPR
 - Vasodilation → ↓ TPR



Blood Pressure Terms

- Systolic Blood Pressure (SBP)
 - Largely determined by stroke volume
- Diastolic Blood Pressure (DBP)
 - Largely determined by TRP
- Pulse pressure
 - SBP - DBP
 - Proportional to SV



Blood Pressure Terms

- **Mean arterial pressure (MAP)**
 - $DBP + 1/3 (SBP - DBP)$
- Example: SBP 120/80
 - $MAP = 80 + 1/3 (40) = 93.3$

Cardiac Output

- Very important physiology parameter
- Must rise to meet demands
- More cardiac output = more work/O₂
 - $CO = HR \times SV$
 - More beats per minute = more work
 - More volume per beat = more work

Cardiac Output

Determinants

1. Preload
2. Afterload
3. Contractility
4. Heart rate

Preload

- **Amount of blood loaded into left ventricle**
- Also how much stretch is on fibers prior to contraction
 - Some books say "length" instead of "stretch"
- More preload = more cardiac output
- More preload = more work the heart must do
 - 10% required

To INCREASE Preload

1. **Add volume** (blood, IVF)
2. **Slow heart rate** → more filling → more volume
3. **Constrict veins**
 - Veins force blood into heart
 - Veins hold LARGE blood volume
 - Response to blood loss → venous constriction
 - Sympathetic stimulation → α_1 receptors in veins

To DECREASE Preload

1. **Remove volume** (bleeding, dehydration)
2. **Raise heart rate** (opposite mechanism above)
3. **Pool blood in veins**
 - Mechanism of action of nitrates
 - Relieve angina
 - Lower preload → less work for heart

Preload

Important Terms

- **LVEDV**
 - Volume of blood in the left ventricle when filled
- **LVEDP**
 - Pressure in the left ventricle when filled

Afterload

- **Forces resisting flow out of left ventricle**
- Heart must squeeze to increase pressure
- Needs to open aortic valve → push blood into aorta
- This is harder to do if:
 - Blood pressure is high
 - Aortic valve is stiff
 - Something in the way: HCM, sub-aortic membrane

To INCREASE Afterload

1. **Raise mean blood pressure**
2. **Obstruct outflow of left ventricle**
 - Aortic stenosis, HCM

To DECREASE Afterload

1. **Lower the mean blood pressure**
2. **Treat aortic valve disease, HCM**
 - More afterload = more work
 - More oxygen required

Contractility

- **How hard the heart muscle squeezes**
- Ejection fraction = index of contractility
- Major regulator: **sympathetic nervous system**
 - Also increases heart rate

To INCREASE Contractility

- Sympathetic nervous system activity
 - Sympathetic innervation to heart
 - Circulating catecholamines (epinephrine, norepinephrine)
 - ↑ calcium release from sarcoplasmic reticulum
 - Triggers: **stress, exercise**
- Sympathomimetic drugs
 - Dopamine, dobutamine, epinephrine, norepinephrine
- Digoxin
 - Inhibits Na-K pump → ↑ calcium in myocytes

To DECREASE Contractility

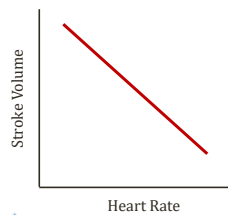
- **Sympathetic system blocking drugs**
 - Beta blockers
- Calcium channel blockers
 - Verapamil, diltiazem
 - Less calcium for muscle contraction
- Heart failure
 - Disease of myocytes

Heart Rate

- Increases cardiac output under physiologic conditions
- Mainly regulated by **sympathetic nervous system**
- Also increased by **sympathomimetic drugs**
- Decreased by beta blockers and calcium blockers

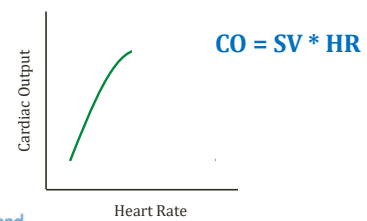
Heart Rate

- $\uparrow \text{HR} = \downarrow \text{stroke volume}$ (less filling time)



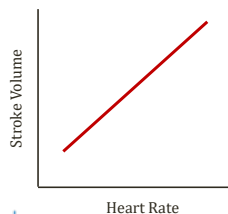
Heart Rate

- $\uparrow \text{HR} = \uparrow \text{cardiac output}$



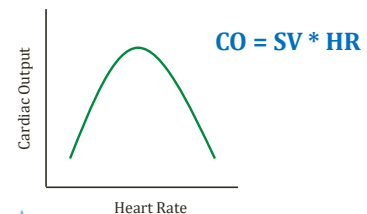
Heart Rate

- Sympathetic nervous system: $\uparrow \text{HR}$ and $\uparrow \text{contractility}$
- Stroke volume rises with increased HR



Heart Rate

- At pathologic heart rates $\uparrow \text{HR} = \downarrow \text{CO}$
- High heart rate with arrhythmia can lead to $\downarrow \text{CO}$



Work of the heart

Myocardial O₂ demand

- Preload (LVEDV/P)
- Afterload (MAP)
- Contractility (EF)
- Heart Rate

Hearts starved for O₂ → Reduce O₂ demand
Low output → Need to increase work

CV Response to Exercise

Cardiovascular Response to Exercise

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Response to Exercise

- Body's overall goal:
 - Maximize perfusion skeletal muscles and heart
 - Minimize perfusion all other areas
- Initiator: **Muscle hypoxia**
- Mediator: **Sympathetic nervous system**

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Response to Exercise

- Process begins with **muscle contraction**
- ATP consumed → oxygen consumed (need more ATP)
- Result: Local hypoxia in muscle tissue
- Vasodilation occurs
 - Multiple mediators released into plasma
 - Adenosine generated from ATP consumption
 - Lactate
 - Carbon dioxide, potassium
- **Lowers total peripheral resistance (TPR)**

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Response to Exercise

- **Sympathetic nervous system** activated
- ↑contractility (stroke volume)
- ↑HR
- Net result: **↑ cardiac output**
- Results in **↑ systolic blood pressure (SBP)**
- Vasoconstriction in some areas (gut, skin)
 - Redistributes blood to important areas (i.e. heart/muscles)

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Response to Exercise

Blood Pressure Summary

- **SBP rises**
 - More CO = more blood in arteries = more pressure
 - Primary determinant systolic BP = cardiac output
- **DBP decreases slightly or stays normal**
 - Local dilation of skeletal muscles
 - Primary determinant diastolic BP = peripheral resistance
- Pulse pressure increases
- TPR goes down

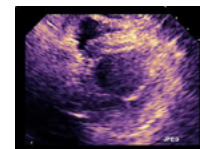
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Response to Exercise

Ejection Fraction

$$EF = \frac{EDV - ESV}{EDV}$$

- **LVEF increases**
 - More vigorous contraction
 - Major impact: ESV decreases
 - EDV effects minor/variable
 - More preload but less filling time at fast heart rates



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Response to Exercise

Coronary Perfusion

- Fast HR shortens diastole
- **LESS** coronary filling time
- Coronary vasodilation → increased blood flow
 - Only way to get more oxygen
 - Cannot extract more O_2
 - Cardiac tissue extracts maximum oxygen from RBCs
 - Cannot extract more to meet increased demand

Response to Exercise

Preload

- **Preload rises with exercise**
- Sympathetic stimulation → **venous contraction**
- Increases preload/EDV
- Contributes to rise in cardiac output
 - Along with increased heart rate and contractility

Lusitropy

- Lusitropy = **myocardial relaxation**
 - Opposite of contractility
- Increased with exercise
- Contributes to increased preload → ↑ cardiac output

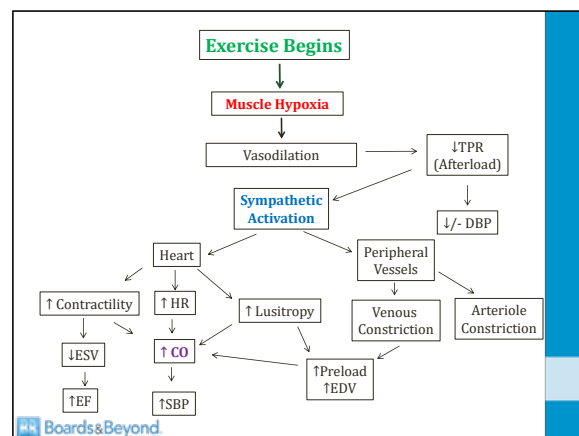
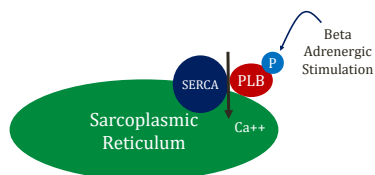
Lusitropy

- Key regulatory protein: Phospholamban
 - **Inhibitor:** sarcoplasmic reticulum Ca^{2+} -ATPase (SERCA)
 - Phosphorylated via beta adrenergic stimulation
 - Stops inhibiting SERCA
 - Result: SERCA takes up calcium → relaxation

SERCA

Sarco/endoplasmic reticulum Ca^{2+} -ATPase

- Sympathetic stimulation → phosphorylates PLB
- Inactivates PLB (relieves inhibitory effect)
- Allows SERCA to uptake more calcium



Blood Flow Mechanics

Blood Flow Mechanics

Jason Ryan, MD, MPH

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Flow Equations

Ohm's Law $V = I R$

For fluids: $\Delta P = Q \times R$

$CO = Q$ for body

TPR = total peripheral resistance

$\Delta P = CO \times TPR$

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Flow Equations



Velocity * Area = Flow

$(m/s) * (m^2) = (m^3/s)$

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Resistance and Compliance



Resistance = resistance to flow
Compliance = distensibility of vessels

Stiff Vessels

↑ resistance
↓ compliance

Stretchy Vessels

↓ resistance
↑ compliance

High resistance = low compliance (vice versa)

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Pulse Pressure

- Systolic BP – diastolic BP
 - Normal = 120 – 80 = 40mmHg
- Older patients = ↑ pulse pressure
- Hypertensive patients = ↑ pulse pressure
- Related to **vessel compliance**
- **↓ compliance = ↑ pulse pressure**

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Pulse Pressure

- Compliance = $\Delta \text{volume} / \Delta \text{pressure}$
- Stiff vessel → ↓ compliance → ↑ pulse pressure
 - Small change in volume for given pressure applied to walls
- Stretchy vessel → ↑ compliance → ↓ pulse pressure
 - Large change in volume for given pressure applied to walls

$$C = \Delta V / \Delta P$$

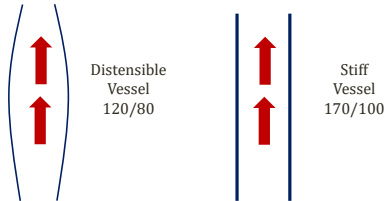


$$\Delta P = \Delta V / C$$

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Pulse Pressure

- **Pulse pressure varies with vessel compliance**
- Stiff vessels → ↓ compliance



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Flow Equation

Total Peripheral Resistance



$$\Delta P = CO * \text{TPR}$$

↑ resistance = ↑ pressure to maintain flow
↑ pressure = ↑ cardiac work

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Total Peripheral Resistance

- Easy to push blood out of heart → less O₂ required
- Resistance to flow → more work for heart
- What resists forward flow out of heart?
 1. Types of vessels (i.e. pipes/tubes)
 2. Thickness of blood (viscosity)

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Types of Vessels

- Aorta: SBP 100mmHg
- Large arteries: Falls few mmHg
- Small arteries: 10-20mmHg
- **Arterioles: 35mmHg**
- Capillaries: 25mmHg

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Types of Vessels

- **Arterioles = “resistance vessels”**
 - Major determinant of total peripheral resistance
 - Large pressure drop
 - Vasoconstriction = ↑ TPR
 - Vasodilation = ↓ TPR

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Viscosity

- Thickness of blood
- Low viscosity
 - Anemia
- High viscosity
 - Polycythemia
 - Multiple myeloma
 - Spherocytosis

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Poiseuille's Law

$$\Delta P = Q \times R$$

$$R = \frac{\Delta P}{Q} = \frac{8 \eta (\text{viscosity}) L (\text{length})}{\pi r (\text{radius})^4}$$

Changes in radius → large change in resistance

Series and Parallel Circuits

Human organs arranged **in parallel**
Resistances add up differently in series than in parallel

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} \quad R_{\text{total}} = R_1 + R_2$$

Parallel

Series

Series and Parallel Circuits

For two resistances (2 and 2), what is total R?

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} \quad R_{\text{total}} = R_1 + R_2$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{2} + \frac{1}{2} \quad R_{\text{total}} = 2 + 2 = 4$$

$$R_{\text{total}} = 1$$

Flow Equation

$$\Delta P = Q \times R$$

- Used to calculate resistance, CO, or ΔP
- Often applied to **body** and **lungs**
 - For both systems Q = Cardiac Output (CO)

Flow Equation

$$\Delta P = Q \times R$$

- Body
 - ΔP = Arterial pressure – right atrial pressure
 - R = Total peripheral resistance (TPR)
 - R = **Systemic vascular resistance** (SVR)
- Lungs
 - ΔP = Pulmonary artery pressure – left atrial pressure
 - R = **Pulmonary vascular resistance** (PVR)

Mean Arterial Pressure

- Diastolic plus 1/3 (Systolic – Diastolic)
- Total body
 - Arterial blood pressure = 120/80 mmHg
 - Mean arterial pressure = $80 + 1/3 (40) = 93$ mmHg
- Lungs
 - Pulmonary artery pressure = 40/20 mmHg
 - Mean pulmonary artery pressure = $20 + 1/3 (20) = 27$ mmHg

Total Body

$$\Delta P = CO * TPR$$

- $R = TPR$
- $\Delta P = MAP - RAP$
 - MAP = mean arterial pressure
 - RAP = right atrial pressure
- CO of 5L/min; BP 155/80 (MAP 105), RA 5

$$TPR = \frac{\Delta P}{CO} = \frac{MAP - RAP}{5} = \frac{105 - 5}{5} = 20$$

Lungs

$$\Delta P = CO * TPR$$

- $R = PVR$
- $\Delta P = PA - LAP$
 - PA = mean pulmonary artery pressure
 - LAP = left atrial pressure
- CO of 5L/min; PA 40/10 (MAP 20), LA 5

$$PVR = \frac{\Delta P}{CO} = \frac{PA - LAP}{5} = \frac{20 - 5}{5} = 3$$

Lung and Body Flow Variables

	Lung	Body
Flow	CO	CO
Resistance	PVR	TPR
Start Pressure	PA	AoP
End Pressure	LA	RA
ΔP	PA - LA	Ao - RA

Velocity and Area

- Flow = Velocity * Area
- Changes as blood moves through vessels
 - Aorta → arterioles → capillaries → veins
 - Cardiac output moves through system (same flow)
 - Different vessels → different area, velocity
 - Area ↑↑, velocity ↓↓

Velocity



Flow = Velocity * Area
 $(m^3/s) = (m/s) * (m^2)$

Flow Properties of Blood Vessels

Property	Highest	Lowest
Flow	--	--
Area	Capillaries	Large arteries
Velocity	Large Arteries	Capillaries
Resistance	Arterioles	Veins
ΔP	Arterioles	Veins

$$\text{Flow} = \text{Vel} * \text{Area}$$

$$\Delta P = Q * R$$

Law of Laplace

- Wall tension or wall stress
- Applies to vessels and cardiac chambers
- ↑ tension → ↑ O2 demand → ischemia/angina

$$\text{Tension} \propto \frac{P * r}{2h}$$

Wall Tension

- Afterload: Increases **pressure** in left ventricle
 - Hypertension, aortic stenosis
 - Will increase wall tension
 - "Pressure overload"

$$\text{Tension} \propto \frac{P * r}{2h}$$

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Wall Tension

- Preload: Increases **radius** of left ventricle
 - Chronic valvular disease (aortic/mitral regurgitation)
 - Will increase wall tension
 - "Volume overload"

$$\text{Tension} \propto \frac{P * r}{2h}$$

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Wall Tension

- Hypertrophy: Compensatory mechanism
 - Will **decrease** wall tension
 - Force distributed over more mass
 - Occurs with chronic pressure/volume overload

$$\text{Tension} \propto \frac{P * r}{2h}$$

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Eccentric Hypertrophy

- Longer myocytes
- Sarcomeres added in series
- Left ventricular mass increased
- Wall thickness NOT increased



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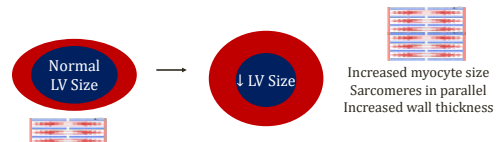
Eccentric Hypertrophy

- Volume overload of left ventricle
 - Aortic regurgitation
 - Mitral regurgitation
- Cardiomyopathy
 - Ischemic and non-ischemic

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Concentric Hypertrophy

- Pressure overload
- Chronic ↑↑ pressure in ventricle
- Sarcomeres added in parallel
- Left ventricular mass increased
- Wall thickness increased



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Concentric Hypertrophy

- Classic causes: Hypertension, Aortic stenosis
 - Both raise pressure in LV cavity
- Decreased compliance (stiff ventricle)
- Often seen in diastolic heart failure

Regulation of Blood Pressure

Regulation of Blood Pressure

Jason Ryan, MD, MPH

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Blood Pressure

- Required for perfusion of tissues
- Varies with sodium/water intake
- Regulated by nervous system

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Baroreceptors

- Blood pressure sensors via **stretch**
- Signal central nervous system (brain)
- Response via **autonomic nervous system**
 - Sympathetic and parasympathetic
- Modify:
 - Heart rate/contractility
 - Arterial tone (vasoconstriction)
 - Venous tone (more tone = more preload to ventricle)
 - Renal renin release

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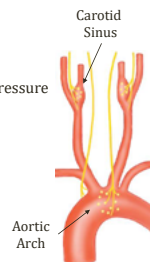
Baroreceptors

- **Aortic arch and carotid sinus**
 - Quick response to changes in blood pressure
 - Rapid response via autonomic nervous system
- Kidneys (renin release)

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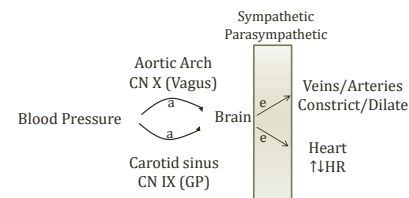
Baroreceptors

- Aortic arch
 - Senses **elevated blood pressure**
 - Poor sensing of low blood pressure
- Carotid sinus
 - Most important baroreceptor
 - Modifies signals over wider range of blood pressure
 - Senses **low and high blood pressure**



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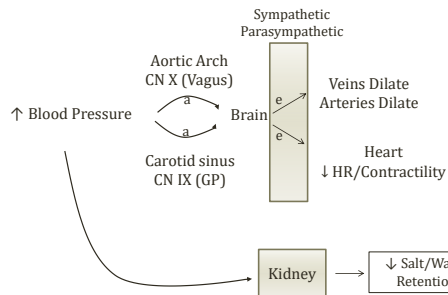
Blood Pressure Control



Afferent = Arrive at the brain
Efferent = Exit the brain
Nucleus Solitarius

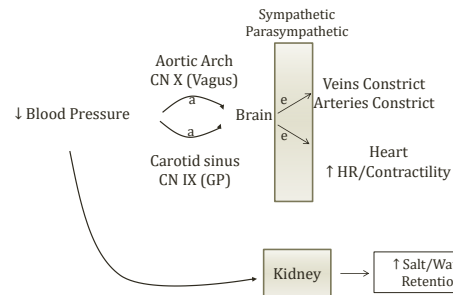
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High Blood Pressure



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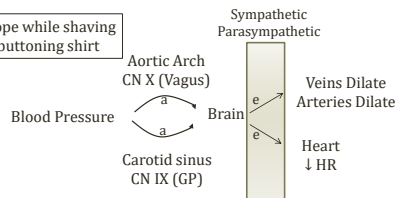
Hemorrhage



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Carotid Massage

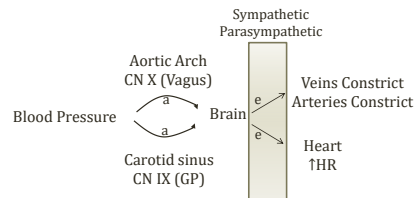
Syncopal while shaving or buttoning shirt



Tricks carotid sinus into thinking ↑ BP
Result: ↓ HR, Vasodilation, ↓ BP

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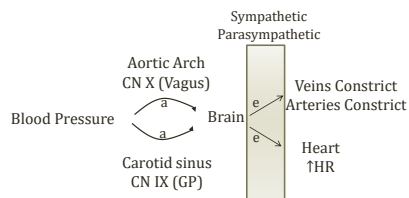
Carotid Occlusion



Tricks carotid sinus into thinking ↓ BP
Result is ↑ HR, Vasoconstriction, ↑ BP

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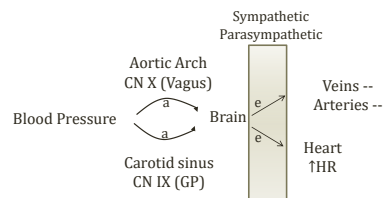
Severing CN IX



Tricks brain into thinking ↓ BP
Result is ↑ HR, Vasoconstriction, ↑ BP

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Severing CN X



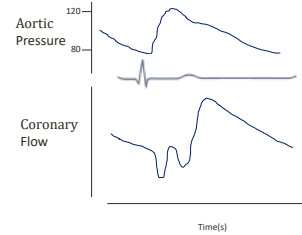
Vagotomy
Unopposed Sympathetic Cardiac Stimulation
Result is ↑ HR

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Summary of Techniques

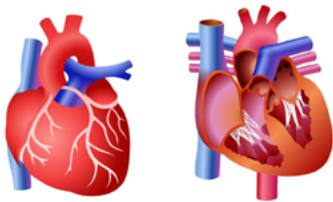
Technique	Interpretation	Result
Carotid Massage	↑BP	↓HR, ↓BP
Carotid Occlusion	↓BP	↑HR, ↑BP
Sever CN IX	↓BP	↑HR, ↑BP
Sever CN X	--	↑HR

Coronary Blood Flow



- In tachycardia, less time in diastole → less flow

Regional Blood Flow



- Epicardium → site of coronary arteries
- Subendocardium receives smallest amount blood flow

Organ Circulation

Organ	Key Features
Lung	100% of Cardiac Output
Liver	Largest Systemic Blood Flow
Kidneys	Highest blood flow by weight
Heart	Largest ΔO_2 (80%) ↑ demand vasodilation

Autoregulation

- Some tissue beds maintain **constant blood flow**
- ↑ BP → ↑ flow → vasoconstriction → ↓ flow (normal)
- Use local metabolites to sense blood pressure

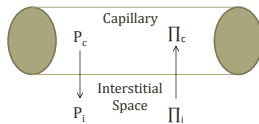
Autoregulation

Organ	Key Control Variables
Heart	CO ₂ , Adenosine, NO
Brain	CO ₂ , pH
Kidneys	BP and NaCl feedback
Lungs	Hypoxia → Vasoconstriction
Skeletal Muscle	Lactate, adenosine, K ⁺
Skin	Sympathetic stimulation

Kidney, brain, heart: Excellent autoregulation systems
Skin: Poor autoregulatory capacity

Capillary Fluid Exchange

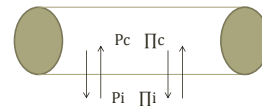
- Two forces drive fluid into or out of capillaries
- Hydrostatic pressure (P)
 - Molecules against capillaries walls
 - Pushes fluid out
- Oncotic pressure (Π)
 - Solutes (albumin) drawing fluid into capillaries



Boards&Beyond

Capillary Fluid Exchange

- Hydrostatic pressure – fluid PUSHING against walls
 - High pressure drives fluid TOWARD low pressure
- Oncotic pressure –solutes PULLING fluid in
 - High pressure draws fluid AWAY from low pressure



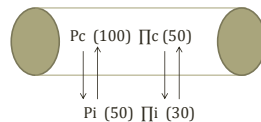
$$\text{Net Pressure (NP)} = (P_c - P_i) - (\Pi_c - \Pi_i)$$

$$\text{Flow} = (\text{NP}) K_f$$

Boards&Beyond

Capillary Fluid Exchange

- Hydrostatic pressure – fluid PUSHING against walls
 - High pressure drives fluid TOWARD low pressure
- Oncotic pressure –solutes PULLING fluid in
 - High pressure draws fluid AWAY from low pressure



$$\text{Net Pressure (NP)} = 50 - 20 = 30$$

$$\text{Flow} = (\text{NP}) K_f$$

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Edema

- Excess fluid movement out of capillaries
- Tissue swelling
- Lungs: Pulmonary edema
- Systemic capillaries: Lower extremity edema

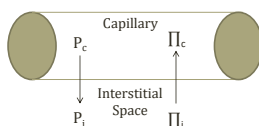


James Hellman, MD

Boards&Beyond

Edema

$$\text{Net Pressure (NP)} = (P_c - P_i) + (\Pi_i - \Pi_c)$$



- ↑ capillary pressure, ↑ P_c (heart failure)
- ↓ plasma proteins, ↓ Π_c (nephrotic syndrome, liver failure)
- ↑ capillary permeability, ↑ K_f (toxins, infections, burns)
- ↑ interstitial osmotic pressure, ↑ Π_i (lymphatic blockage)

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3rd Spacing

- Intracellular fluid – 1st space
 - About 2/3 body fluid
- Extracellular fluid – 2nd space
 - About 1/3 body fluid
- Third spacing - fluid where it should NOT be
 - Pleural effusions
 - Ascites
 - Cerebral edema
 - Low intravascular volume/High total volume
- Occurs post-op, sepsis

Boards&Beyond

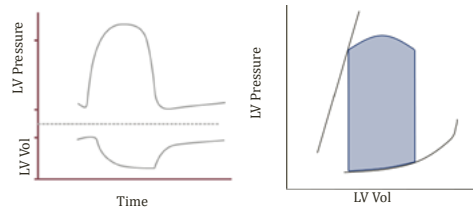
PV Loops

Pressure Volume Loops

Jason Ryan, MD, MPH

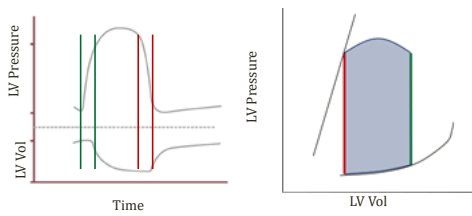
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PV Loops



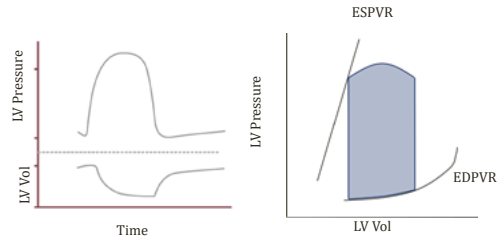
Boards&Beyond

PV Loops



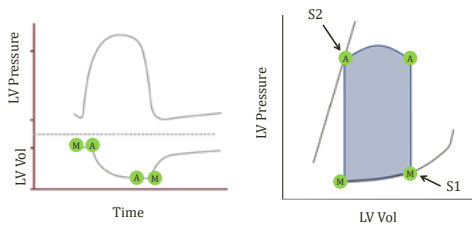
Boards&Beyond

PV Loops

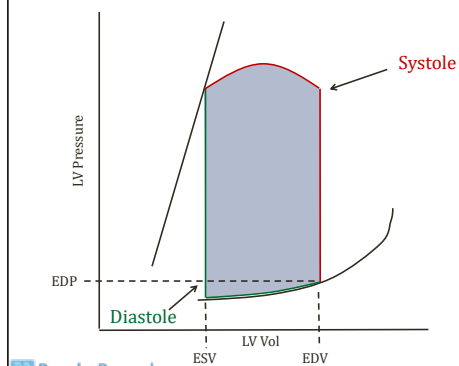


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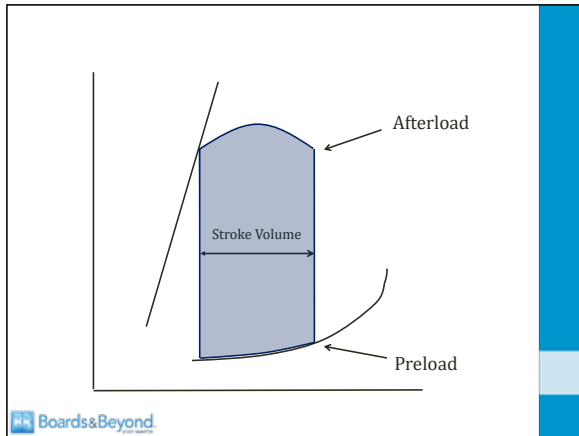
PV Loops



Boards&Beyond

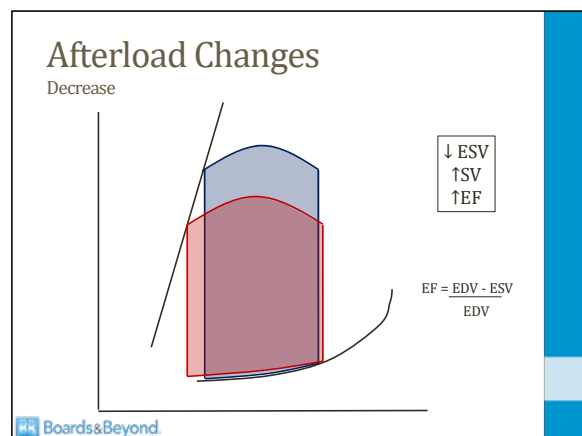
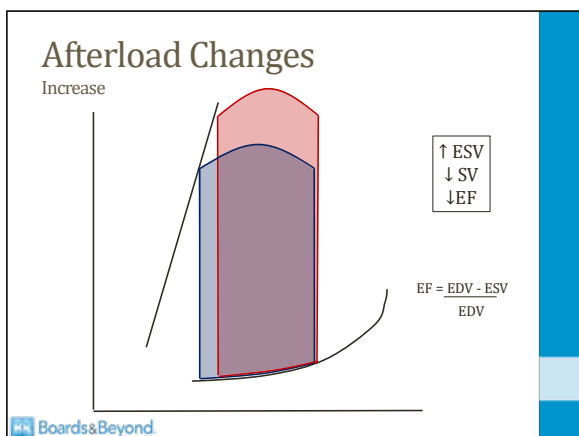
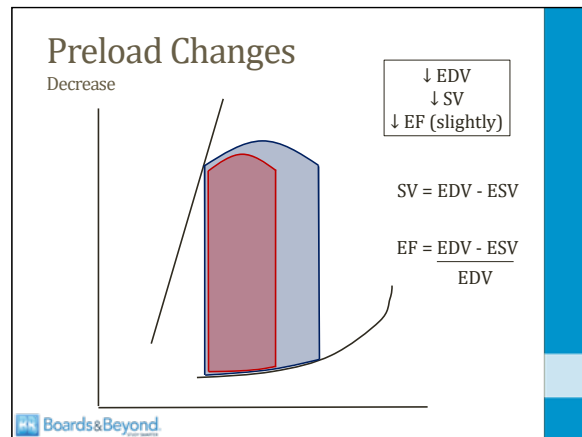
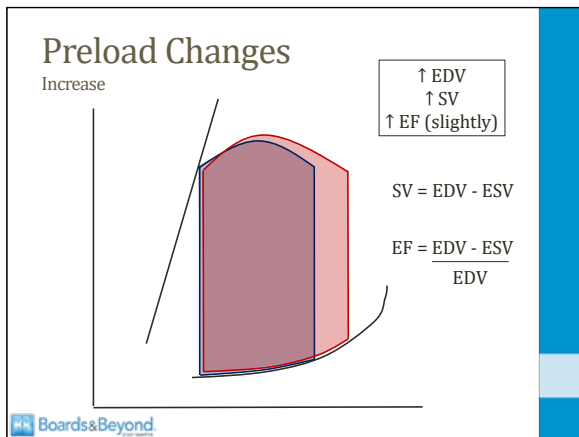


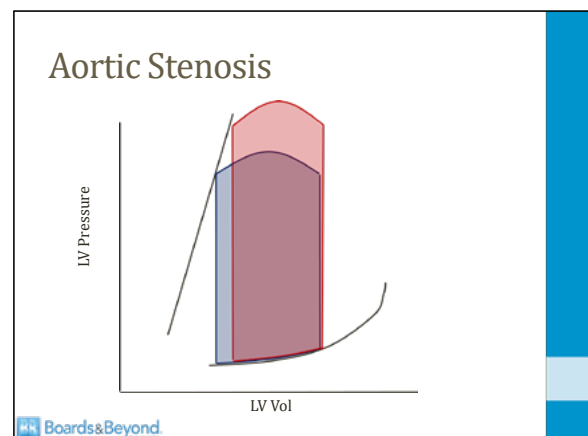
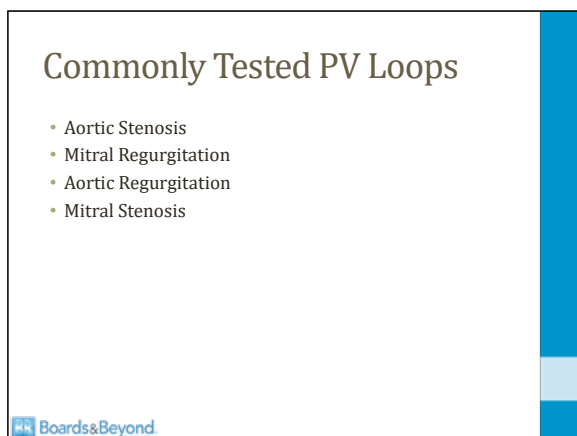
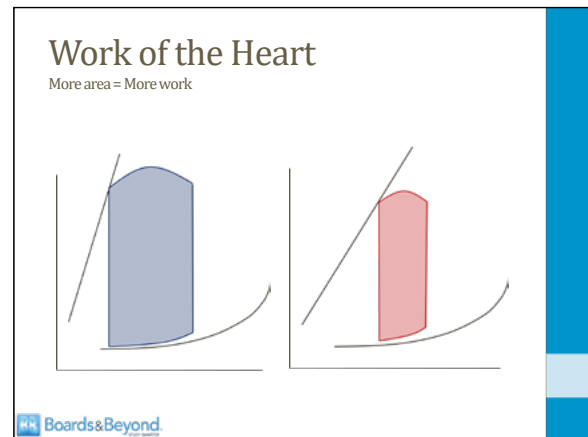
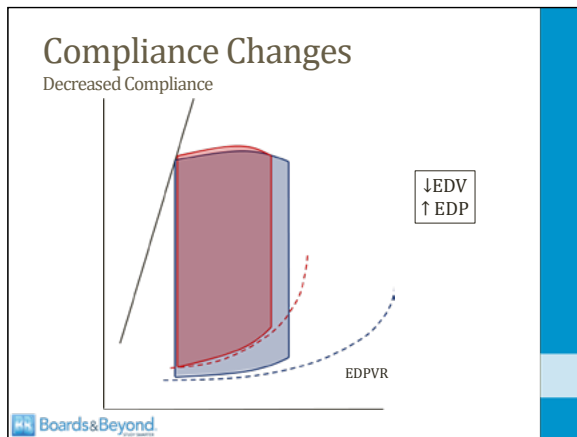
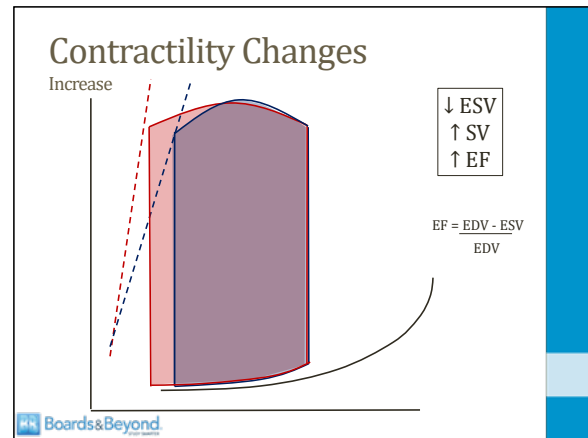
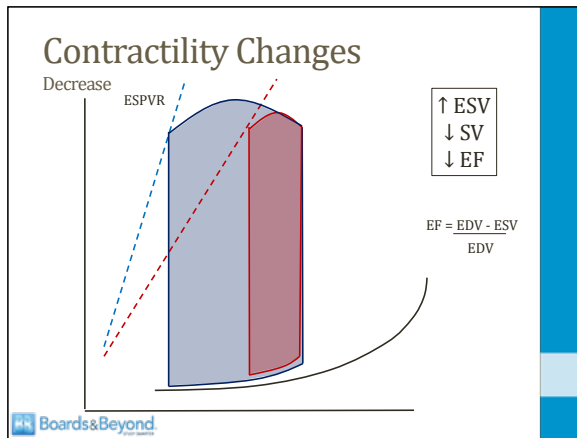
Boards&Beyond



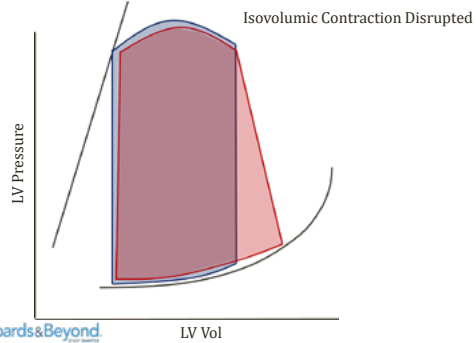
PV Loop Parameters

- Changes in preload
- Changes in afterload
- Changes in contractility
- Changes in compliance
- In reality, these are inter-related
 - Example: $\uparrow$ preload $\rightarrow$ $\uparrow$ contractility (Frank-Starling)

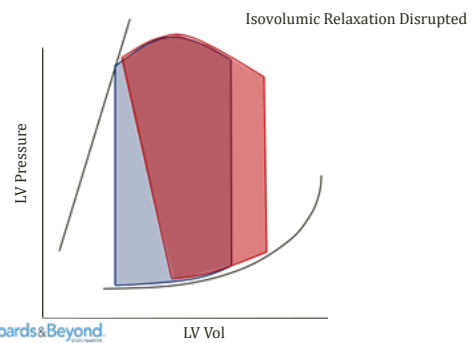




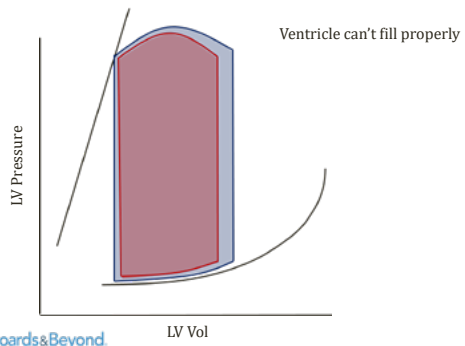
Mitral Regurgitation



Aortic Regurgitation



Mitral Stenosis



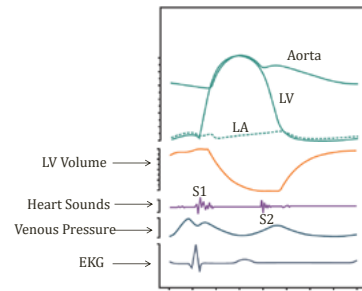
Wiggers' Diagram

Wiggers' Diagram

Jason Ryan, MD, MPH

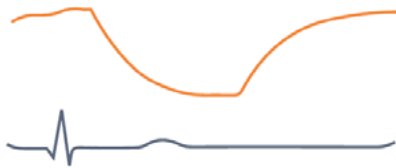
Boards&Beyond

Wiggers' Diagram

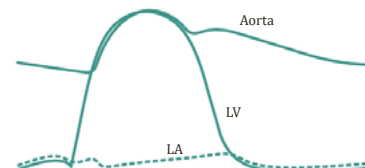


Boards&Beyond

Left Ventricular Volume

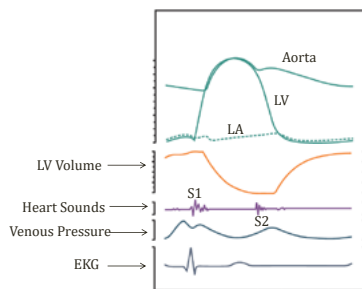


Boards&Beyond

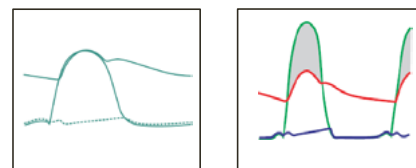


Boards&Beyond

Wiggers' Diagram



Boards&Beyond

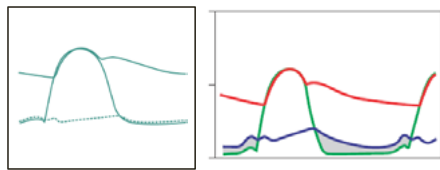


Normal Heart

Diseased Heart

Aortic Stenosis

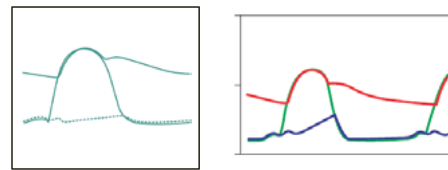
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Normal Heart

Diseased Heart

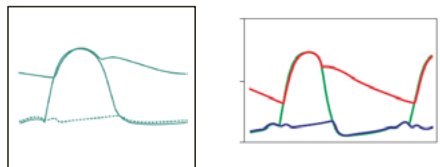
Mitral Stenosis



Normal Heart

Diseased Heart

Mitral Regurgitation



Normal Heart

Diseased Heart

Aortic Regurgitation

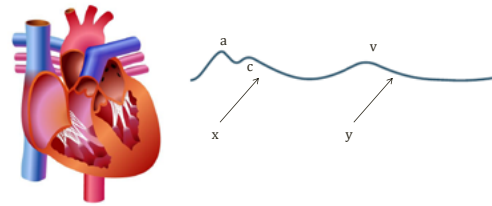
Venous Pressure Tracings

Venous Pressure Tracings

Jason Ryan, MD, MPH

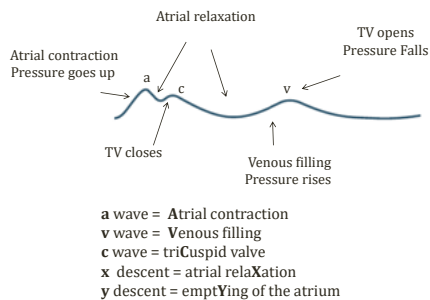
Boards&Beyond

Venous Pressure



Boards&Beyond

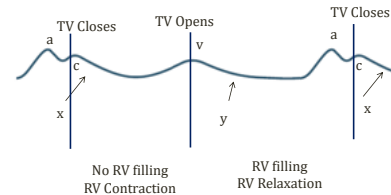
Venous Pressure



Boards&Beyond

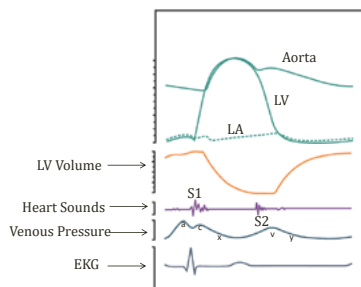
Venous Pressure

Tricuspid valve



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Wiggers' Diagram



Boards&Beyond

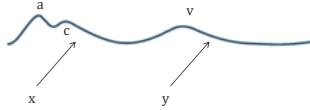
Classic Findings

- Large a wave
- Cannon a wave
- Absent a waves
- Large v waves

Boards&Beyond

Large a wave

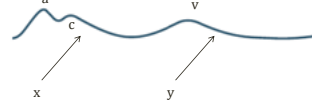
Tricuspid stenosis



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Cannon a wave

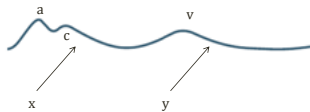
AV dissociation



Boards&Beyond

Absent a wave

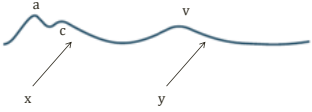
Atrial fibrillation



Boards&Beyond

Giant v wave

Tricuspid regurgitation



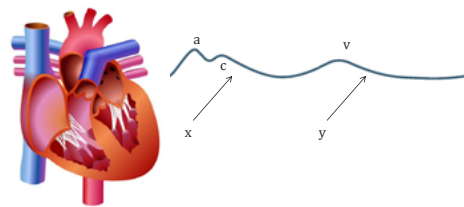
Boards&Beyond

High Yield Findings

- Large a wave (increased atrial contraction pressure)
 - Tricuspid stenosis
 - Right heart failure/Pulmonary hypertension
- Cannon a wave (atria against closed tricuspid valve)
 - Complete heart block
 - PAC/PVC
 - Ventricular tachycardia
- Absent a wave (no organized atrial contraction)
 - Atrial fibrillation
- Giant V waves
 - Tricuspid regurgitation

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Left Atrial Pressure



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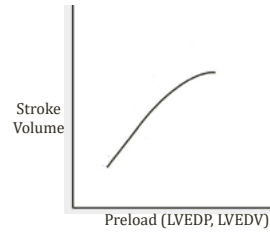
Starling Curve

Starling Curves

Jason Ryan, MD, MPH

Boards&Beyond

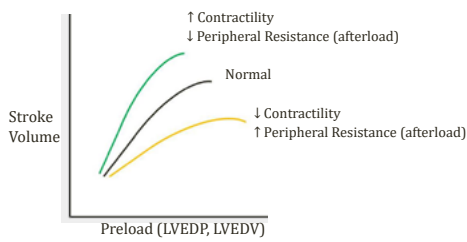
Frank-Starling Curve



Boards&Beyond

Frank-Starling Curve

Left and Right Shifts



Boards&Beyond

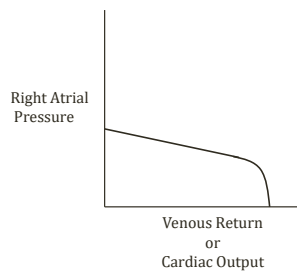
Frank-Starling Curve

Left and Right Shifts

- **Contractility**
 - Increase: Exercise, inotropes
 - Decrease: Myocardial infarction, heart failure
- **Peripheral resistance:**
 - Total peripheral resistance (TPR)
 - Systemic vascular resistance (SVR)
 - Increase: Vasopressors
 - Decrease: Vasodilators, sepsis

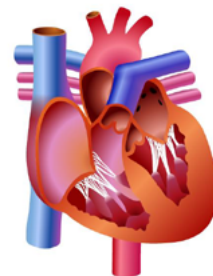
Boards&Beyond

Venous Return Curve



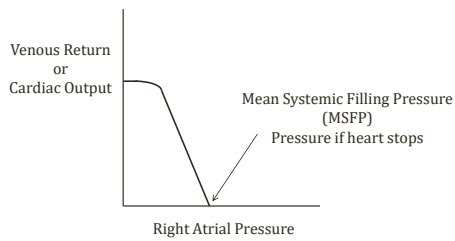
Boards&Beyond

Right Atrial Pressure



Boards&Beyond

Venous Return Curve

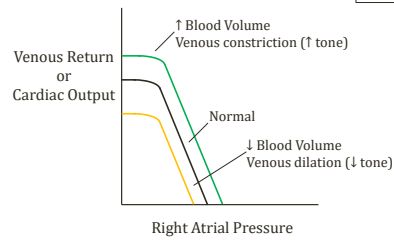


Boards&Beyond

Venous Return Curve

Left and Right Shifts

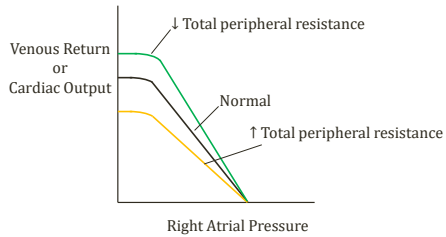
**Volume
Venous Tone**



Boards&Beyond

Venous Return Curve

Changes in Slope



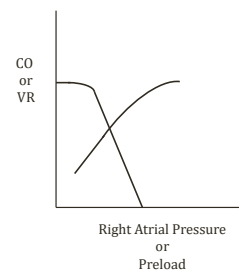
TPR change shifts curve right/left
No change in MSFP

Result: change in **slope** of Venous Return curve

Boards&Beyond

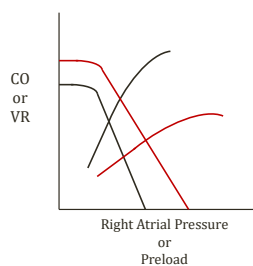
Combined Curves

Starling and Venous Return



Boards&Beyond

Heart Failure

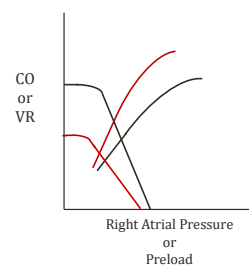


↓ contractility
↑ TPR
↑ Fluid volume

* Black = normal

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Hemorrhage



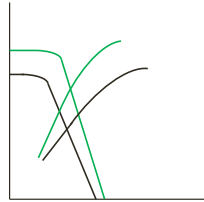
Blood loss
↑ TPR
↑ Contractility

* Black = normal

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Exercise

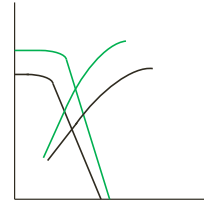
- Decreased afterload (TPR)
- Venous contraction
- Increased contractility
- Net result = increased CO



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AV Fistulas

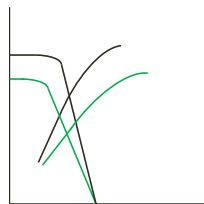
- Decreased afterload (TPR)
- Increased contractility
- Venous contraction
- Net result = increased CO



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Vasopressors

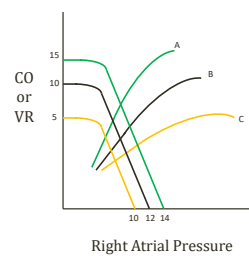
- Increased afterload (TPR)
- Alters VR and Starling Curves
- Net result = decreased CO



Boards&Beyond

Combined Curves

Starling and Venous Return



* Black = normal

For a patient on starling curve A with a MSFP of 10 what is the cardiac output?

Boards&Beyond

Atherosclerosis

Atherosclerosis

Jason Ryan, MD, MPH

Boards&Beyond

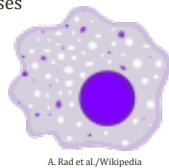
Vocabulary

- **Arteriosclerosis**
 - Hardening of arteries
 - Hyaline
 - Hyperplastic
- **Atherosclerosis**
 - Form of arteriosclerosis
 - Most common type

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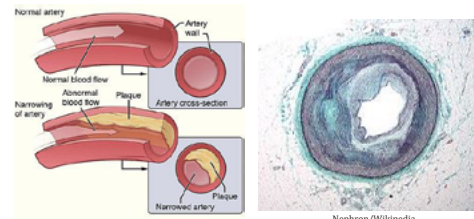
Atherosclerosis

- **Plaque accumulation** in arterial walls
- Chronic inflammatory process
- Involves **macrophages**, T-cells
- Accumulation of **lipoproteins especially LDL**
- Underlying cause of many diseases
 - Myocardial infarction
 - Stroke
 - Peripheral vascular disease



A. Rad et al./Wikipedia

Boards&Beyond



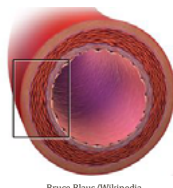
OpenStax College/Wikipedia

Nephron/Wikipedia

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Arterial Structure

- **Intima**
 - Single layer of endothelial cells
 - Basement membrane
- **Media**
 - Smooth muscle cells
 - Elastin
- **Adventitia**
 - Connective tissue
 - Vasa vasorum (blood supply to artery wall)
 - Nerve fibers



Bruce Blaus/Wikipedia

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Type of Arteries

- **Elastic**
 - Large amounts of **elastin** in media layer
 - Expansion in systole, contraction in diastole
 - Aorta, carotid arteries, iliac arteries
- **Muscular**
 - Layers of smooth muscle cells
 - Vasoconstriction/vasodilation to modify blood flow
 - Arterioles: smallest muscular vessels (most flow resistance)

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Atherosclerosis

- **Large elastic arteries**
 - Aorta, carotid arteries, iliac arteries
- **Medium-sized muscular arteries**
 - Coronary, popliteal



Atherosclerosis

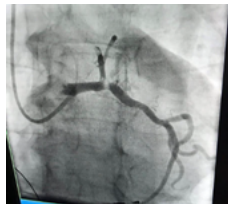
Pathogenesis

- **Endothelial injury or dysfunction**
 - Details incompletely understood
 - Believed to be related to risk factors
 - Cigarette smoke
 - High blood pressure
 - High cholesterol

Atherosclerosis

Pathogenesis

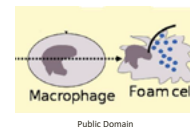
- **Branch points** and **vessel origins** (ostia)
 - Common sites of plaque
 - Turbulent flow → endothelial stress



Atherosclerosis

Pathogenesis

- **Lipids**
 - **LDL accumulation** in intima
 - Oxidized by free radicals
 - Oxidized LDL scavenged by macrophages
 - Cannot be degraded
 - Macrophages become foam cells



Atherosclerosis

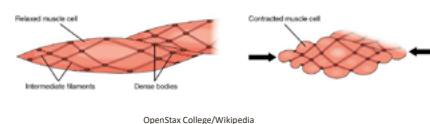
Pathogenesis

- **Chronic inflammation**
 - LDL oxidized from free radicals
 - Damages endothelium, smooth muscle
 - Macrophages release cytokines

Atherosclerosis

Pathogenesis

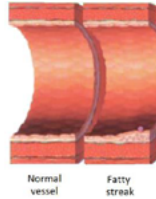
- **Smooth muscle cells** proliferate in intima
- Lay down extracellular matrix
- Key growth factor: **PDGF**
 - Platelet-derived growth factor



Atheroma Growth

- **Fatty streaks**

- Macrophages filled with lipids
- Form line (streak) along vessel lumen
- Do not impair blood flow
- Can be seen in children, adolescents
- Not all progress



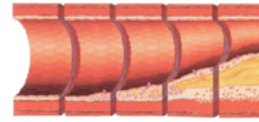
Npachett/Wikipedia

Boards&Beyond

Atheroma Growth

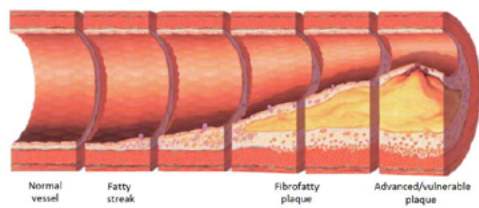
- **Atherosclerotic plaques**

- Intima thickens
- Lipids accumulate
- Usually patchy along vessel wall
- Rarely involve entire vessel wall
- Usually eccentric



Npachett/Wikipedia

Boards&Beyond

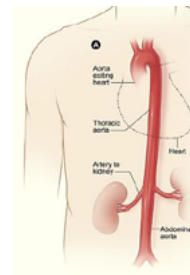


Npachett/Wikipedia

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Locations

- Abdominal aorta (large vessel)
- Coronary arteries
- Popliteal arteries
- Internal carotid
- Circle of Willis



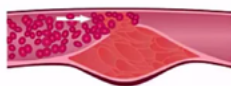
Public Domain

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Atherosclerosis Complications

- **Ischemia**

- Plaque rupture
 - Exposes thrombogenic substances
 - Clot formation
 - May cause acute vessel closure (STEMI)
 - Thrombus may embolize (stroke from carotid plaque)



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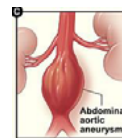
Atherosclerosis Complications

- **Hemorrhage into plaque**

- Lesions: proliferating small vessels ("neovascularization")
- Contained rupture may suddenly expand lesion

- **Aneurysm**

- Lesions may damage underlying media
- Plaque associated with abdominal aortic aneurysms



Public Domain

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Dystrophic Calcification

- Commonly seen in atheroma
- Result of chronic inflammation
- Basis for “coronary CT scans”



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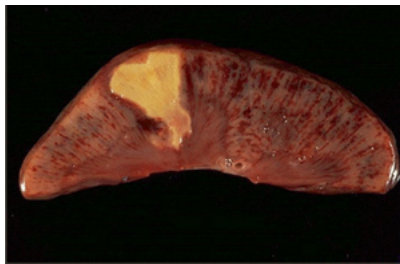
Infarction

- Area of ischemic necrosis
- Two types: white and red
- **White infarcts**
 - Occlusion of arterial supply to a solid organ
 - Common in heart, kidneys, spleen
 - Limited blood seepage from healthy tissue
 - Tissue becomes pale (white)

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White Infarct

Renal Infarction



Ryan Johnson/Flickr

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Red Infarcts

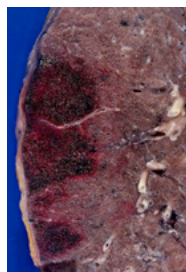
Hemorrhagic Infarct

- Blood enters ischemic tissue
- Blockage of venous drainage
 - Testicular torsion
- Tissues with dual circulation
 - Blood flow from 2nd supply floods ischemic area
 - Classic location: Lung (diffuse blood supply)
 - Small intestine
- Flow re-established to necrotic area
 - Angioplasty restores flow in coronary artery

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Red Infarct

Lung Infarction



Yale Rosen/Wikipedia

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Cardiac Ischemia

Cardiac Ischemia

Jason Ryan, MD, MPH



Cardiac Ischemia

- Caused by coronary atherosclerosis
- **O₂ SUPPLY << O₂ DEMAND = ISCHEMIA**
- Typical symptoms
 - Chest pain (angina)
 - Dyspnea
 - Diaphoresis

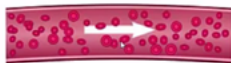


Freestocks.org



Stable Angina

- Stable atherosclerotic plaque
 - No plaque ulceration
 - No thrombus
- Must occlude ~75% of lumen to cause symptoms



NO SYMPTOMS



SYMPTOMS WITH EXERTION
(EXERTIONAL ANGINA)



Acute Coronary Syndromes

- Plaque rupture → thrombus formation
- Subtotal occlusion
 - Unstable angina
 - Non-ST elevation myocardial infarction
- Total occlusion (100%)
 - ST-elevation myocardial infarction (STEMI)



Subtotal Occlusion



Total Occlusion



Sudden Death

- Common complication of CAD
- Plaque rupture → **arrhythmias**
- CAD is **most common cause of sudden death adults**
 - Younger patients: Hypertrophic cardiomyopathy (HCM)



Risk Factors

- Major risk is **prior coronary disease**
- Coronary risk equivalents
 - Diabetes
 - Peripheral artery disease
 - Chronic kidney disease



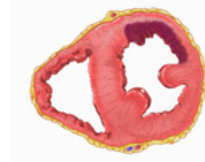
Risk Factors

- Hypertension
- Hyperlipidemia
- Family History (1st relative, M<50, F<60)
- Smoking
- Obesity, sedentary lifestyle

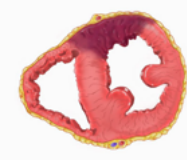
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Extent of Ischemia

- **Transmural ischemia**
 - Occurs with complete 100% flow obstruction (STEMI)
- **Subendocardial ischemia**
 - Occurs with flow obstruction but some distal blood flow
 - Stable angina, unstable angina, NSTEMI



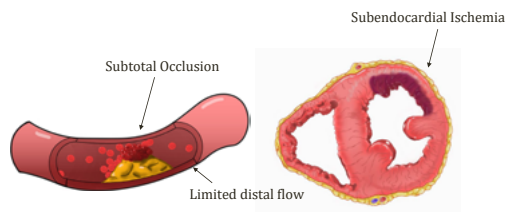
Subendocardial Ischemia



Transmural Ischemia

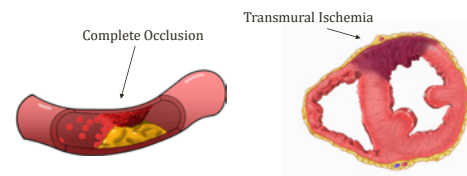
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Subendocardial Ischemia



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Transmural Ischemia



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Ischemic EKG changes

ST depressions



Subendocardial Ischemia

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Ischemic EKG changes

T wave inversions

- Many causes other than CAD
 - Cerebral T waves
- Resolving pericarditis
- Bundle branch blocks
- Ventricular hypertrophy



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Ischemic EKG changes

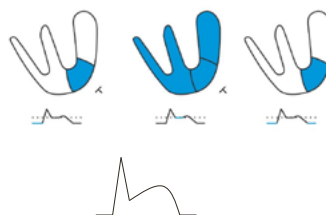
T wave inversions



Subendocardial Ischemia

Ischemic EKG changes

ST Elevations

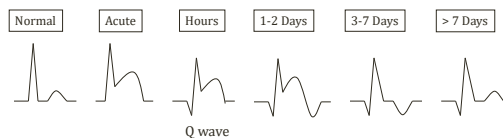


Seen only with transmural ischemia



Evolution of EKG changes

STEMI



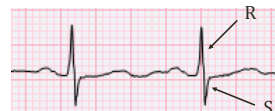
Hyperacute T waves

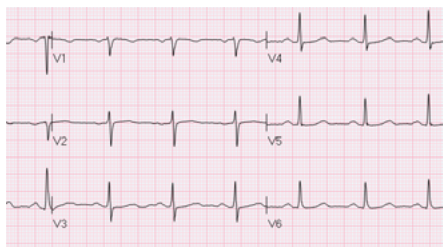
- Seen in transmural ischemia
- Early sign of ischemia
- Seen before ST elevations



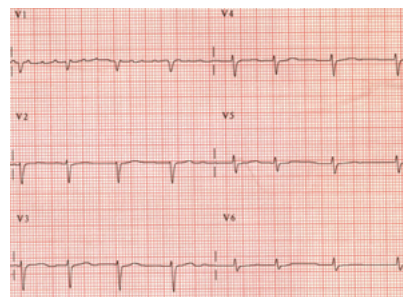
Poor R Wave Progression

- R wave increases (progresses) in size V1-V6
- Normally R>S waves seen by lead V3
- Poor progression seen in **anterior ischemia**
 - Acute or prior infarction





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Terminology

- Revascularization
- Angioplasty
- Coronary stenting
- Coronary bypass surgery

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Revascularization



Coronary angiogram: Before



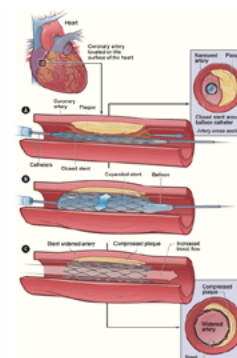
Coronary angiogram: After

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Coronary Stents

- Angioplasty: Reshape vessel
- Balloon angioplasty: Balloon inflation to open vessel
- Percutaneous Coronary Intervention (PCI)
- Stent placement
- About 600,000 stents/year implanted US

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Wikipedia/Public Domain

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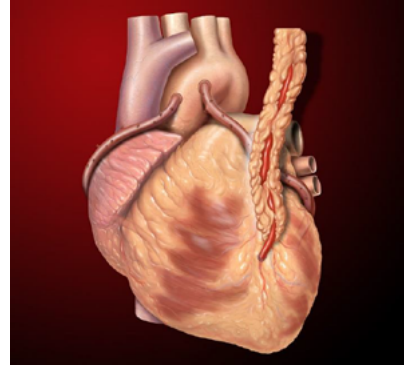
CABG

Coronary Artery Bypass Surgery

- "Bypass Surgery"
- Left Internal Mammary Artery (LIMA) Graft
- Saphenous (leg) Vein Grafts
- Radial (arm) Artery Grafts



Patrick J. Lynch/Wikipedia



Patrick J. Lynch/Wikipedia

Revascularization

Major Indications

- Angina
- Myocardial infarction
- Systolic dysfunction
 - Hibernating myocardium

Ischemic Pathologic Changes

Myocardium

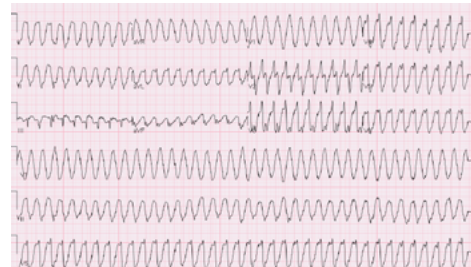
- Zero to 4 hrs
 - **No changes!**
- 4 – 12 hrs
 - Gross: Mottled
 - Micro: Necrosis, edema, hemorrhage
- 12-24 hrs
 - Gross: Hyperemia
 - Micro: Surrounding tissue inflammation
- 5 – 10 days
 - Gross: Central yellowing
 - Micro: Granulation tissue
- 7 weeks
 - Gross: Gray-white scar
 - Micro: Scar

Complications of Ischemia

- First 4 days
 - Arrhythmia
- 5 – 10 days
 - Free wall rupture
 - Tamponade
 - Papillary muscle rupture
 - VSD (septal rupture)
- Weeks later
 - Dressler's syndrome
 - Aneurysm
 - LV Thrombus/CVA

Cause of Death

0 – 4 days after MI



Cause of Death

5-10 days after MI

- **Free wall rupture**
 - Usually fatal – sudden death
 - May lead to tamponade
- **Papillary muscle rupture**
 - Acute mitral regurgitation (holosystolic murmur)
 - Heart failure, respiratory distress
 - More common inferior MIs
- **Septal rupture – VSD**
 - Loud, holosystolic murmur (thrill)
 - Hypotension, right heart failure (↑ JVP, edema)

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Ventricular Aneurysm

Weeks after MI

- More common **anterior infarction**
- Risk of thrombus → stroke, peripheral embolism



Patrick J. Lynch, medical illustrator/Wikipedia

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Ventricular Pseudoaneurysm

- **Rupture contained by pericardium/scar tissue**
- Not a true aneurysm
 - No endocardium or myocardium
- May rupture
- Presents as chest pain or dyspnea
- Often seen in the **inferior wall**
- Occurs earlier (<2 weeks) than true aneurysm

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Dressler's Syndrome

Weeks to months after MI

- Form of pericarditis
 - Chest pain
 - Friction rub
- Immune-mediated (details not known)
- Treatment: NSAIDs or steroids

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Fibrinous Pericarditis

- Occurs **days** after MI
 - Sometimes called "post-MI" pericarditis
 - Not autoimmune
 - Extension of myocardial inflammation
- Dressler's occurs **weeks** after MI
 - Sometimes called "post cardiac injury" pericarditis
- Rarely life-threatening

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Secondary Prevention

- Any CAD → ↑ risk of recurrent events
 - STEMI, NSTEMI, stable angina
- Preventative therapy used to lower risk
- Even in asymptomatic patients

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Secondary Prevention

- Several proven therapies for risk reduction
- **Aspirin**
- **Statins**
 - Atorvastatin, Rosuvastatin
- **Beta blockers**
 - Used in patients with prior infarction (STEMI/NSTEMI)

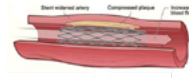


Ragesoss/Wikipedia

Stent Complications

Restenosis

- Slow, steady growth of scar tissue over stent
- "Neo-intimal hyperplasia"
- Re-occlusion of vessel
- Rarely life-threatening
- Slow, steady return of angina
- Most stents coated "drug eluting stents"
 - Metal stent covered with polymer
 - Polymer impregnated with drug to prevent tissue growth
 - Sirolimus



Stent Complications

Thrombosis

- Acute closure of stent
- Same as STEMI: life-threatening event
- Dual anti-platelet therapy for prevention
- Associated with **missed medication doses**



Stent Thrombosis Prevention

- "Dual antiplatelet therapy"
- Typically one year of:
 - Aspirin
 - Clopidogrel, Prasugrel or Ticagrelor
- After one year, stent metal no longer exposed to blood
 - "Endothelialization"
 - Risk of thrombosis is lower (but not zero)
 - Most patients take aspirin only

STEMI

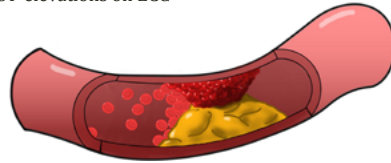
ST-Elevation Myocardial Infarction (STEMI)

Jason Ryan, MD, MPH

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STEMI

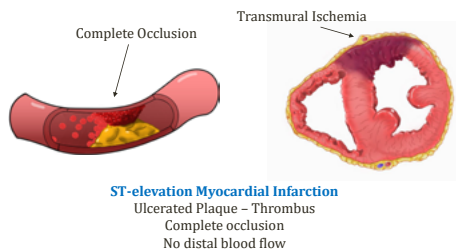
- Atherosclerotic plaque rupture
- Thrombus formation
- Complete (100%) vessel occlusion
- Ischemic chest pain
- ST-elevations on ECG



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STEMI

- **Transmural ischemia**



ST-elevation Myocardial Infarction
Ulcerated Plaque - Thrombus
Complete occlusion
No distal blood flow

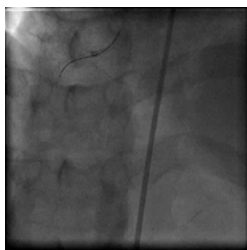
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STEMI



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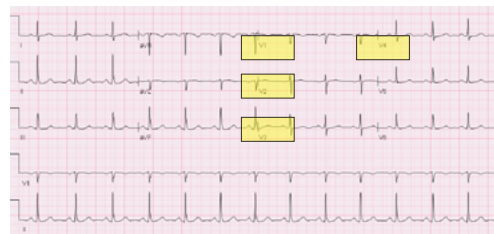
STEMI



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Leads go together

ST Elevations - Anterior



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Leads go together

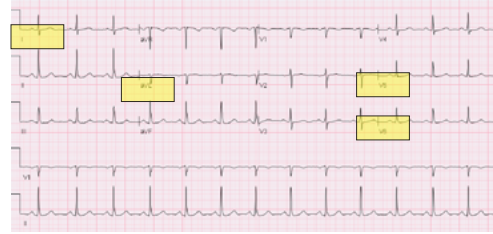
ST Elevations - Anterior



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Leads go together

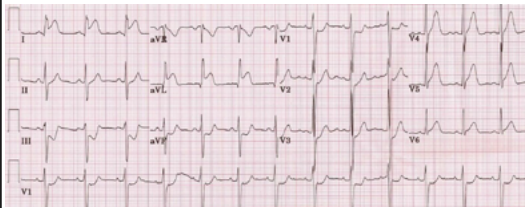
ST Elevations - Lateral



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Leads go together

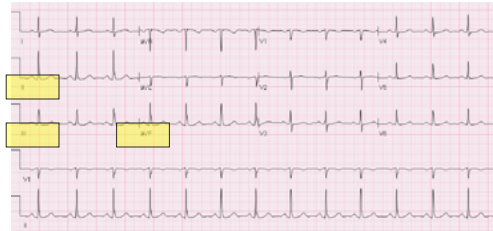
ST Elevations - Lateral



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Leads go together

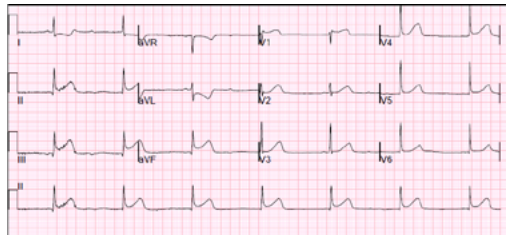
ST Elevations - Inferior



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Leads go together

ST Elevations - Inferior



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Coronary Artery Territories

- Left anterior descending artery
 - Anterior → V1-V4
- Left circumflex artery
 - Lateral → I, L, V5, V6
- Posterior descending artery
 - Inferior → II, III, F
 - Branch of **right coronary artery 90%**
 - LCX 10%

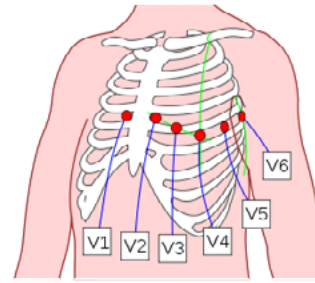
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Special Complications

Inferior MI

- Right ventricular infarction
 - Loss of right ventricular contractility
 - **Elevated jugular venous pressure**
 - Decreased preload to left ventricle → **hypotension**
 - Diagnosis: Right sided chest leads

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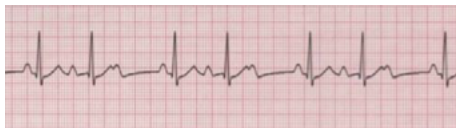
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Special Complications

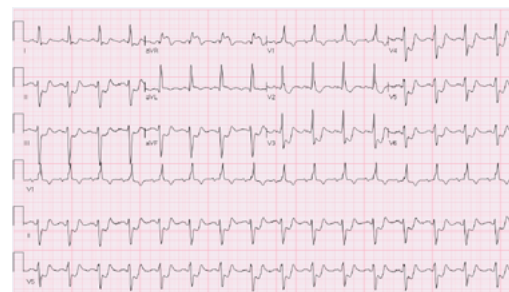
Inferior MI

- Sinus bradycardia and heart block
 - **Vagal stimulation** from inferior wall ischemia



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Left Main Occlusion



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Posterior Myocardial Infarction



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STEMI

Special Subtypes

- Left main
 - **ST-elevation aVR**
 - Diffuse ST depressions
- Posterior
 - Anterior ST depressions with standard leads
 - ST-elevation in **posterior leads (V7-V9)**

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Treatment of STEMI

- "Time is muscle"
- Coronary artery occluded by thrombus
- Longer occlusion → more muscle dies
 - More likely the patient may die
 - More heart failure symptoms
 - More future hospitalization for heart disease
- **Medical emergency**

Treatment of STEMI

- Main objective is to open the artery
 - "Revascularization"
- Option 1: Emergency angioplasty
 - Mechanical opening of artery
 - Should be done <90min
- Option 2: Thrombolysis
 - Lysis of thrombus with drug
 - Should be done <30min
- "Door to balloon" or "door to needle"

Treatment of STEMI

- Time matters
 - Medical therapy is supportive
 - Given while working to open artery
- Remember: this is a **thrombotic** problem
 - Aspirin to inhibit platelet aggregation
 - Heparin to inhibit clot formation
- This is also an **ischemic** problem
 - Beta blockers to reduce O2 demand
 - Nitrates to reduce O2 demand

Cautions

- Beta blockers
 - Inferior MI stimulates vagal nerve
 - **Bradycardia and AV block** can develop
- Nitrates
 - Occlusion of RCA can cause **RV infarct**
 - RV infarction → ↓ preload
 - Nitrates ↓ preload → **hypotension**

Other STEMI Treatments

- Clopidogrel
 - ADP receptor blocker
 - Inhibits platelets
- Eptifibatide
 - IIB/IIIA receptor blocker
 - Inhibits platelets
- Bivalirudin
 - Direct thrombin inhibitor
 - Inhibits clot formation

Typical STEMI Course

- Arrival in ER with chest pain 5:42pm
- EKG done 5:50pm
 - STEMI identified
- Cardiac cath lab activated for emergent angioplasty
- Meds given in ER
 - Aspirin
 - Metoprolol
 - Nitro drip
 - Heparin bolus
 - Transport to cath lab 6:15pm
- Artery opened with balloon 6:42pm
 - DTB time 60 minutes (ideal <90min)

Typical STEMI Course

- Arrival in ER with chest pain 5:42pm
- EKG done 5:54pm
 - STEMI identified
- Meds given in ER
 - Aspirin
 - Metoprolol
 - Nitro drip
 - Heparin bolus
- tPA given based on weight 6:07pm
 - IV push
 - Door to needle time 25min (ideal <30)

Unstable Angina/NSTEMI

NSTEMI and Unstable Angina

Jason Ryan, MD, MPH

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NSTEMI

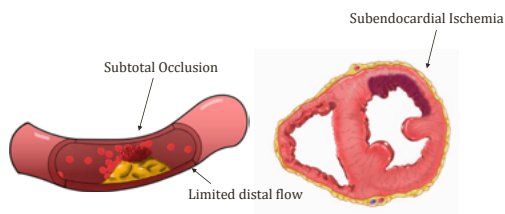
Non-ST-Elevation Myocardial Infarction

- Atherosclerotic plaque rupture
- Thrombus formation
- Subtotal (<100%) vessel occlusion
- Ischemic chest pain



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Subendocardial Ischemia

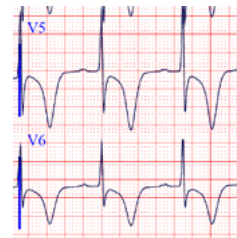


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NSTEMI

ECG Changes

- ST depressions
- T-wave inversions



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Cardiac Biomarkers

- Biomarkers spill into blood with cardiac injury
- Most common marker used: **Troponin I or T**
 - Increase 2-4 hours after MI
 - Stay elevated for weeks
- **CK-MB also used**
 - Increase 4-6 hours after MI
 - Normalize within 2-3 days

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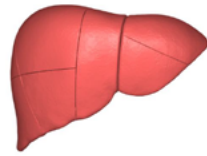
Cardiac Biomarkers

- Several types of CK
 - MM - Skeletal muscle
 - MB - Cardiac
 - BB - Brain
- Most tissues have some of all three
- Ratio of MB to total CK can be used in ischemia
 - Helpful when total CK also up due to muscle damage

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Cardiac Biomarkers

- Some AST found in cardiac cells
 - Abdominal pain with isolated ↑ AST could be MI



Bodyparts3D/Wikimedia Commons

Treatment of NSTEMI

- Thrombotic and ischemic syndrome (like STEMI)
- Unlike STEMI: No “ticking clock”
 - Subtotal occlusion
 - Some blood flow to distal myocardium
 - No emergency angioplasty
 - No benefit to thrombolysis
- Aspirin
- Beta blocker
- Heparin
- Angioplasty (non-emergent)

Typical NSTEMI Course

- Presents to ER with chest pain
- Biomarkers elevated
- Medical Therapy
 - Aspirin
 - Metoprolol
 - Heparin drip
- Admitted to cardiac floor
- Hospital day 2 → angiography
- 90% blockage of LAD → Stent

Unstable Angina

- Atherosclerotic plaque rupture
- Thrombus formation
- Subtotal (<100%) vessel occlusion
- Ischemic chest pain
- **Normal biomarkers**



Unstable Angina

- Diagnosis largely based on **patient history**
 - Chest pain increasing in frequency/intensity
 - Chest pain at rest
- ECG may show ST depressions or T wave inversions
- Treatment is same as for NSTEMI
- Condition often called “UA/NSTEMI”

Stable Angina

Stable Angina

Jason Ryan, MD, MPH

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Stable Angina

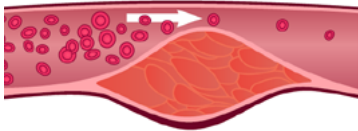
- Ischemic chest pain with exertion
- Relieved by rest
- Stable pattern over time
- **Stable coronary atherosclerotic plaque**
- No plaque rupture/thrombus



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Stable Angina

- Symptoms generally absent until ~75% occlusion
- Distal arteriolar dilation → normal flow if <75%



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Stable Angina

- Diagnosis: cardiac stress test
- Increases demand for O₂



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Wikipedia/Public Domain

Stable Angina

- NOT a thrombotic problem
- No role for heparin or antithrombotic drugs
- In US usually treated with revascularization
 - Most common indication PCI, CABG is stable angina
 - Recent clinical trials suggest medical therapy may work just as well as PCI/CABG in some patients

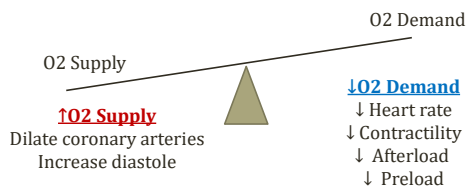
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Stable Angina: Typical Case

- 65-year old man with chest pain while walking
- Relieved with rest
- Presents to ED
 - EKG normal
 - Biomarkers normal
- Stress test
 - Walks on treadmill → chest pain, EKG changes
- Cardiac catheterization performed
- 90% LAD artery blockage
- Stent placed → angina resolved

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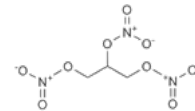
Medical Therapy for Ischemia



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Nitrates

- Converted to nitric oxide → vasodilation
- Predominant mechanism is **venous dilation**
 - Bigger veins hold more blood
 - Takes blood away from left ventricle
 - **Lowers preload (LVEDV)**
- Also arterial vasodilation (art << veins)
 - Increase coronary perfusion
 - Some peripheral vasodilation



Nitroglycerine

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Nitrates

- ↓ preload → ↓ cardiac output
- **Sympathetic nervous system activation**
- Increased heart rate/contractility
 - Increases O2 demand
 - Opposite of what we want to do for angina

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Nitrates

- Rare patients with complex CAD → angina
- In most patients, preload reducing effects dominate
 - Nitrates alone often improve angina
- **Co-administer beta-blocker or Ca channel blocker**
 - Blunts "reflex" effect

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Nitrates

Forms

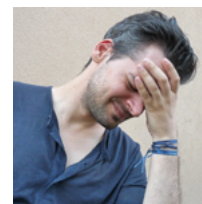
- Nitroglycerin Tablets/Spray
 - Rapid action ~5 minutes
 - Take during angina attack, before exercise
- Isosorbide Dinitrate
 - Effects last ~6hrs
- Isosorbide Mononitrate
 - Once daily drug
- Topical Nitroglycerin
 - Topical cream, patches

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Nitrates

Adverse Effects

- Headache (meningeal vasodilation)
- Flushing
- Hypotension
- Angina
 - Reflex sympathetic activation



phree/Pixabay/Public Domain

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Nitrate Tolerance

- Drug stops working after frequent use
- Avoid continuous use for more than 24 hours
- Does not occur with daily isosorbide mononitrate



Jpeterson101/Wikipedia

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Nitrate Withdrawal

- Nitrate withdrawal (rebound) after discontinuation
- Occurs when using large doses of long-acting nitrates
- Angina frequency will increase



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Monday Disease

- Workers in nitroglycerin manufacturing facilities
- Regular exposure to NTG in the workplace
- Leads to the development of **tolerance**
- Over the weekend workers lose the tolerance
- "Monday morning headache" phenomenon
 - Re-exposed on Monday
 - **Prominent vasodilation**
 - Tachycardia, dizziness, and a headache

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Beta Blockers

- **Slow heart rate and decrease contractility**
- Increase preload (LVEDV)
 - Slower heart rate = more filling time
 - Increase O₂ demand
 - Blunts some beneficial effect
- Reduced blood pressure (↓ afterload)
- **Net effect = less O₂ demand**

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Beta Blockers

- For angina, generally use cardioselective (β_1) drugs
 - Metoprolol, atenolol
- Some beta blockers are partial agonists
 - Pindolol, Acebutolol
- Don't use in angina

Boards&Beyond

Calcium Channel Blockers

- Three major classes of calcium antagonists
 - dihydropyridines (nifedipine)
 - phenylalkylamines (verapamil)
 - benzothiazepines (diltiazem)
- Vasodilators and negative inotropes

Boards&Beyond

Calcium Channel Blockers

- **Nifedipine: vasodilator**
 - Lower blood pressure
 - Reduce afterload
 - Dilate coronary arteries
 - May cause reflex tachycardia
- **Verapamil/diltiazem: negative inotropes**
 - Similar to beta blockers
 - Reduced heart rate/contractility
 - Can precipitate **acute heart failure** if LVEF very low

Antianginal Therapy

Nitrates/Beta Blockers

	Nitrates	Beta blockers	Nitrates + Beta blockers
Supply			
Coronary vasodilation	Increase	--	Increase
Duration diastole	↓ reflex	Increase	--
Demand			
Preload	Decrease	Increase	Decrease
Afterload	Decrease	Decrease	Decrease
Contractility	↑ reflex	Decrease	--/↓
Heart Rate	↑ reflex	Decrease	--/↓

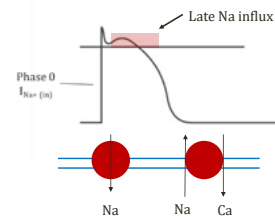
Antianginal Therapy

Calcium Channel Blockers

	Verapamil	Diltiazem	Nifedipine
Supply			
Coronary vasodilation	--	--	Increase
Duration diastole	Increase	Increase	↓ reflex
Demand			
Preload	Increase	Increase	--
Afterload	Decrease	Decrease	Decrease
Contractility	Decrease	Decrease	↑ reflex
Heart Rate	Decrease	Decrease	↑ reflex

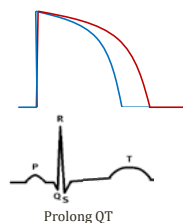
Ranolazine

- Inhibits **late sodium current**
- Reduces **calcium overload** → high wall tension
- Reduces **wall tension** and O2 demand



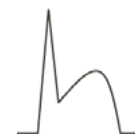
Ranolazine

- Constipation, dizziness, headache
- QT prolongation (blockade of K channels)



Variant (Prinzmetal) Angina

- Ischemia from **vasospasm**
 - Not caused by atherosclerotic narrowing
 - Often artery is "clean" with no stenosis
 - May also occur near sites of mild atherosclerosis
- **Spontaneous** episodes of angina
- Transient myocardial ischemia
- **ST-segment elevation on ECG**



Variant (Prinzmetal) Angina

- Episodes usually **at rest**
- Midnight to **early morning**
- Sometimes symptoms improve with exertion
- Associated with smoking

Variant (Prinzmetal) Angina

Diagnosis

- Usually based on history
- Intracoronary **ergonovine**
 - Acts on smooth muscle serotonergic (5-HT₂) receptors
 - Can be administered during angiography
 - Vasospasm visualized on angiogram
- Intracoronary **acetylcholine**
 - Acts on endothelial muscarinic receptors
 - Healthy endothelium → vasodilation via nitric oxide
 - Endothelial dysfunction → vasoconstriction
 - Vasospasm visualized on angiogram

Variant (Prinzmetal) Angina

Treatment

- Quit smoking
- Calcium channel blockers, nitrates
 - Vasodilators
 - Dilate coronary arteries, oppose spasm
- **Avoid propranolol**
 - Non selective blocker
 - Can cause unopposed alpha stimulation
 - Symptoms may worsen



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Coronary Steal

- Mechanism of angina
- Induced by drugs
- Blood flow increased to healthy vessels
- Blood flow decreased in stenotic vessels
- Blood “stolen” from diseased coronary vessels

Coronary Steal

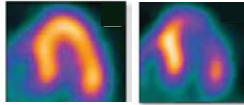
- Stenotic vessels
 - Significant (>75%) narrowing
 - **Arterioles maximally dilated** to maintain flow
- Normal vessels
 - No or minimal narrowing
 - **Arterioles NOT maximally dilated**

Coronary Steal

- **Vasodilator** administered
- Stenotic vessels → no response
 - Arterioles already maximally dilated
- Normal vessels → vasodilation
- Flow increases to normal vessels
- Flow decreases to abnormal vessels
- Results: ischemia due to coronary steal

Coronary Steal

- Rarely seen with nitrates, nifedipine
- Key principle for **chemical stress tests**
 - **Adenosine, persantine, regadenoson**
 - **Potent, short-acting vasodilators**
- Brief ↓ in blood flow to stenotic vessels → ischemia
- Nuclear tracers can detect ↓ blood flow

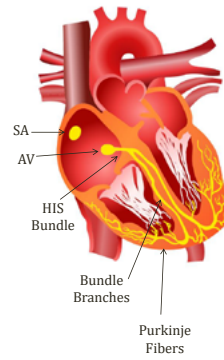


EKG Basics

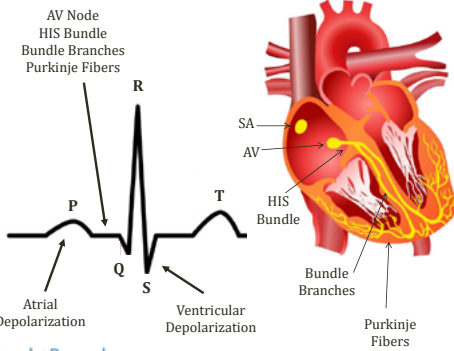
EKG Basics

Jason Ryan, MD, MPH

Boards&Beyond

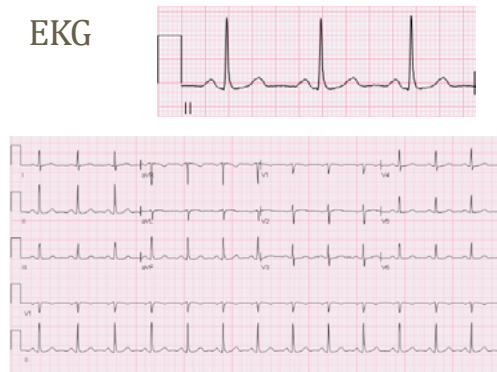


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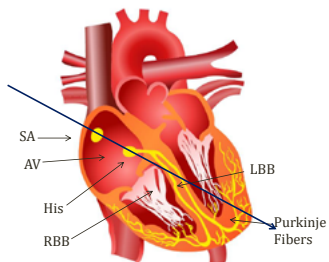
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EKG



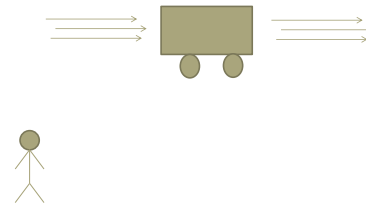
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EKG Electrical Activity



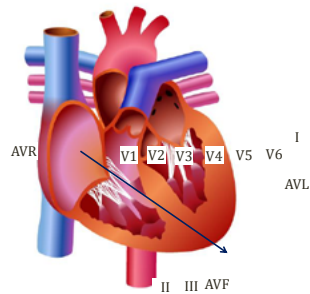
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EKG Electrical Activity



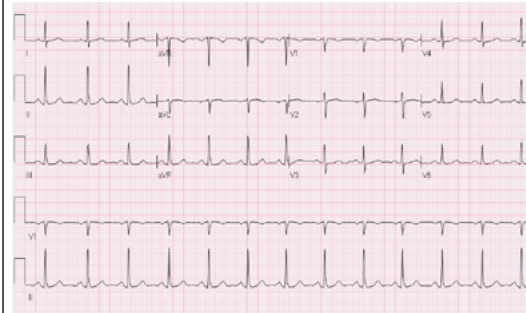
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EKG Electrical Activity



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EKG



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EKGs

Key Principles

- #1: Waves represent repolarization/depolarization
- #2: EKGs have 12 leads
 - Each lead watches the same thing
 - Each lead watches from different vantage point
 - Electrical activity toward lead = upward deflection
 - Electrical activity away from lead = negative deflection

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Pacemakers

- SA node is dominant pacemaker of the heart
- Other pacemakers exist but are *slower*
- If SA node fails, others takeover
 - SA node (60-100 bpm)
 - AV node (40-60 bpm)
 - HIS (25-40 bpm)
 - Bundle branches (25-40 bpm)
 - Purkinje fibers (25-40 bpm)

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Conduction Velocities

- SLOWEST conduction is through AV node
 - Very important so ventricle has time to fill
- Purkinje fibers → fastest conduction
- Purkinje > Atria > Vent > AV node

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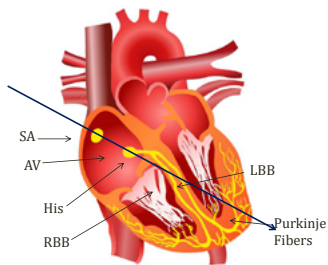
Determining Heart Rate

- 3 – 5 big boxes between QRS complex



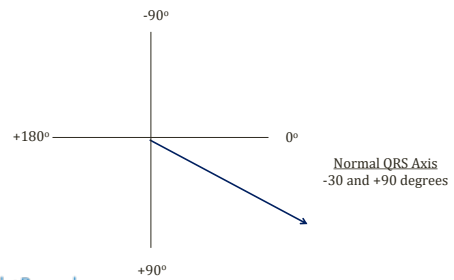
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QRS Axis



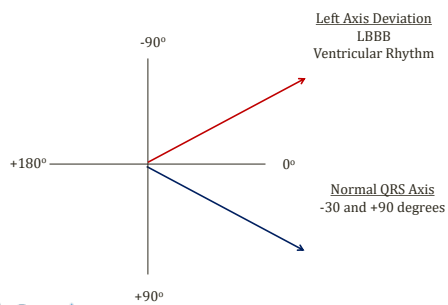
Boards&Beyond

QRS Axis



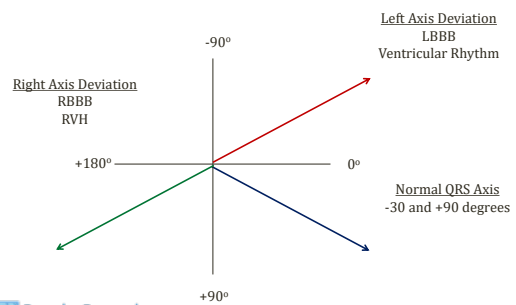
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QRS Axis



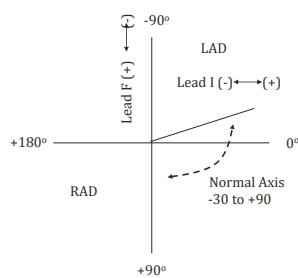
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QRS Axis



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Determining Axis



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Axis Quick Method

- First, glance at aVr.
- It should be negative
- If upright, suspect limb lead reversal

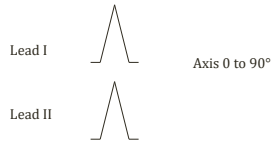


Normal

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Axis Quick Method

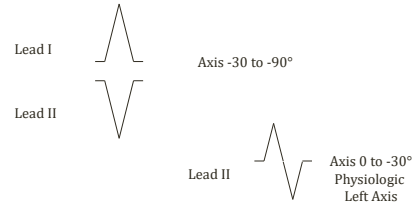
- If leads I and II are both positive, axis is normal



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Axis Quick Method

- For left axis deviation:
 - All you need is lead II



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Axis Quick Method

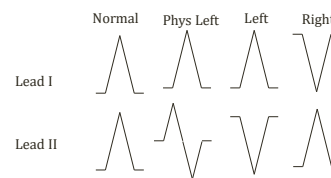
- For right axis deviation:
 - All you need is lead I
 - Negative = RAD



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Axis Quick Method

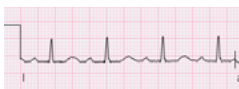
- Look at aVr: Make sure its negative
- Look at I, II: If both positive, axis is normal
- If II is negative: LAD
- If I is negative: RAD



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PR Interval

Normal PR
120-200ms



Prolonged PR
1° AV block



Short PR - WPW

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QRS Interval

Normal QRS
<120ms



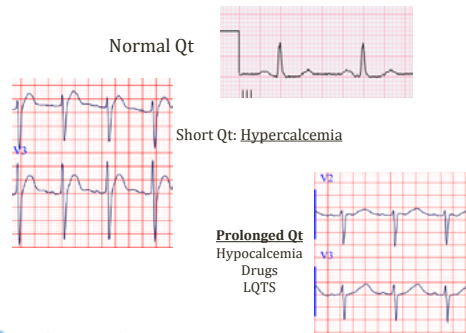
Right Bundle
Branch Block

Left Bundle
Branch Block



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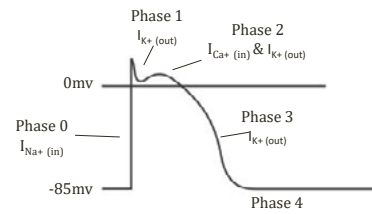
Qt Interval



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Calcium

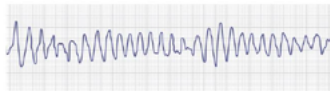
Myocyte Action Potential



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Torsade de Pointes

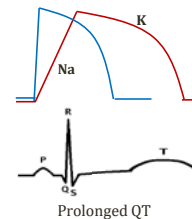
- Feared outcome of Qt prolongation
- Results in cardiac arrest
- **Antiarrhythmic drugs**
- **Hypokalemia, hypomagnesemia**
- Rarely from hypocalcemia



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Congenital Long Qt Syndrome

- Rare genetic disorder
- Abnormal K/Na channels



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Congenital Long Qt Syndrome

- Family history of sudden death (torsades)
- Classic scenario: Young patient recurrent "seizures"
 - EKG shows long Qt interval
- Jervell and Lange-Nielsen Syndrome
 - Norway and Sweden
 - Congenital deafness

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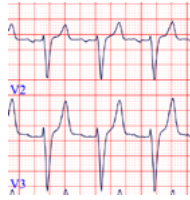
Acquired Long Qt Syndrome

- Antiarrhythmic drugs
- Levofloxacin (antibiotic)
- Haldol (antipsychotic)
- Many other drugs
- Congenital LQTS: need to avoid these drugs

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T waves

Peaked T waves
↑K
Early ischemia
(hyperacute)

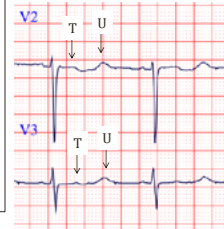


U waves

Origin unclear

May represent
repolarization of
Purkinje fibers

Can be normal but
also seen in
hypokalemia



High Yield EKGs

High Yield EKGs

Jason Ryan, MD, MPH

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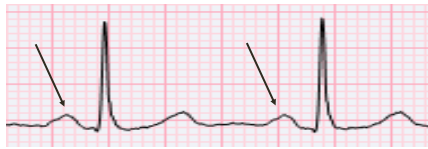
EKGs You Should Know

1. Sinus rhythm
2. Atrial Fibrillation/Flutter
3. Ischemia: ST elevations, ST depressions
4. Left bundle branch block
5. Right bundle branch block
6. PAC/PVC
7. 1st, 2nd, 3rd degree AV block
8. Ventricular tachycardia
9. Ventricular fibrillation/Torsades

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Step 1: Find the p waves

- Are p waves present?



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Sinus p waves

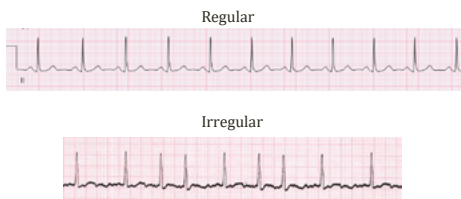
- Originate in sinus node
- Upright in leads II, III, F



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Step 2: Regular or Irregular

- Distance between QRS complexes (R-R intervals)



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Steps 1 & 2

- P waves present, regular rhythm
 - **Sinus rhythm**
 - Rare: atrial tachycardia, atrial rhythm
- No p waves, irregular rhythm
 - **Atrial fibrillation - irregularly irregular**
 - Atrial flutter with variable block

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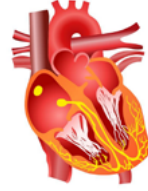
Steps 1 & 2

- P waves present, irregular rhythm
 - Sinus rhythm with PACs
 - Multifocal atrial tachycardia
 - Sinus with AV block
- No p waves, regular rhythm
 - Hidden p waves: retrograde
 - Supraventricular tachycardias (SVTs)
 - Ventricular tachycardia

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Step 3: Wide or narrow

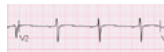
- Narrow QRS (<120ms; 3 small boxes)
 - His-Purkinje system works
 - No bundle branch blocks present
- Wide QRS
 - Most likely a **bundle branch block**
 - Ventricular rhythm (i.e. tachycardia)



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QRS Interval

Normal QRS



Right Bundle Branch Block

Left Bundle Branch Block



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Step 4: Check the intervals

- PR (normal <210ms; ~5 small boxes; ~1 big box)
 - Prolonged in AV block
 - Lengthens with vagal tone, drugs
 - Shortens with sympathetic tone
- QT (normal <1/2 R-R interval)
 - Prolonged with ↓ Ca (tetany; numbness; spasms)
 - Prolonged by antiarrhythmic drugs
 - Shortened with ↑ Ca (confusion, constipation)

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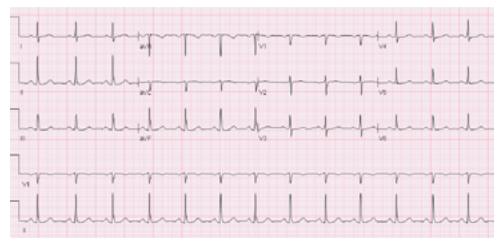
Step 5: ST segments

- T wave abnormalities
 - Inverted: ischemia
 - Peaked: Early ischemia, hyperkalemia (↑K)
 - Flat/U waves: Hypokalemia (↓K)
- ST Depression
 - Subendocardial ischemia
- ST Elevation
 - Transmural ischemia



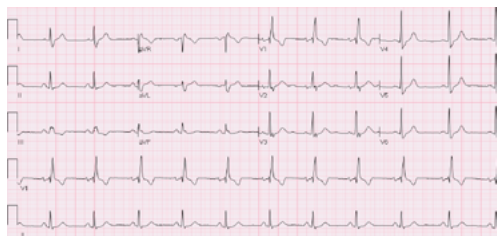
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Normal Sinus Rhythm



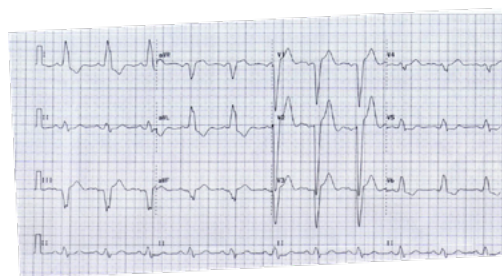
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Right Bundle Branch Block



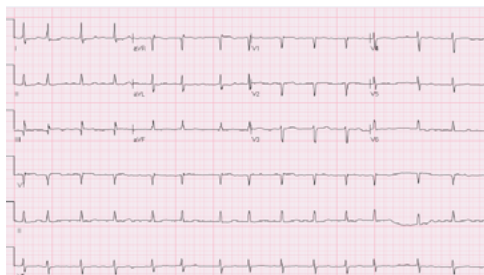
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Left Bundle Branch Block



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Atrial Fibrillation



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Atrial Flutter



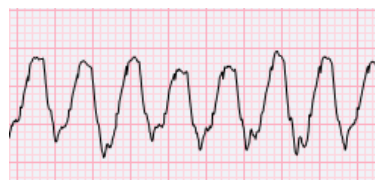
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Ventricular Tachycardia



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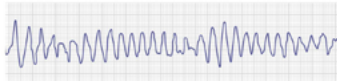
Ventricular Tachycardia



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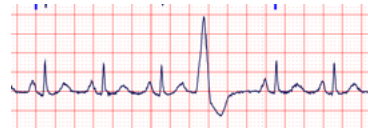
Torsades de pointes

- ↑ risk with **prolonged Qt interval**
 - Antiarrhythmic drugs
 - Congenital long Qt syndrome
 - Antibiotics (erythromycin, quinolones)
- Hypokalemia
- Hypomagnesemia
- Rarely hypocalcemia



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PAC and PVC



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Action Potentials

Cardiac Action Potentials

Jason Ryan, MD, MPH

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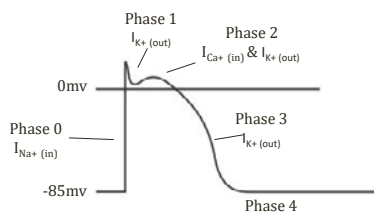
Cardiac Action Potential

- Changes in membrane voltage of cell
- Transmit electrical signals through heart
- Triggers contraction of myocytes

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Myocyte Action Potential

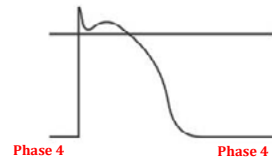
Atrial/ventricular myocytes



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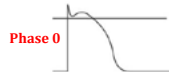
Phase 4

- Resting potential: about -85mV
- Constant outward leak of K^+
- "Inward rectifier channels"
- Na^+ and Ca^{2+} channels are closed



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Phase 0

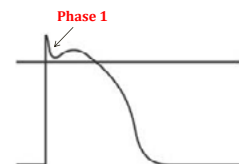


- Nearby myocyte raise membrane potential
- **Gap junctions**
- Rising potential opens "**Fast**" Na^+ channels
- Threshold potential reached (about -70mV)
- Large Na^+ current $\rightarrow$ rapid depolarization
- Membrane potential overshoots ($>0\text{mV}$)
- Fast Na^+ channels close
- **Class I antiarrhythmic drugs**: block Na channels

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Phase 1

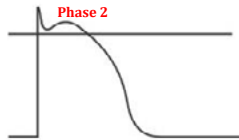
- Membrane potential is positive
- **K^+ channels open**
- Outward flow of K^+ returns membrane to $\sim 0\text{mV}$



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Phase 2

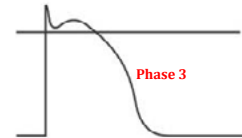
- L-type Ca^{2+} channels open $\rightarrow$ inward Ca^{2+} current
- Contraction trigger: **excitation-contraction coupling**
- K^+ leaks out (down concentration gradient)
- Delayed rectifier K^+ channels
- **Balanced flow in/out** = plateau of membrane charge
- **Verapamil/Diltiazem** = block L-type Ca channels



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Phase 3

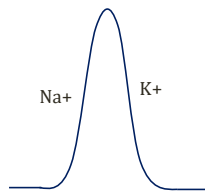
- Ca^{2+} channels inactivated
- **Persistent outflow of K^+**
- Resting potential back to -85 mV
- **Class III antiarrhythmic drugs**: block K channels



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Skeletal Muscle

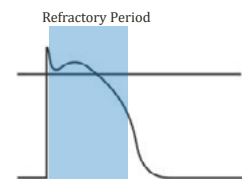
- No plateau (phase 2)
- No gap junctions
- Each cell has its own NMJ



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Refractory Period

- Phase 0 until next possible depolarization
- Determines how fast myocyte can conduct
- Many antiarrhythmic drugs prolong refractory period

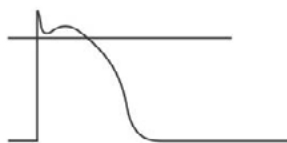


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Myocyte Action Potential

Atrial/ventricular myocytes

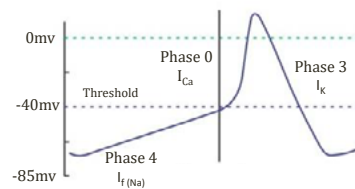
- Similar AP in HIS, bundle branches, Purkinje fibers



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Pacemaker Action Potential

SA node, AV node



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Pacemaker Action Potential

SA node, AV node

- **Funny current** (pacemaker current)
 - Spontaneous flow of Na^+
- About -40 mV : threshold potential
- L-type Ca^{2+} channels open $\rightarrow$ depolarize cell
- Delayed rectifier K^+ channels open
- Return cell to -60 mV

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Pacemaker Action Potential

SA node, AV node

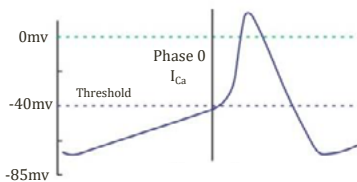
- Automaticity
 - Do not require stimulation to initiate action potential
 - Capable of self-initiated depolarization
- No fast Na^+ channel activity
 - Fewer inward rectifier K^+ channels
 - Membrane potential never lower than -60 mV
 - Fast Na^+ channels need -85 mV to function

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Pacemaker Action Potential

SA node, AV node

- Any drug that slows this AP may cause two effects:
 - **Slower heart rate** (sinus node)
 - **Slower AV conduction** (AV node)

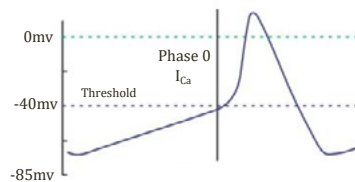


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Pacemaker Action Potential

SA node, AV node

- Two key drug classes that affect pacemaker AP
 - **Calcium channel blockers** (Verapamil/Diltiazem)
 - **Beta blockers**

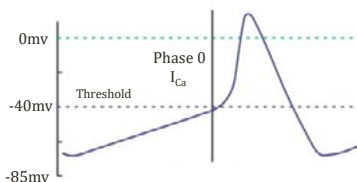


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Pacemaker Action Potential

SA node, AV node

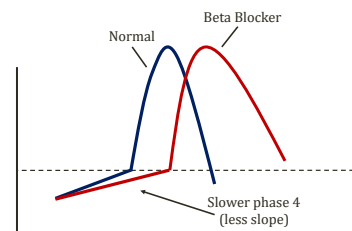
- **Verapamil/Diltiazem** = block L-type Ca channels
 - Slow rate of sinus depolarization (slow heart rate)
 - Slow AV node conduction



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Beta Blockers

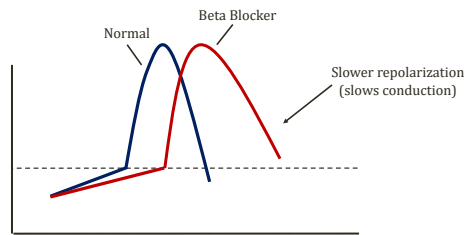
- Modify slope of phase 4
- Less slope $\rightarrow$ longer to reach threshold $\rightarrow$ $\downarrow$ HR



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Beta Blockers

- Also prolong repolarization
- Slow AV node conduction

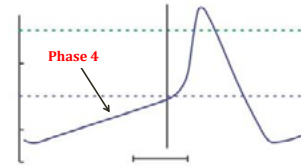


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Slope of Phase 4

Sinus Node

- Changes in slope modify heart rate
- Decrease slope (slower rise)
 - Parasympathetic NS, beta blockers, adenosine
- Increased slope (faster rise)
 - Sympathetic NS, sympathomimetic drugs



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Pacemakers

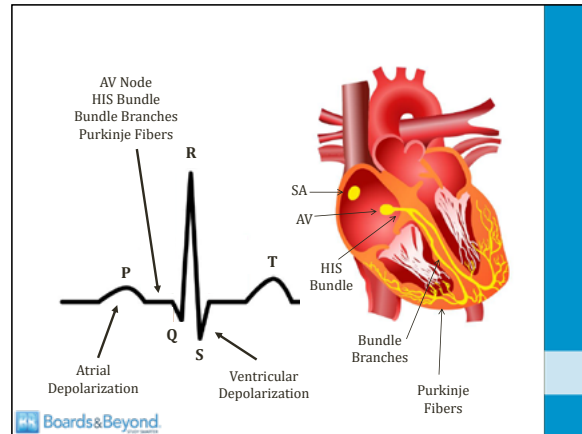
- Many cardiac cells capable of automaticity
- SA node normally dominates
 - Fastest rise in phase 4
 - Controls other pacemaker cells
- Pacemakers: SA Node > AV Node > Bundle of HIS
 - SA node (60-100 bpm)
 - AV node (40-60 bpm)
 - HIS (25-40 bpm)

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AV and Bundle Branch Blocks

AV and Bundle Branch Blocks

Jason Ryan, MD, MPH



AV Blocks

- Slowed or blocked conduction atria → ventricles
- Can cause prolonged PR interval
- Can cause non-conducted p wave



Prolonged PR Interval



Non-conducted P wave



AV Blocks

Symptoms

- Often incidentally noted on EKG
 - Especially milder forms with few/no non-conducted p waves
- Can cause bradycardia
 - Occurs when many or all p waves not conducted
 - Fatigue, dizziness, syncope
 - Symptomatic AV block often treated with a pacemaker



AV Blocks

Anatomy

- Caused by disease in AV conduction system
 - AV node → HIS → Bundle Branches → Purkinje fibers
- Divided into two causes
 - **AV node disease**
 - **HIS-Purkinje disease**



AV Blocks

Anatomy

- AV node disease
 - Usually less dangerous
 - Conduction improves with exertion (sympathetic activity)
- HIS-Purkinje disease
 - **More dangerous**
 - Usually does not improve with exertion
 - Often progresses to complete heart block
 - Often requires a pacemaker



AV Blocks

Four Types

- Type I
 - Prolongation of PR interval only
 - All p waves conducted
- Type II
 - Some p waves conducted
 - Some p waves NOT conducted
 - Two sub-types: Mobitz I and Mobitz II
- Type III
 - No impulse conduction from atria to ventricles

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1st degree AV Block



Prolonged PR (normal <200ms)
Block usually in AV Node
 Beta blockers
 Calcium channel blockers
 Well-trained athletes

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2nd degree AVB

Mobitz I/Wenckebach

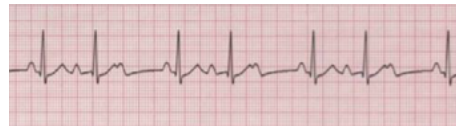


Block usually in AV Node
 Progressive PR prolongation
 Grouped Beating
 RR intervals NOT regular
 Similar causes as 1st degree AV block

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2nd degree AVB

Mobitz I/Wenckebach



Block usually in AV Node
 Progressive PR prolongation
 Grouped Beating
 RR intervals NOT regular
 Similar causes as 1st degree AV block

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2nd degree AVB

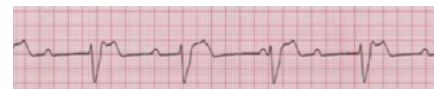
Mobitz II



Block usually in the HIS-Purkinje System
 Often seen with bundle branch block
 Usually symptomatic
 Dizziness, syncope

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3rd degree AVB



Block usually in the HIS-Purkinje System
 Regular RR intervals excludes Wenckebach

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3rd degree AVB



Block usually in the HIS-Purkinje System
Regular RR intervals excludes Wenckebach

Lyme Disease

- Spirochete infection with **Borrelia burgdorferi**
- Stage 2: Lyme carditis
- Varying degrees of AV block
 - 1st, 2nd, 3rd
- AV block improves with antibiotics



Image courtesy of Wikipedia/Public Domain

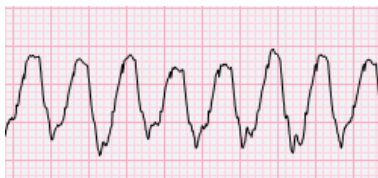
Vocabulary

- **Complete heart block**
 - Impulses cannot be transmitted from atria to ventricle
- **AV dissociation**
 - Atria and ventricular depolarization uncoupled ("dissociated")
 - Can be caused by complete heart block
 - Also occurs if ventricular rate > sinus rate (no heart block)
 - Seen in ventricular tachycardia and other rhythms

Ventricular Tachycardia



Ventricular Tachycardia



Escape Rhythm

- SA node: Dominant (fastest) pacemaker
- Heart block: SA cannot send impulses to ventricles
- Other pacemakers exist but are slower
 - SA node (60-100 bpm)
 - AV node (40-60 bpm)
 - HIS (25-40 bpm)
 - Bundle branches (25-40 bpm)
 - Purkinje fibers (25-40 bpm)

Escape Rhythm

- Heart block: lower pacemaker depolarizes ventricles
 - "Escape rhythm"
- Rate of lower pacemaker determines symptoms
 - Very slow: dizziness, syncope, hypotension
 - Less slow: fatigue, exercise intolerance

Sites of AV Block

Disorder	Common Site of Block
1 st Degree	AV node
Mobitz I	AV Node
Mobitz II	His-Purkinje System
3 rd degree (Complete)	His-Purkinje System

Causes of Heart Block

- Drugs
 - Beta blockers, calcium channel blockers
 - Digoxin
- High vagal tone
 - Athletes
- Fibrosis and sclerosis of conduction system

Pacemaker

- Treatment for "high grade" AV block
- Usually 3rd degree or Mobitz II
- Often in patients with symptoms (syncope, dizziness)

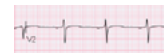


Bundle Branch Blocks

- Both bundle branches blocked
 - Results in AV block
 - Form of HIS-Purkinje system disease
- **ONE** bundle branch blocked
 - Does not cause AV block
 - Normal PR interval
 - **QRS will be prolonged**

Bundle Branch Blocks

Normal QRS

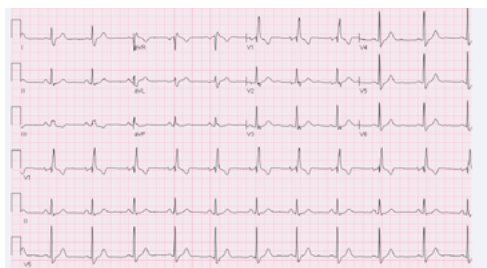


Right Bundle Branch Block

Left Bundle Branch Block



Right Bundle Branch Block



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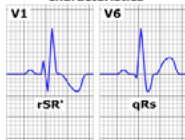
Left Bundle Branch Block



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Bundle Branch Blocks

Right bundle branch block characteristics



Left bundle branch block characteristics



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A. Rad/Wikipedia

Bundle Branch Blocks

- Symptoms: **None**
 - Identified incidentally on ECG
- May progress to AV block (need for pacemaker)
- Interfere with detection of ischemia
 - ST elevations, T-wave inversions can be normal

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Bundle Branch Blocks

Causes

- Often caused by slowly progressive fibrosis/sclerosis
- More common in older patients
- Can result from "structural heart disease"
- LBBB: Prior MI, cardiomyopathy
- RBBB: Right heart failure

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Atrial Fibrillation

Atrial Fibrillation and Flutter

Jason Ryan, MD, MPH

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Atrial Fibrillation

- Cardiac arrhythmia
- Results in an **irregularly, irregular** pulse
- Can cause palpitations, fatigue, dyspnea
- Diagnosis: EKG

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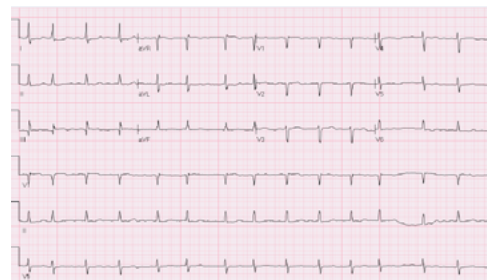
Atrial Fibrillation



J. Heuser/Wikipedia

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Atrial Fibrillation



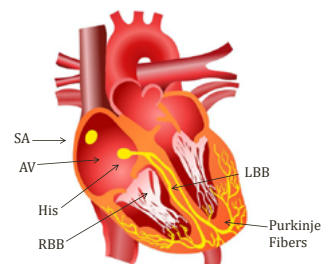
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Atrial Fibrillation



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Atrial Fibrillation



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Atrial Fibrillation

Terminology

- Paroxysmal
 - Comes and goes; spontaneous conversion to sinus rhythm
- Persistent
 - Lasts days/weeks; often requires cardioversion
- Permanent

Atrial Fibrillation

Symptoms

- Wide spectrum of symptoms



Asymptomatic
Heart Rate <100bpm

Palpitations, Dyspnea, Fatigue
Heart Rate >100bpm

Cardiomyopathy

- Caused by untreated, **rapid** atrial fibrillation
- “Tachycardia-induced cardiomyopathy”
- ↓ LVEF
- Systolic heart failure

Heart Rate

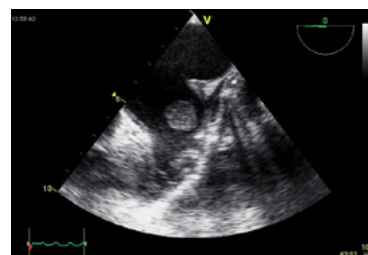
- **AV node refractory period** determines heart rate
- Young, healthy patients → rapid heart rate
- Older patients → slower heart rate
- Atrial rate in fibrillation: 300-500bpm
- Ventricular rate: 70-180bpm

Preload

- Atrial fibrillation eliminates ventricular pre-filling
- “Loss of atrial kick”
- **Decreases preload**
- Can lead to low cardiac output and hypotension
- Especially in “preload dependent” patients
 - Aortic stenosis
 - LVH or diastolic heart failure (stiff ventricle)

Atrial Fibrillation

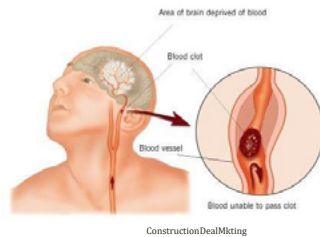
Thrombus in Left Atrial Appendage



Atrial Fibrillation

Cardiac Embolism

- Brain (stroke)
- Gut (mesenteric ischemia)
- Spleen



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Valvular Atrial Fibrillation

- Associated with rheumatic heart disease
- Usually **mitral stenosis**
- Often refractory to treatment
- VERY high risk of thrombus
- Non-valvular: not associated with rheumatic disease

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Atrial Fibrillation

Risk Factors

- **Age**
 - ~10% of patients >80
 - <1% of patients <55
- More common in women
- Most common associated disorders: **HTN, CAD**
- Anything that dilates the atria → atrial fibrillation
 - Heart failure
 - Valvular disease
- Key diagnostic test: Echocardiogram

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Hyperthyroidism

- Commonly leads to atrial fibrillation
- Reversible with therapy for thyroid disease
- Atrial fibrillation therapies less effective
- Key diagnostic test: **TSH**

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Atrial Fibrillation

Triggers

- Often no trigger identified
- Binge drinking ("holiday heart")
- Increased catecholamines
 - Infection
 - Surgery
 - Pain

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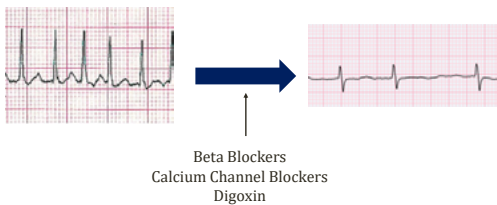
Atrial Fibrillation

Treatment

- Heart Rate
 - "Rate control"
 - Ideally <110bpm
- Heart Rhythm
 - "Rhythm control"
 - Restoration of sinus rhythm
- Anticoagulation

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Rate Control



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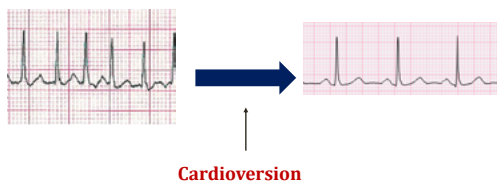
Rate Control

- Use drugs that **slow AV node conduction**
- Beta blockers
 - Usually β_1 selective agents
 - Metoprolol, Atenolol
- Calcium channel blockers
 - Verapamil, Diltiazem
- Digoxin
 - Increases parasympathetic tone to heart

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Rhythm Control

- Goal: restore sinus rhythm



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Cardioversion

- **Electrical**
 - Deliver "synchronized" shock at time of QRS
 - Administer anesthesia
 - Deliver electrical shock to chest
 - All myocytes depolarize
 - Usually sinus node first to repolarize/depolarize



Pollo/Wikipedia

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Cardioversion

- **Chemical**
 - Administration of antiarrhythmic medication
 - Often Ibutilide (class III antiarrhythmic)
 - Less commonly used due to drug toxicity

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Cardioversion

- **Spontaneous**
 - Often occurs after hours/days

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Cardioversion

Risk of Stroke

- Chemical/electrical cardioversion may cause stroke
- 48hours required for thrombus formation
- Symptoms <48hours: cardioversion safe
- Symptoms >48hours (or unsure)
 - Anti-coagulation 3 weeks → cardioversion
 - Transesophageal echocardiogram to exclude thrombus
- Exception: Hypotension/shock
 - Emergent cardioversion performed

Rhythm Control

- **Antiarrhythmic medications**
- Administered before/after cardioversion
- Class I drugs
 - Flecainide, propafenone
- Class III drugs
 - Amiodarone, sotalol, dofetilide

Anticoagulation

- Warfarin
 - Requires regular INR monitoring
 - Goal INR usually 2-3
- Rivaroxaban, Apixaban
 - Factor X inhibitors
- Dabigatran
 - Direct thrombin inhibitor
- Aspirin
 - Less effective
 - Only used if risk of stroke is very low
 - Less risk of bleeding

Anticoagulation

- Whether atrial fibrillation persists or sinus rhythm restored anticoagulation **MUST** be administered
- Studies show similar stroke risk for rate control versus rhythm control

Stroke Risk

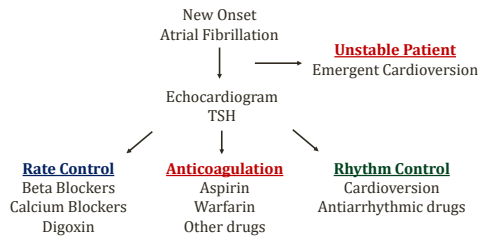
- **CHADS Score**
 - CHF (1 point)
 - HTN (1 point)
 - Age >75yrs (1 point)
 - Diabetes (1 point)
 - Stroke (2 points)
- Score ≥ 2 = Warfarin or other anticoagulant
- Score 0 -1 = Aspirin

Stroke Risk

- **CHADS VASC Score**
 - CHF (1point)
 - HTN (1point)
 - Diabetes (1point)
 - Stroke (2points)
 - Female (1point)
 - Age 65-75 (1point)
 - Age >75yrs (2points)
 - Vascular disease (1point)
- Score ≥ 2 = Warfarin or other anticoagulant
- Score 0 -1 = Aspirin

Atrial Fibrillation

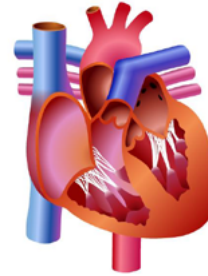
Summary



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Pulmonary Vein Isolation

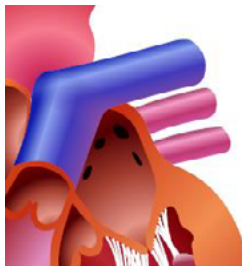
Surgical Therapy for Atrial Fibrillation



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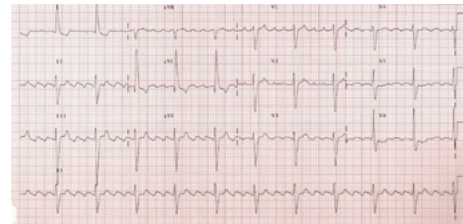
Pulmonary Vein Isolation

Surgical Therapy for Atrial Fibrillation



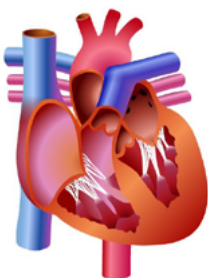
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Atrial Flutter



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Atrial Flutter



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Atrial Flutter

Symptoms

- Generally the same as atrial fibrillation
- May be asymptomatic
- Palpitations, dyspnea, fatigue

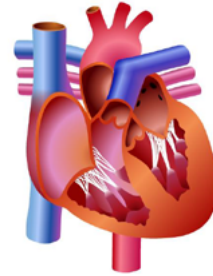
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Atrial Flutter

Treatment

- Generally the same as atrial fibrillation
- Rate or rhythm control
- Rate-slowing drugs
- Cardioversion
- Anticoagulation based on stroke risk

Atrial Flutter Ablation



AVNRT

AVNRT

Jason Ryan, MD, MPH

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PSVT

Paroxysmal Supraventricular Tachycardia

- Intermittent tachycardia (HR > 100bpm)
- **Sudden onset/offset**
 - Contrast with sinus tachycardia
- Electrical activity originates above ventricle
 - "Supraventricular"
 - Contrast with ventricular tachycardia
 - Produces narrow QRS complex (<120ms)



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PSVT

Paroxysmal Supraventricular Tachycardia

- Often causes **sudden-onset palpitations**
- Chest discomfort
- Rarely syncope



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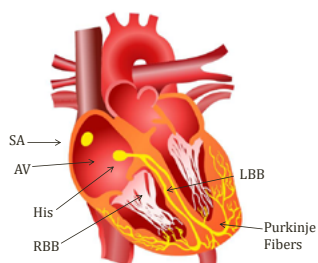
AVNRT

Atrioventricular nodal reentrant tachycardia

- **Most common cause of PSVT**
- More common in young women
- Mean age onset: 32 years old
- Requires **dual AV nodal pathways**

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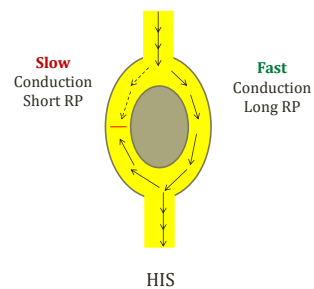
Normal Conduction



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Dual Pathways

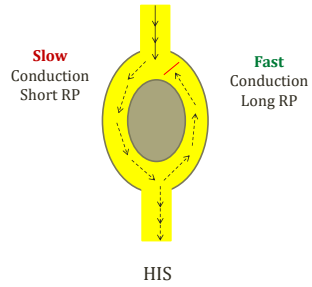
Sinus Rhythm



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Dual Pathways

PAC



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Retrograde P Waves



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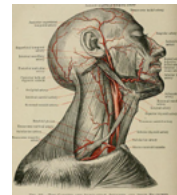
AVNRT

- Recurrent episodes of palpitations
- Many episodes spontaneously resolve
- ↓ conduction in AV node breaks arrhythmia
 - Will halt conduction in slow pathway
- Carotid massage
- Vagal maneuvers
- Adenosine

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Carotid Massage

- Examiner presses on neck near carotid sinus
- **Stretch of baroreceptors**
- CNS response as if **high blood pressure**
- Increased vagal tone
- ↓ AV node conduction



Wikipedia/Public Domain

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Vagal Maneuvers

- **Valsalva**
 - Patient bears down as if moving bowels
 - Increased thoracic pressure
 - Aortic pressure rises → ↓ heart rate and AV conduction
- Breath holding
- Coughing
- Deep respirations
- Gagging
- Swallowing

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AVNRT

Chronic Treatment

- Many patients need no therapy
- Beta blockers, Verapamil/Diltiazem
 - Slow conduction in slow pathway
- Surgical ablation of slow pathway

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WPW

Wolff-Parkinson White

Jason Ryan, MD, MPH

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WPW Syndrome

Wolff-Parkinson White Syndrome

- Cardiac electrical disorder
- **"Accessory atrioventricular pathway"**
 - Conducts impulses from atria to ventricles
 - Bypasses AV node
 - "Bundle of Kent"
- Causes **"pre-excitation"**
 - Ventricular depolarization before AV nodal impulse
- May lead to **arrhythmias**

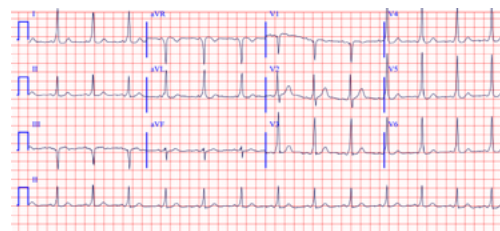
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EKG in WPW



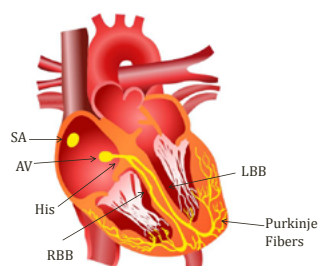
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WPW EKG



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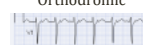
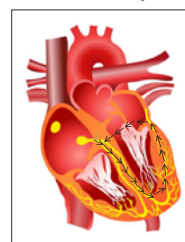
Cardiac Electrical System



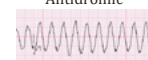
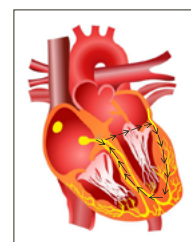
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AVRT

AV Re-entrant Tachycardia



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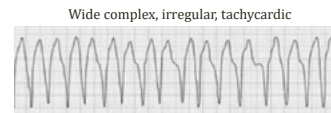


Bypass Tract Consequences

- Most patients **asymptomatic**
 - EKG with delta wave only
 - Called **WPW "pattern"**
- Some have tachycardias
 - Presents as **palpitations**
 - Called **WPW syndrome**
 - AVRT (anti or orthodromic)
- Rarely causes syncope or sudden death
- Treatment: Ablation of accessory pathway

Atrial Fibrillation in WPW

- Atrial fibrillation can be **life threatening**
- Atrial depolarization rate 300-500/min
- AV node conducts <200/min
- Impulses may conduct rapidly over bypass tract



Atrial Fibrillation in WPW

- Slowing AV node conduction is dangerous
- Allows more impulses over bypass tract
- **Usual atrial fibrillation therapies contraindicated**
 - Beta blockers
 - Calcium channel blockers
 - Digoxin
 - Adenosine
- Acute treatment: Cardioversion or antiarrhythmics
 - Ibutilide, procainamide
 - Slow conduction in bypass tract

Antiarrhythmic Drugs

Antiarrhythmic Drugs

Jason Ryan, MD, MPH

Boards&Beyond

Vaughan Williams

Class I	Class II
Quinidine } Ia Procainamide } Lidocaine } Ib Mexiletine } Flecainide } Ic Propafenone }	Beta Blockers
Class III	Class IV
Amiodarone Sotalol Dofetilide Ibutilide	Ca-channel Blockers (Verapamil/Diltiazem)

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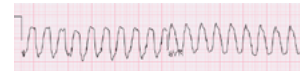
Use of Antiarrhythmic Drugs

- Drugs used to "suppress" arrhythmias
- Prevent formation of aberrant impulses
- Most also **cause** arrhythmias
- Can lead to cardiac arrest and death
- Used in dangerous arrhythmias
- Also used in recurrent symptomatic arrhythmias

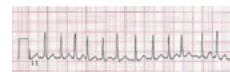
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Use of Antiarrhythmic Drugs

- Persistent/recurrent ventricular tachycardia
- Recurrent atrial fibrillation



Ventricular Tachycardia



Rapid Atrial Fibrillation

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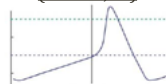
Mechanisms

- All drugs slow cardiac electrical activity
- Class I drugs → Block Na channel
- Class III drugs → Block K channels
- Class II, IV: Slow sinus and AV node conduction

Class I, III



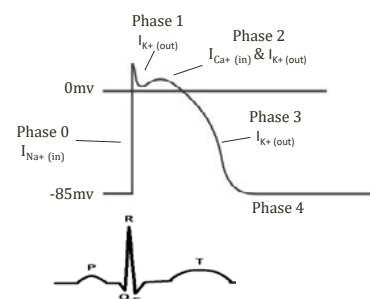
BB, CCB
(Class II, IV)



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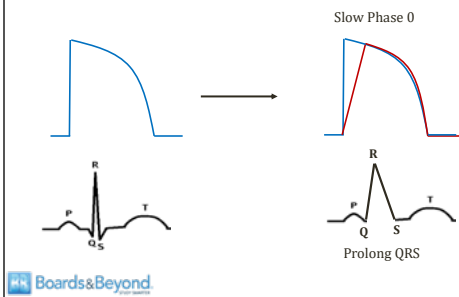
Myocyte Action Potential

Atrial/ventricular myocytes

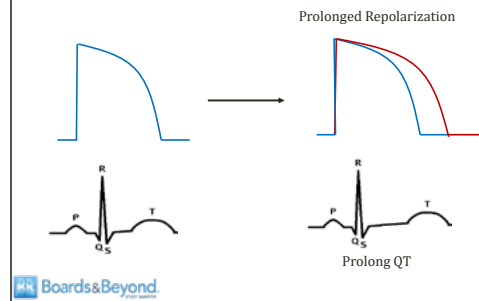


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Na Channel Blockade

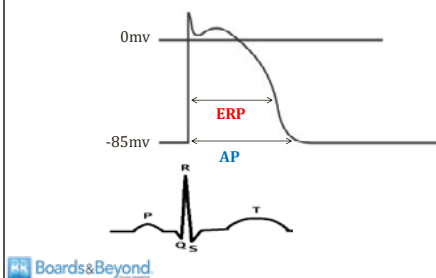


K Channel Blockade



Myocyte Action Potential

Atrial/ventricular myocytes



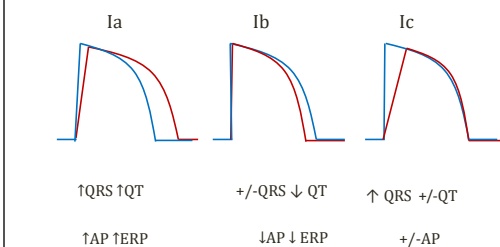
Class I drugs

- Block sodium channels → prolong QRS
- Some also affect K⁺ channels → prolong Qt
- Can prolong action potential duration
- Can prolong effective refractory period

Boards&Beyond

Class I drugs

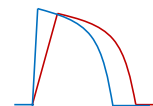
Effects on Resting Action Potential



Class Ia Drugs

Quinidine, procainamide

- **Prolong QRS**
- **Can also prolong Qt** (↓K⁺ outflow)
- Quinidine
 - Oral drug
 - Can decrease recurrence rate of atrial fibrillation
 - Associated with increased mortality
- Procainamide
 - Intravenous drug
 - Slows conduction in accessory pathways (WPW)
 - Used in arrhythmias associated with bypass tracts



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Procainamide

- Associated with **drug-induced lupus**
 - Classic drugs: **INH, hydralazine, procainamide**
- Often rash, arthritis, anemia
- Antinuclear antibody (ANA) can be positive
- Key features: **anti-histone antibodies**
- Resolves on stopping the drug



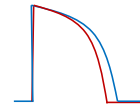
Pixabay/Public Domain

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Class Ib Drugs

Lidocaine, Mexiletine

- Little/no effect on QRS at normal HR
- Slight decrease in Qt interval (minimal)
- Least effect on action potential of class 1 drugs**

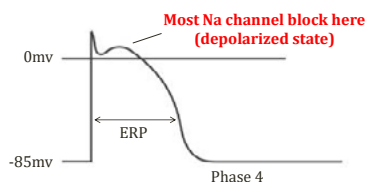


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Class Ib Drugs

Lidocaine, Mexiletine

- Most Na channel binding in **depolarized state**
- Ischemia → more depolarized myocytes
- Effective drugs in ischemic arrhythmias



Boards&Beyond

Class Ib Drugs

Lidocaine, Mexiletine

- Drug **rapidly unbinds**
- Slow heart rates: little drug effect by next heart beat
- More effective in fast heart rates
- Less time to unbind before Na channels open again
- Main use: ischemic ventricular tachycardia
 - Fast heart rates
 - Depolarized Na channels

Boards&Beyond

Class Ib Drugs

Lidocaine, Mexiletine

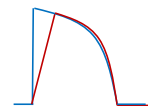
- Lidocaine also a local anesthetic
 - Na channel nerve block
- May cause **CNS stimulation**
 - Tremor, agitation
 - Tremor in patient on Mexiletine = toxicity
- Cardiovascular side effects
 - From excessive block of Na channels
 - Bradycardia, heart block, hypotension

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Class Ic Drugs

Flecainide, Propafenone

- Block open Na channels
- Very slow unbinding
- Result: **QRS can markedly prolong**
- Limited use due to concern of toxicity
- Especially proarrhythmic effects



Boards&Beyond

CAST Trial

The Cardiac Arrhythmia Suppression Trial

- Landmark clinical trial of antiarrhythmic drugs
- Tested the suppression hypothesis
 - Suppression of arrhythmias with drugs is a good thing
- Patients with asymptomatic arrhythmias after MI
- Encainide and flecainide administered
- Patients taking drugs had less arrhythmias
- Also: **3.6-fold increased risk of arrhythmic death**
- Result: Major ↓ antiarrhythmic drugs
- Now used only with compelling indication

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Class Ic Drugs

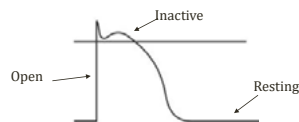
Flecainide, Propafenone

- Only used in patients with structurally normal hearts
- Effective in reducing recurrence of atrial fibrillation
- Must monitor for **QRS prolongation**
- Prolonged QRS → Risk of cardiac arrest

Boards&Beyond

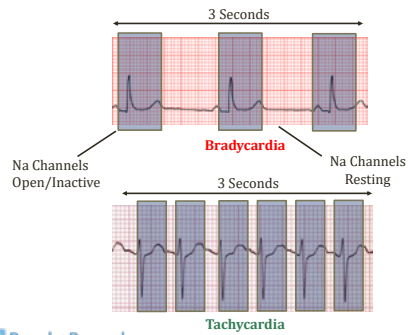
Use Dependence

- Na channels fluctuate between 3 different states
- Resting, Open and Inactivated
- Drugs bind more in certain states
- Class I drugs bind best in **open/inactivate states**
 - States when Na channel is in "use"
- These drug exhibit "use dependence"



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Use Dependence



Boards&Beyond

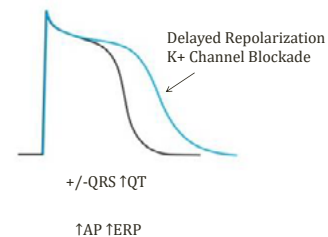
Use Dependence

- Use dependent drugs: **more binding fast heart rates**
- All class I drugs have some use dependence
- Seen most frequently IC drugs
- Practical implication:
 - Flecainide and propafenone (IC drugs)
 - Marked use dependence
 - Toxicity (QRS prolongation) at high heart rates
 - Stress testing often done to screen for toxicity

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Class III drugs

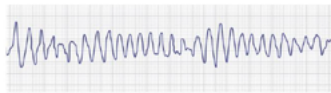
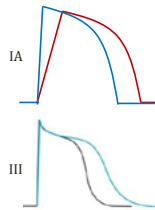
Amiodarone, Sotalol, Dofetilide, Ibutilide



Boards&Beyond

Torsade de Pointes

- Feared outcome of Qt prolongation
- Results in cardiac arrest
- **Class IA, III drugs** prolong Qt



Amiodarone

- Class III drug
 - K channel blocker: Prolongs Qt interval
 - Lowest incidence TDP of all class IIIs
- **Also has class I, II, and IV effects**
 - Class I: Prolongs QRS
 - Class II, IV: Slow HR, delay AV conduction
- Very effective drug
- Suppresses atrial fibrillation
- Suppresses ventricular tachycardia

Amiodarone

- Highly **lipid soluble**
- Accumulates in liver, lungs, skin, other tissues
- Half-life **about 58 days**
- Once steady state reached, very long washout
- Safe in renal disease (biliary excretion)

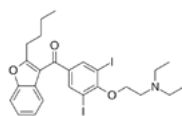
Amiodarone

- Many potential side effects related to accumulation
- Less likely at lower dosages
- Risk accumulates over time
- Young patients on indefinite therapy at greatest risk
- Often used in older patients

Amiodarone

Side Effects

- **Hyper and hypothyroidism**
 - Contains iodine
- **Increased LFTs**
 - Usually asymptomatic and mild
 - Drug stopped if elevation is marked
- **Skin sensitivity to sun**
 - Patients easily sunburn



Wikipedia/Public Domain

Amiodarone

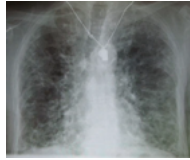
Side Effects

- **Blue-gray discoloration**
 - Less common skin reaction
 - "Blue man syndrome"
 - Most prominent on face
- **Corneal deposits**
 - Secretion of amiodarone by lacrimal glands
 - Accumulation on corneal surface
 - Appearance of "cat whiskers" on cornea
 - Does not usually cause vision problems
 - See in many patients on chronic therapy

Amiodarone

Side Effects

- Pulmonary fibrosis
- Most common cause of death from amiodarone
- Foamy macrophages seen in air spaces
- Filled with amiodarone and phospholipids
- "Honeycombing" pattern on chest x-ray



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Amiodarone

Side Effects

- When starting amiodarone
 - Chest X-ray
 - Pulmonary function tests (PFTs)
 - Thyroid function tests (TFTs)
 - Liver function tests (LFTs)

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Sotalol and Dofetilide

- Both drugs block K channels (class III)
- Can prolong QT interval → **torsade de pointes**
- Practical consideration:
 - Patients often admitted to hospital to start therapy
 - Rhythm monitored on telemetry
 - QT segment checked by EKG each day
- Sotalol: Also has beta blocking properties
- Can be used in patients with cardiomyopathy
- "Reverse use dependence"

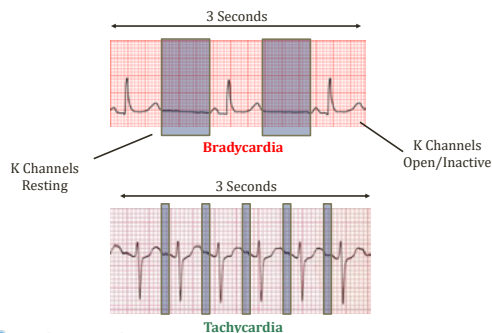
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Reverse Use Dependence

- K channels also fluctuate between 3 different states
- Class III drugs bind best in **resting state**
- These drug exhibit "reverse use dependence"

Boards&Beyond

Reverse Use Dependence



Boards&Beyond

Sotalol and Dofetilide

- Reverse use dependence: **more binding slow rates**
- Practical implication:
 - Bradycardia in patient on sotalol/dofetilide
 - QT interval may prolong
 - Increased risk of torsade de pointes

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Sotalol and Dofetilide

- Commonly used in patients with **atrial fibrillation**
- Typical case
 - Recurrent episodes symptomatic atrial fibrillation
 - Sotalol/Dofetilide started
 - Cardioversion to restore sinus rhythm
 - Sinus rhythm persists on therapy
- Other antiarrhythmic also used in this manner
 - Amiodarone
 - Propafenone
 - Flecainide

Ibutilide

- Intravenous drug
- Half life of 2 to 12 hours
- Used for “chemical cardioversion”

Cardioversion

- Termination of arrhythmias
- Often atrial fibrillation or flutter



Electrical Cardioversion
Requires sedation



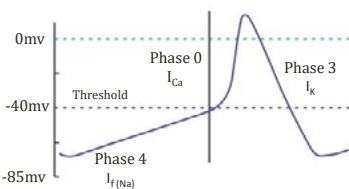
Ibutilide
Chemical Cardioversion
No sedation
May cause Torsade



Beta Blockers

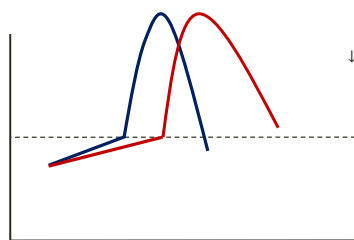
Class II Antiarrhythmics

- Main effect: **Pacemaker cells** (SA and AV node)
- **Decrease slope of phase 4**
- **Prolong repolarization (phase 3)**



Beta Blockers

Class II Antiarrhythmics

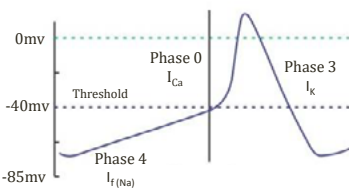


↓HR
↓Cond Velocity
↑PR Interval

Calcium Channel Blockers

Verapamil and Diltiazem

- Block **calcium channels**
- Slow heart rate
- Slow AV node conduction



Verapamil and Diltiazem

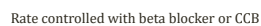

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 A Division of The McGraw-Hill Companies

- Beta blockers/Ca channel blockers → ↓ AV conduction



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- Beta blockers and CCBs commonly used
- Control ventricular rate



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- Increased risk among systolic heart failure patients
- Lower rates among patients on **beta blockers**
- Improved mortality



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- Nucleoside base
- Used to make ATP
- Receptors in many locations (purinergic receptors)
 - **AV nodal tissue**
 - **Vascular smooth muscle**


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- AV nodal cells:
 - **Activates K⁺ channels**
 - Drives K⁺ out of cells
 - Hyperpolarizes cells: Takes longer to depolarize
 - Also **blocks Ca influx**
- Result: **Slowing** of conduction through AV node

 Boards&Beyond

Adenosine

- Short half life
- Given IV for acute therapy of SVT
- Slows AV node conduction



Narrow Complex
Originates above HIS bundle

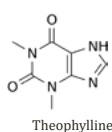
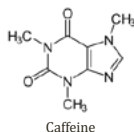
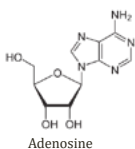
Adenosine

- Most common SVT: **AVNRT**
 - AV node reentrant tachycardia
- Slow and fast circuits in AV node → arrhythmia
- Adenosine slows AV node conduction
- Arrhythmia with terminate



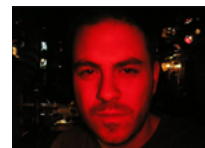
Adenosine

- Effects blocked by **theophylline** and **caffeine**
- Block adenosine receptors



Adenosine

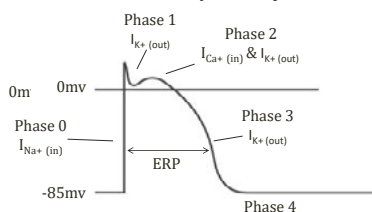
- Also a vasodilator
- Causes skin flushing, hypotension
- Some patients also develop dyspnea, chest pain
- Effects quickly resolve
- Must warn patients before administration for SVT



Jorge González/Flickr

Magnesium

- Acute management of **torsade de pointes**
- Mg **blocks influx of Ca** into cells
- Ca influx leads to early afterdepolarizations



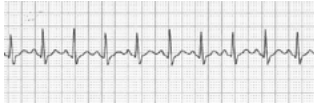
Atropine

- **Muscarinic receptor antagonist**
 - Parasympathetic block → ↑ HR and AV conduction
- Used in bradycardia → **↑ heart rate**
- Also speeds conduction through AV node
- Useful for bradycardia especially from AV block

Atropine



Before Atropine



After Atropine

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Atropine

- May **side effects** related to muscarinic block
- Toxicity:
 - Dry mouth
 - Constipation
 - **Urinary retention**
 - Confusion (elderly)

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Digoxin

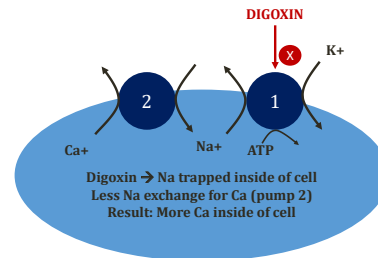
- Two cardiac effects
- #1: Increases contractility
 - Used in systolic heart failure with ↓ LVEF
- #2: Slows AV node conduction
 - Used in atrial fibrillation to slow ventricular rate

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Digoxin

Increased Contractility

- Inhibits Na-K-ATPase



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Digoxin

AV Nodal Slowing

- Suppresses **AV node conduction**
 - Increased **vagal (parasympathetic) tone**
 - Separate effect from blockade of Na-K-ATPase
- Can be used to **↓ heart rate** in rapid atrial fibrillation
 - Continued atrial fibrillation
 - Fewer impulses to ventricle → slower heart rate
- Effects similar to BB and CCB in AV node

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Digoxin Toxicity

- **Renal** clearance
 - Risk of toxicity in patients with chronic kidney disease
- **Hypokalemia** promotes toxicity
 - Caused by many diuretics, especially loop diuretics
 - Digoxin patient on furosemide → toxicity
- Levels often need to be monitored

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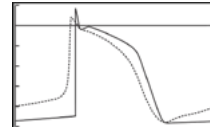
Digoxin Toxicity

- Gastrointestinal
 - Anorexia, nausea, vomiting, abdominal pain
- Neurologic
 - Lethargy, fatigue
 - Delirium, confusion, disorientation
 - Weakness
- Visual changes
 - Alterations in color vision, scotomas, blindness
- Cardiac arrhythmias

Digoxin Toxicity

Cardiac Toxicity

- More Na inside of cell
- ↑ resting potential atrial/ventricular cells
- Increased automaticity
- Dig toxic rhythms:
 - **Extra beats**: atrial, junctional, ventricular
 - Evidence of **AV node block**



Digoxin Toxicity

Treatment

- Digibind
 - Digoxin antigen binding fragments (Fab)
 - Produced in animals (sheep)
 - Dig bound to albumin (haptens) → antibodies
 - Antibody converted to fragments
- Corrects hyperkalemia, symptoms

Heart Murmurs

Heart Murmurs

Jason Ryan, MD, MPH

Boards&Beyond

Heart Murmurs

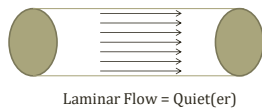
- Cardiac sound heard with stethoscope
- Caused by **turbulent blood flow**
- May be normal or pathologic



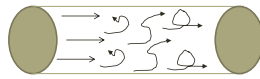
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Laminar vs. Turbulent Flow



Laminar Flow = Quiet(er)



Turbulent Flow = Loud
High Flow Rates
Narrow Flow Areas

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Murmurs

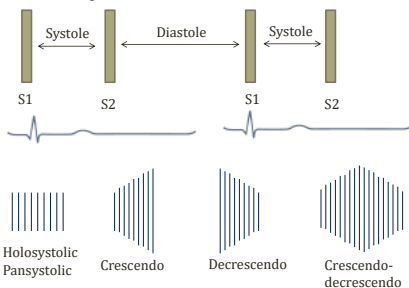
Grading

- I - barely audible on listening carefully
- II - faint but easily audible
- III - loud and easily audible, no thrill
- IV - loud murmur with a thrill
- V - heard with scope barely touching chest
- VI - audible with scope not touching the chest

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Murmurs

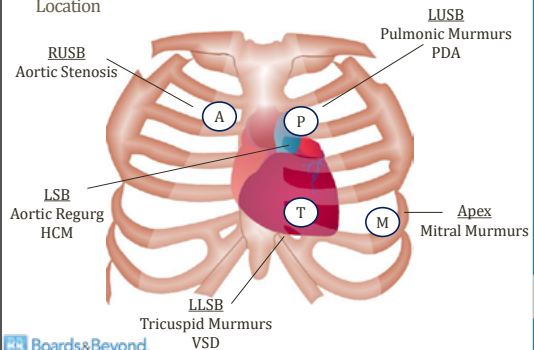
Other Descriptors



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Murmurs

Location

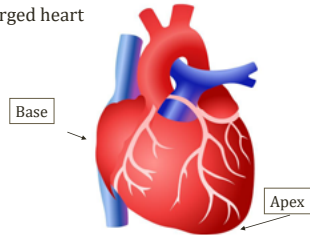


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Murmurs

Location

- Point of maximal impulse (apical impulse)
 - Left 5th intercostal space
 - Mid-clavicular line
- Lateral shift = Enlarged heart
- Hyperdynamic



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Innocent/Functional Murmurs

- Caused by normal flow of blood
- Common in children
- Also young, thin patients
- Generally soft murmurs
- No signs/symptoms of heart disease
- Still's murmur
- Pulmonic flow murmur
- Venous hum

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Systolic Murmurs

- Occur when heart contracts/squeezes
- Between S1-S2
- Aortic stenosis
- Mitral regurgitation
- Pulmonic stenosis
- Tricuspid regurgitation
- Hypertrophic cardiomyopathy
- Ventricular septal defect (VSD)

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Diastolic Murmurs

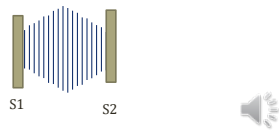
- Occur when heart relaxes/fills
- Between S2-S1
- Aortic regurgitation
- Mitral stenosis
- Pulmonic regurgitation
- Tricuspid stenosis

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Aortic Stenosis

Murmur

- Systolic crescendo-decrescendo murmur
- Also called an "ejection murmur"



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Aortic Stenosis

Severe Disease Findings

- **Late-peaking murmur**
 - Slow flow across stenotic valve
- **Soft/quiet S2**
 - Stiff valve can't slam shut
- **Pulsus parvus et tardus**
 - Weak and small carotid pulses
 - Delayed carotid upstroke

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HCM

Hypertrophic Cardiomyopathy

- Same murmur as aortic stenosis
- Differentiated by maneuvers
- **Valsalva**
 - Decreases venous return/preload
 - Increase HCM murmur
 - Decrease AS murmur



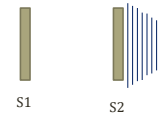
HCM

Boards&Beyond

Aortic Regurgitation

Murmur

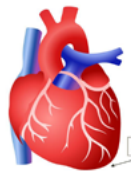
- Decrescendo, **blowing** diastolic murmur



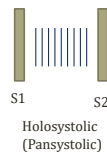
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Mitral Regurgitation

- **Holosystolic murmur heard best at the apex**
 - 5th intercostal space, mid-clavicular line



Apex



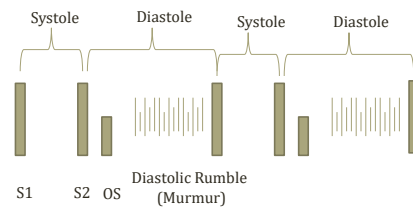
Holosystolic (Pansystolic)



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Mitral Stenosis

- **Diastolic rumbling murmur**
- Preceded by **opening snap**



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Mitral Stenosis

- No left sided S3, S4 in mitral stenosis
- **Time to opening snap** associated with severity
 - **High left atrial pressure** in severe disease
 - Higher left atrial pressure → ↓ time to opening snap
 - Short time to opening snap seen in severe disease

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Tricuspid/Pulmonic Disease

- Valve lesions sound like left sided-counterparts
- Heard in different locations
- Left upper sternal border
 - Pulmonic stenosis/regurgitation
- Left lower sternal border
 - Tricuspid stenosis/regurgitation

Boards&Beyond

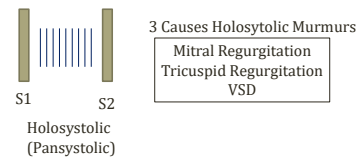
Carvallo's Sign

- Most right sided murmurs louder with inspiration
- Inspiration draws blood volume to lungs
- Louder right sided murmurs
- Softer left sided murmurs
- **Right sided** murmurs increase with Inspiration
- **Left sided** murmurs increase with Exhalation

VSD

Ventricular Septal Defect

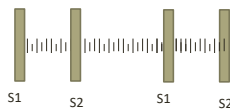
- Holosystolic murmur similar to MR
- Small VSD → more turbulence → loud murmur



PDA

Patent Ductus Arteriosus

- **Continuous, "machine-like" murmur**



Maneuvers

- Performed at bedside with patient
- May increase or decrease murmur
- Used to make diagnosis



Maneuvers

Preload/Venous Return

- Increase preload/venous return
 - **Leg raise** – blood falls back toward heart
 - **Squatting** – blood in legs forced back toward heart
- Decrease preload/venous return
 - **Valsalva** – ↑ intra-thoracic pressure → vein compression → ↓ VR
 - **Standing** – Blood falls toward feet, away from heart
- Most murmurs INCREASE with more preload except:
 - HCM
 - MVP

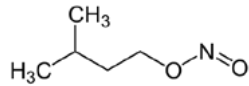
Valsalva Maneuver

- Bear down as if moving bowels
- Phase I (few seconds)
 - ↑ thoracic pressure
 - ↓ venous return (compression of veins → ↑ RA pressure)
 - Transient rise in aortic pressure (compression)
 - ↓ heart rate and AV node conduction (baroreceptors)
- Phase II
 - ↓ preload → ↓ cardiac output
 - ↑ heart rate and AV node conduction (baroreceptors)

Maneuvers

Afterload

- Increase Afterload
 - **Hand grip** - clench fist
- Decrease Afterload
 - **Amyl Nitrate** - vasodilator



Amyl Nitrate

Maneuvers

Afterload

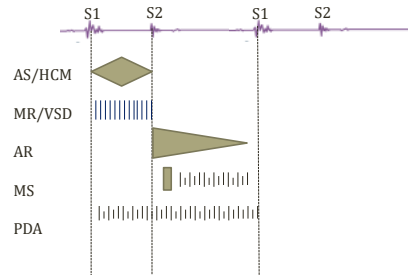
- Backward Valve Disorders
 - AR, MR, VSD
 - Louder with more afterload
 - More force pushing blood backward
- Forward Valve Disorders
 - MS, AS
 - Softer with more afterload
 - Less pressure difference moving blood forward
- MVP, HCM
 - Softer
 - Increased LV cavity size

Clues to Diagnosis

- Young female, otherwise healthy → MVP
- Healthy, young athlete, syncope → HCM
- Immigrant or pregnant → Mitral stenosis
- IV drug abuser → Tricuspid regurgitation
- Turner Syndrome or Aortic Coarctation
 - Bicuspid AV
 - Early stenosis
 - Aortic regurgitation
- Marfan → MVP

Summary

Commonly Tested Murmurs



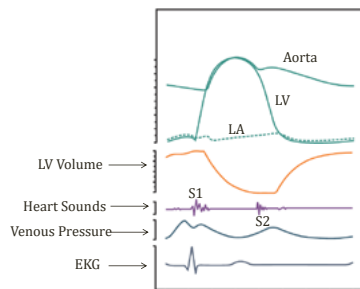
Heart Sounds

Heart Sounds

Jason Ryan, MD, MPH



The Cardiac Cycle



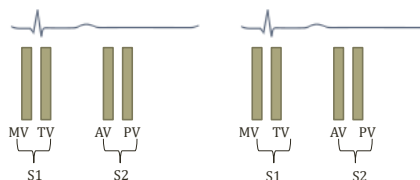
S1 and S2

- **Normal** heart sounds
- Each has **two components**
 - One from left sided valves (aortic, mitral)
 - One from right sided valves (tricuspid, pulmonic)
- S1 usually "single"
 - Two components close together
 - Cannot distinguish separate sounds
- S2 can be "split"
 - Two components far enough apart to be audible

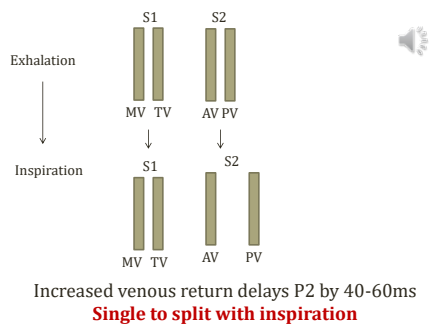


S1 and S2

- S1
 - Mitral and tricuspid valves close
- S2
 - Aortic and pulmonic valves close

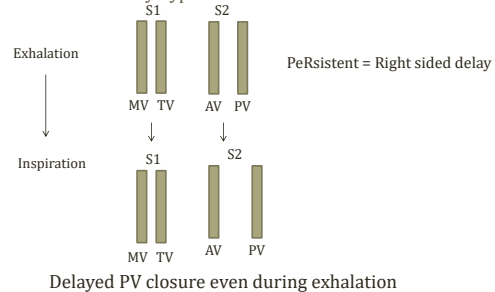


Physiologic S2 splitting



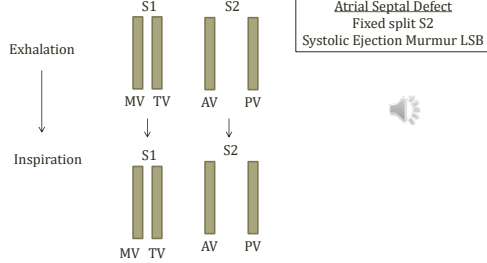
Persistent S2 splitting

RBBB or Pulmonary Hypertension



Fixed S2 splitting

Atrial septal defect

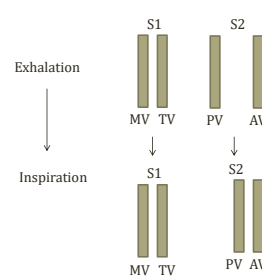


Flow across ASD → **increased right sided flow**

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Paradoxical S2 splitting

Delayed closure of aortic valve



Boards&Beyond

Paradoxical Splitting

- Electrical causes → delayed LV activation
 - LBBB
 - RV pacing
- Mechanical causes → delayed LV outflow
 - LV systolic failure
 - Aortic stenosis
 - Hypertrophic cardiomyopathy

Paradoxical = Left sided delay

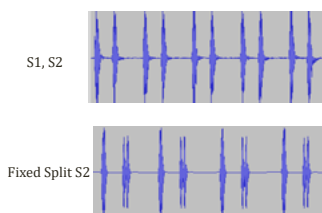
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Summary of S2 Splitting

- Physiologic = normal respiratory variation
- PeRsistent = RBBB, pulmonary HTN
- Fixed = Atrial septal defect
- Paradoxical = LBBB, cardiomyopathy

Boards&Beyond

Cardiac Phonography



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Loud P2

- Loud pulmonic component of S2
- **Pulmonary hypertension**
- Forceful closure of pulmonary valve
- Normally P2 not heard at apex
 - If you hear it here, it's "loud"

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S3 and S4

- Pathologic/abnormal heart sounds
- Occur in **diastole** during filling of left ventricle
- **Low-pitched** sounds heard best with bell
- S3: Early filling sound
- S4: Late filling sound



Tama988/Wikipedia

Boards&Beyond

S3

- Commonly seen in **acute heart failure**
 - High LA pressure → rapid early filling of LV → S3
 - Associated with ↑ LAP & ↑LVEDP
 - "Pushers" → push blood into LV
 - Very specific sign of high left atrial pressure
- **May be heard in normal hearts**
 - Young patients (<30), pregnant women
 - These patients are "suckers"
 - Vigorous LV relaxation lowers pressure rapidly

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S3

- Low frequency **best heard with bell**
- Louder in left **lateral decubitus position**
- Loudest at **apex**



Rama/Wikipedia

Boards&Beyond

S4



- Heard in patients with **stiff left ventricle**
 - Long-standing hypertension
 - Hypertrophic cardiomyopathy
 - Diastolic heart failure
- Rapid late filling of LV due to atrial kick
- Not heard in **atrial fibrillation**



Atrial Fibrillation

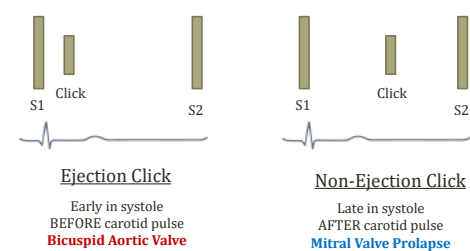
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Right Sided S3 & S4

- Both sounds can occur in right ventricle
- Same mechanisms as left sounds
- Right heart failure → right sided S3
- Right ventricular hypertrophy → right sided S4

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Systolic Clicks

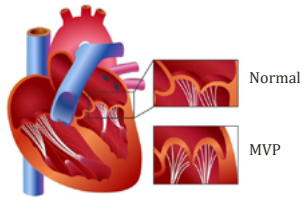


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Mitral Valve Prolapse

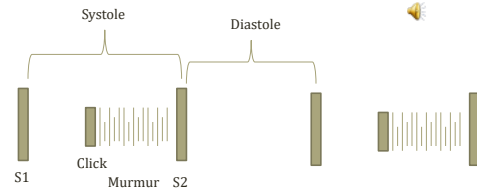
- **Billowing** of mitral valve leaflets above annulus
- Common cause of **mitral regurgitation**
- Causes a **systolic click**
 - Don't confuse with opening snap of mitral stenosis

Mitral Valve Disorders
 Proclick
 Stenosnap



Boards&Beyond

Mitral Valve Prolapse



Boards&Beyond

Heart Failure Basics

Heart Failure Basics

Jason Ryan, MD, MPH



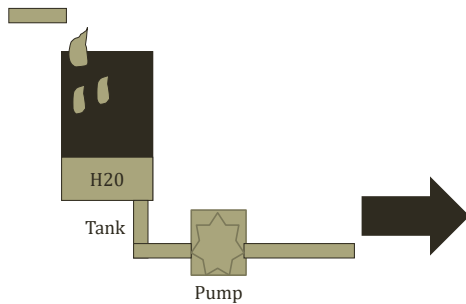
Heart Failure

- Impaired ability of the heart to pump blood
- Hallmark: **Low cardiac output**

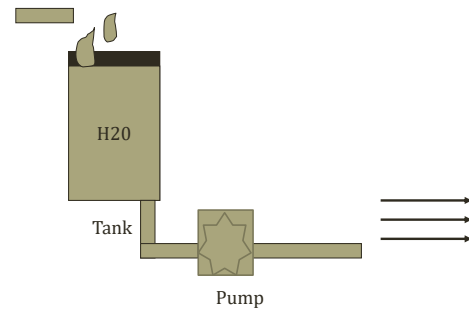
↓ CO



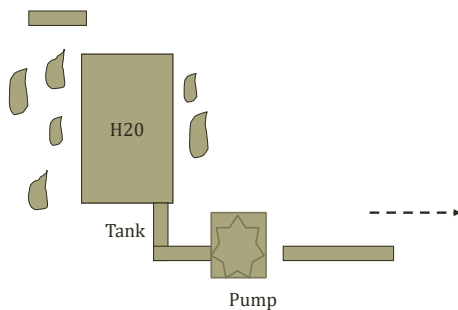
Heart Failure



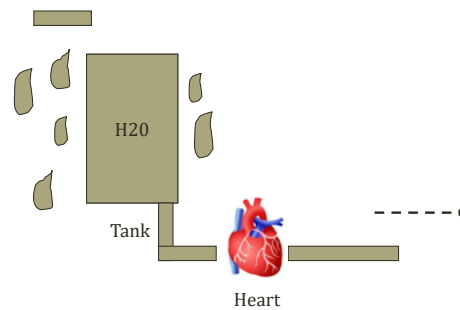
Heart Failure



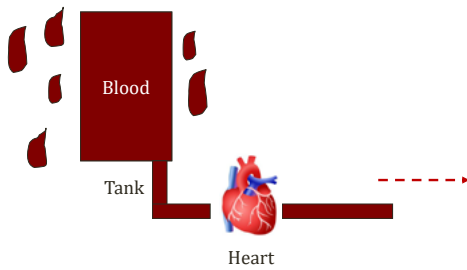
Heart Failure



Heart Failure

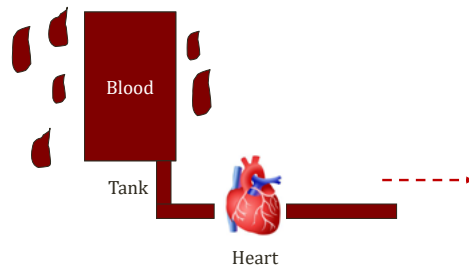


Heart Failure



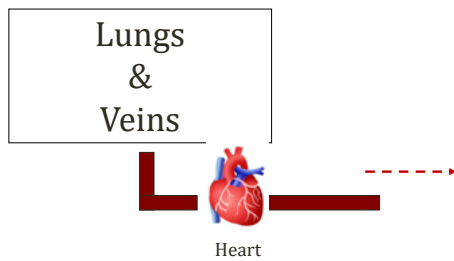
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Heart Failure



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Heart Failure



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Heart Failure

Pathophysiology

- “Failing” chambers → Increased pressures
- **Pressures** rise in cardiac chambers

Boards&Beyond

Heart Failure

Pathophysiology

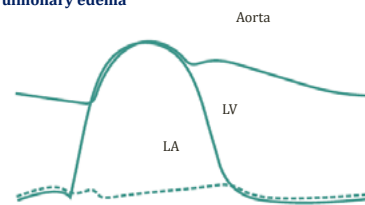
- Left ventricular failure → ↑ LV pressure
 - LV systolic pressure: depends on contractility (can be low)
 - **LVEDP** = always **high** in left heart failure
 - Hallmark of left heart failure
 - Less blood pumped out → more left behind → more pressure
 - Stiff ventricle (diastolic HF) → high pressure

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Heart Failure

Pathophysiology

- ↑ LVEDP → ↑ LA pressure
- ↑ pulmonary capillary pressure
 - **Dyspnea**
 - **Pulmonary edema**



Boards&Beyond

Heart Failure

Pathophysiology



Heart Failure

Pathophysiology

- ↑ pulmonary capillary pressure → ↑ PA pressure
- ↑ PA pressure → ↑ RV pressure
- ↑ RV pressure → ↑ RA pressure
 - Right atrial pressure = central venous pressure
 - **High pressure in venous system**
 - ↑ jugular venous pressure (neck veins)
 - Capillary leak → pitting edema

Heart Failure

Signs/Symptoms

- Physiologic effects of lying flat (supine)
 - Increased venous return
 - Redistribution of blood volume
 - From lower extremities and splanchnic beds to lungs
- Little effect in normal individuals
- Impaired ventricle cannot tolerate changes
- Worsens pulmonary congestion and breathing

Heart Failure

Signs/Symptoms

- Left heart failure
 - Dyspnea especially on exertion
 - Paroxysmal nocturnal dyspnea (wake up SOB)
 - Orthopnea (can't breathe lying flat)
- Right heart failure
 - Increased jugular venous pressure
 - Lower extremity edema
 - Liver congestion (rarely can cause cirrhosis)
- "Backward failure"

Heart Failure

Right Heart Failure

- Most common cause R heart failure: Left heart failure
- Occasionally right heart failure occurs in isolation
 - Normal left atrial pressure
 - High pulmonary artery, right ventricular, right atrial pressure
 - Usually secondary to a lung process
 - Pulmonary hypertension
 - COPD
 - This is often called "cor pulmonale"

Heart Failure

Signs/Symptoms

- Low flow signs/symptoms ("forward failure")
 - Loss of appetite
 - Weight loss (cachexia)
 - Confusion
 - Cool extremities
 - "Narrow pulse pressure"
- Seen only with very low cardiac output (systolic HF)
- Not seen in diastolic heart failure

Heart Failure

Lung Findings

- Classic finding is **rales**
 - Fluid filled alveoli "pop" open with inspiration
- Chest X-ray shows congestion
- Lungs/CXR can be clear in chronic heart failure
 - ↑lymphatic drainage

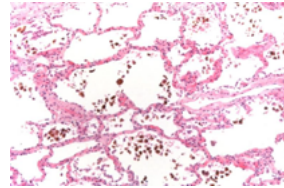


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Heart Failure

Lung Findings

- Heart failure cells
- Hemosiderin (iron) laden macrophages
- Brown pigment in macrophages



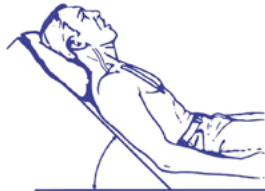
Zorkun/Wikipedia

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Heart Failure

Signs/Symptoms

- Elevated **jugular venous pressure** (normal 6-8cmH₂O)
- Look for height of double bounce (cause by a and v waves)



Boards&Beyond

Heart Failure

Hepatojugular Reflux

- Pressure on abdomen raises JVP 1-3cm normally
- With failing RV, increase is greater



Boards&Beyond

Heart Failure

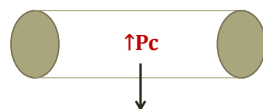
Signs/Symptoms

- **Lower extremity pitting edema**
- Increased capillary hydrostatic pressure
- Fluid leak from capillaries → tissues
- Gravity pulls fluid to lower extremities



James Heilman, MD

Boards&Beyond



Heart Failure

Abnormal Heart Sounds

- **S3** (associated with high left atrial pressure)
- **S4** (associated with stiff left ventricle)
- Displaced apical impulse – enlarged heart



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Heart Failure

Pathophysiology

- All forms of heart failure lead to **↓ cardiac output**
- Activates two physiologic systems
 - Activation of **sympathetic nervous system**
 - Activation of **renin-angiotensin-aldosterone system**
 - All RAAS hormone levels will rise
- Both systems lead to two key effects:
 - Increased peripheral vascular resistance (vasoconstriction)
 - Retention of sodium/water (kidneys)

Heart Failure

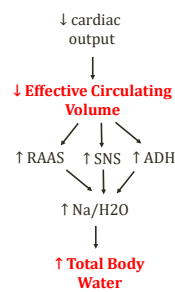
Total Peripheral Resistance

- Cardiac output falls → vasoconstriction
- Angiotensin II, sympathetic nervous system
- **TPR always high**
- Blood pressure often high but may be low
- Depends on combined changes CO and TPR

$$BP = CO \times TPR$$

Heart Failure

Sodium/Water Retention



Heart Failure

Other Hormones

- **ANP (Atrial natriuretic peptide)**
- Atrial stretch (pressure/volume) → ANP release
- Vasodilator (↓TPR)
- Constricts renal efferents/dilates afferents
- ↑ diuresis
- Opposite effects of RAAS system



Heart Failure

Other Hormones

- ANP released by atrial myocytes
- **BNP (brain natriuretic peptide):** Ventricles
- Both rise with volume/pressure overload
- Both counter effects of RAAS system
- BNP sometimes used for diagnosis in dyspnea
 - High levels suggest heart failure
 - Low levels suggest other causes of dyspnea



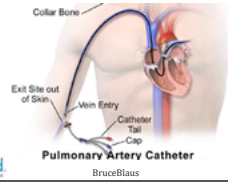
Nesiritide

- Recombinant **BNP**
- Vasodilation
- ↓ afterload, ↑CO
- Failed to show benefit in clinical trials

Heart Failure

Diagnosis

- Most common: typical signs/symptoms
- Elevated BNP level
- **Heart catheterization**
 - Increased LVEDP = left heart congestion/failure
 - Increased RA, RVEDP = right heart congestion/failure



Systolic and Diastolic Heart Failure

Systolic and Diastolic Heart Failure

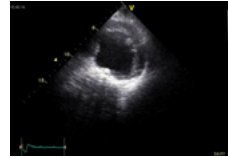
Jason Ryan, MD, MPH



Heart Failure

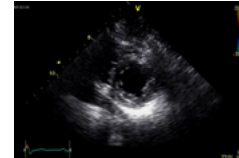
Systolic and Diastolic

Systolic Heart Failure



Ejection fraction is reduced

Diastolic Heart Failure



EF is normal (55-65%)



Heart Failure

Systolic and Diastolic

- Same congestive signs/symptoms
 - Dyspnea, orthopnea, paroxysmal nocturnal dyspnea
 - Rales, $\uparrow$ JVP, pitting edema
- Exception: **Low flow symptoms** in systolic only
 - Cool extremities
 - Cachexia
 - Confusion



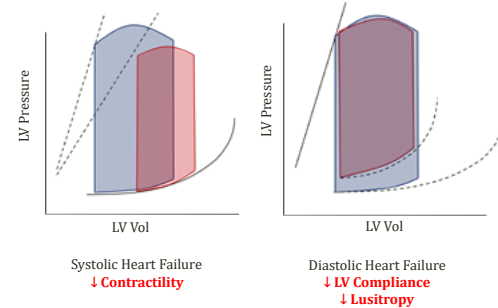
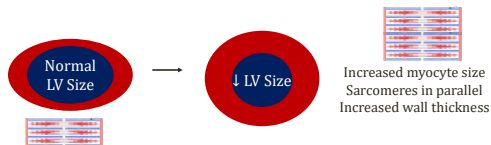
Dilated Cardiomyopathy

- Systolic heart failure with LV cavity dilation
- **"Eccentric" hypertrophy**
 - Volume overload (chronic retention of fluid in cavity)
 - Longer myocytes
 - Sarcomeres added in series



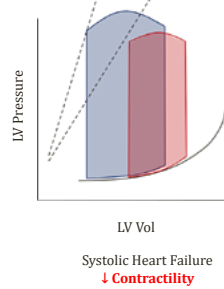
Concentric Hypertrophy

- Pressure overload
- Chronic $\uparrow\uparrow$ pressure in ventricle: HTN, Aortic stenosis
- Decreased compliance (stiff ventricle)
- Often seen in diastolic heart failure



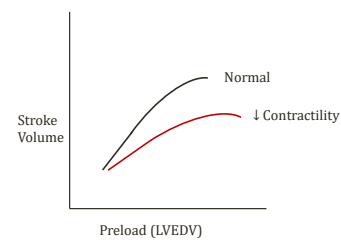
Systolic Heart Failure

- ↓ Cardiac output
- Problem in SYSTOLE
- Can't get blood out
- ↓ Stroke volume
 - $SV = EDV - ESV$
 - ↑↑ ESV (↓ contractility)
 - ↑ EDV (↑ $ESV + VR$)
 - ↑ $LVEDP$



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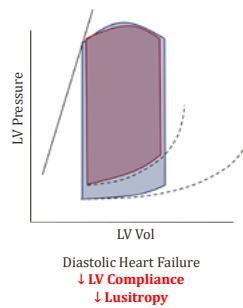
Frank-Starling Curve



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Diastolic Heart Failure

- ↓ Cardiac output
- Problem in DIASTOLE
- Can't get blood in
- Small ↓ stroke volume
 - ↓ EDV (↓ filling)
- ↑↑ $LVEDP$ (stiff ventricle)



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Systolic vs. Diastolic

	Normal	Systolic/Dilated	Diastolic
EDV	100	200	90
ESV	40	160	40
SV/CO	60	40	50
EF %	60	20	56

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Systolic Heart Failure

- Most common cause: **Myocardial infarction**
 - Myocytes replaced by scar tissue
 - "Ischemic" cardiomyopathy
- Many causes of "non-ischemic" cardiomyopathy
 - About 50% idiopathic
 - Many other causes: viral, familial, peri-partum, chemotherapy toxicity, HIV, alcoholic, sarcoidosis, tachycardia-mediated

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Diastolic Heart Failure

- Exact cause unknown
- Many cases have **concentric hypertrophy**
- Many associated conditions
 - Age, diabetes, hypertension
- Terms:
 - Heart failure preserved EF
 - HFpEF
 - Diastolic dysfunction

Boards&Beyond

Nonischemic Cardiomyopathy

Viral

- May follow upper respiratory infection
- Many associated viruses
 - **Coxsackie**
 - Influenza, adenovirus, others
- Virus enters myocytes
- Causes myocarditis → cardiomyopathy
- Myocarditis phase may go undiagnosed
- No specific therapy for virus



Pixabay/Public Domain

Nonischemic Cardiomyopathy

Peri-partum

- Late in pregnancy or early post-pregnancy
- Exact cause unknown (likely multifactorial)
- Women often advised to avoid future pregnancy

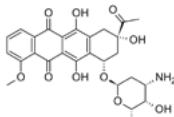


Øyvind Holmstad/Wikipedia

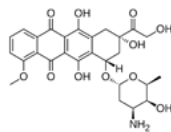
Nonischemic Cardiomyopathy

Chemotherapy

- Usually after treatment with anthracyclines
 - Antitumor antibiotics
 - Doxorubicin and daunorubicin



Daunorubicin



Doxorubicin
(Adriamycin)

Nonischemic Cardiomyopathy

Familial

- Mutations
 - Often sarcomere proteins
 - Beta myosin heavy chain
 - Alpha myosin heavy chain
 - Troponin
- Many autosomal dominant
- X-linked, autosomal recessive also described



Wikipedia/Public Domain

Nonischemic Cardiomyopathy

Tachycardia-mediated

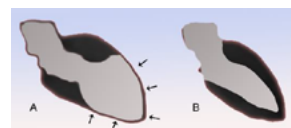
- Constant, rapid heart rate for weeks/months
- Leads to depression of LV systolic function
- **Reversible** with slower heart rate



Nonischemic Cardiomyopathy

Takotsubo/Apical ballooning

- **Stress-induced** cardiomyopathy
- Occurs after severe emotional distress
- Markedly reduced LVEF
- Increase CK, MB, Troponin; EKG changes
- Looks like anterior MI (but no coronary disease)
- Usually recovers 4-6 weeks



Jheuser/Wikipedia

Alcohol

- Chronic consumption can cause cardiomyopathy
- Believed to be due to toxic metabolites
- Can recover with cessation of alcohol



Pixabay/Public Domain

High Output Heart Failure

- Heart in overdrive
 - Severe anemia
 - Thyroid disease
 - Thiamine (B1) vitamin deficiency (beriberi)
 - A-V fistulas (post-surgical)
- Exact mechanism unclear
 - Decreased LV filling time
- Defining characteristic: **HIGH cardiac output**
 - Heart failure symptoms in absence of low output
 - ↑JVP, pulmonary edema



John Liu/Flickr

Restrictive Cardiomyopathy

Restrictive Cardiomyopathy

Jason Ryan, MD, MPH



Restrictive Heart Disease

- Something “infiltrates” the myocardium
 - Granulomas (Sarcoid)
 - Amyloid protein (Amyloidosis)
- Heart cannot relax and fill
- SEVERE diastolic dysfunction



MarkBuckawicki/Wikipedia



Restrictive Heart Disease

- **LVEF = normal**
- Left ventricular volume = normal (not dilated)
- Restricted filling = $\uparrow$ atrial pressure
- Dilated left and right atria
- Classic imaging findings:
 - Normal left ventricular function/size
 - Bi-atrial enlargement



Restrictive Heart Disease

Clinical Features

- Dyspnea
- Prominent **right heart failure**
 - Markedly elevated jugular venous pressure
 - Lower extremity edema
 - Liver congestion
 - May lead to cirrhosis (“nutmeg liver”)



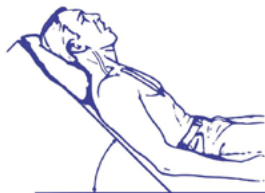
David Monniaux/Wikipedia



Restrictive Heart Disease

Classic signs

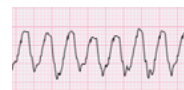
- Kussmaul's sign
 - Inspiration causes rise in JVP



Restrictive Heart Disease

Rhythm Disturbances

- Myocardial infiltration may disrupt electrical activity
- Arrhythmias (sudden death)
- AV block



Ventricular Tachycardia



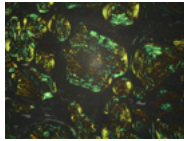
3rd Degree Heart Block



Restrictive Heart Disease

Major Causes

- **Amyloidosis**
 - Amyloid protein deposits in heart
 - Various forms (primary, secondary, etc.)

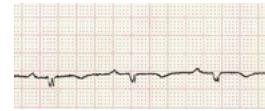


Ed Uthman, MD

Restrictive Heart Disease

Classic signs

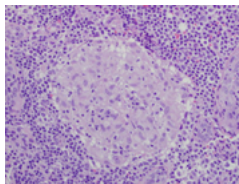
- Can see thickened myocardium
- **Low** voltage on EKG
- Classic finding in amyloidosis and Fabry's disease



Restrictive Heart Disease

Major Causes

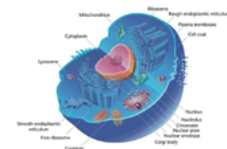
- **Sarcoidosis**
 - Granuloma formation
 - Usually involves lungs
 - Extra-pulmonary organs include heart



Restrictive Heart Disease

Major Causes

- **Fabry disease**
 - Lysosomal storage disease
 - Deficiency of α -galactosidase A
 - Accumulation of ceramide trihexoside



Mediran/Wikipedia

Restrictive Heart Disease

Major Causes

- **Hemochromatosis**
 - **Iron** excess
 - Commonly causes dilated cardiomyopathy
 - Rarely may cause restrictive



Tomihahndorf

Restrictive Heart Disease

Major Causes

- **Post-radiation**
 - Acutely: May cause inflammation
 - Fibroblast recruitment
 - **Extra-cellular matrix deposition**
 - Collagens and fibronectin



Dina Wakulchik/Wikipedia

Restrictive Heart Disease

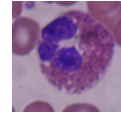
Major Causes

- Pericarditis may occur acutely after therapy
- Long term effects
 - Pericardial disease
 - Coronary artery disease
 - Valvular disease
 - Conduction abnormalities
- Restrictive cardiomyopathy
 - **Fibrous tissue** accumulation
 - Diastolic dysfunction

Restrictive Heart Disease

Major Causes

- Loeffler's syndrome
 - **Hypereosinophilic syndrome**
 - High eosinophil count
 - Eosinophilic infiltration of organs
- Skin (eczema)
- Lungs (fibrosis)

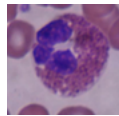


Bobigalindo/Wikipedia

Restrictive Heart Disease

Major Causes

- Primary HES
 - Neoplastic disorder
 - Stem cell, myeloid, or eosinophilic neoplasm
- Secondary HES
 - Reactive process
 - Eosinophilic overproduction due to cytokines
 - Occurs in parasitic infections (ascaris lumbricoides)
 - Some tumors/lymphomas
- Idiopathic HES

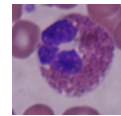


Bobigalindo/Wikipedia

Restrictive Heart Disease

Major Causes

- Eosinophilic infiltration of myocardium
 - Common mode of death
- Acute phase
 - Myocarditis (often asymptomatic)
- Chronic phase
 - Endomyocardial fibrosis and myocyte death
 - Can see restrictive heart disease
 - **Thrombus** formation common (embolic stroke)

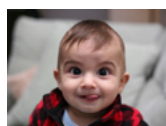


Bobigalindo/Wikipedia

Restrictive Heart Disease

Major Causes

- Endocardial fibroelastosis
 - **Endocardial** thickening (innermost myocardium)
 - Infants in first year of life
 - Thick myocardium
 - Proliferation of **fibrous (collagen) and elastic fibers**
- Restrictive cardiomyopathy



Avsar Aras

Acute Heart Failure

Acute Heart Failure

Jason Ryan, MD, MPH

Boards&Beyond

Heart Failure

Acute vs. Chronic

Acute

- Congested/Swollen
- Pulmonary Edema
- Pitting Edema
- ↑JVP

Chronic

- Euvolemic
- Clear lungs
- No pitting edema
- JVP flat

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Acute Exacerbations

Causes

- #1: Dietary indiscretion
 - High salt intake
- #2: Poor medication compliance



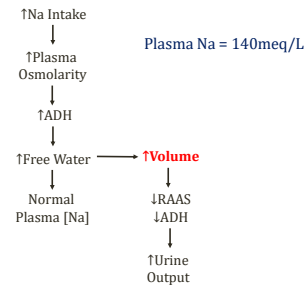
LukeB20161933/Wikipedia



Pixabay/Public Domain

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Dietary Indiscretion



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Acute Exacerbations

Causes

- Infection/trauma/surgery
 - Activation of sympathetic nervous system
- Ischemia (rare)
 - Decreased cardiac output
- **NSAIDs**
 - Inhibit cyclooxygenase (COX) → ↓ prostaglandins
 - Prostaglandins maintain renal perfusion
 - Result: Less renal perfusion → salt/water retention

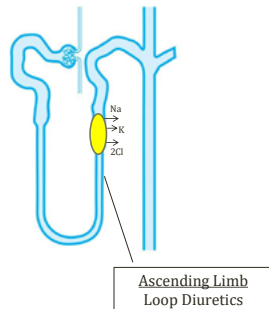
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Acute Heart Failure Therapy

- Often treated in the hospital
- Goal: Symptom relief
 - Contrast with chronic HF: reduce mortality/hospitalizations
- Often same therapies for diastolic versus systolic

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Loop Diuretics



Boards&Beyond

Loop Diuretics

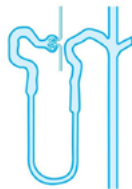
Furosemide, Bumetanide, Torsemide, Ethacrynic Acid

- Inhibit Na-K-Cl pump in ascending loop of Henle
- Result in salt-water excretion
- Relieve congestion
- IV better than PO (gut is swollen)
- Key side effects
 - Hypokalemia
 - Volume depletion (Renal failure; hypotension)
- Sulfonamide drugs: allergy (except ethacrynic acid)

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Metolazone

- Thiazide-like diuretic
- Inhibits Na-Cl reabsorption distal tubule
- Gives loop diuretics a "kick"
- Vigorous diuresis
- Side effects: additional fluid, K⁺ loss



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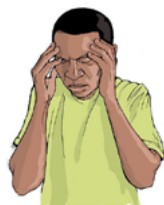
Nitrates

- Predominant mechanism is **venous dilation**
 - Bigger veins hold more blood
 - Takes blood away from left ventricle
 - Lowers LVEDV (preload), LA pressure
 - Less pulmonary edema → improved dyspnea

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Nitrates

- Side effects
 - Headache (meningeal vasodilation)
 - Flushing
 - Hypotension



Wikipedia/Public Domain

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Vasodilators

"Afterload reduction"

- ACE inhibitors
- Hydralazine
- Cause peripheral vasodilation
- Reduced afterload → increased cardiac output

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Nitrates plus Hydralazine

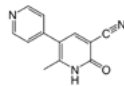
- Combination therapy for **acute and chronic HF**
 - Studied in systolic heart failure
 - Reduction in preload (nitrates) and afterload (hydralazine)
 - Acute therapy: Improves symptoms
 - Chronic therapy: Lowers mortality in some studies
- Largely replaced by ACE inhibitors

Inotropes

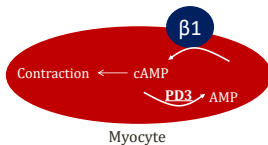
- Increase contractility
- Only for **systolic heart failure**
 - No role in diastolic heart failure (normal contractility)
- All activate **$\beta 1$ pathways in myocytes**
 - Increased HR and contractility
- Can also active **$\beta 2$ pathways in smooth muscle**
 - Vasodilation $\rightarrow$ hypotension

Inotropes

Milrinone

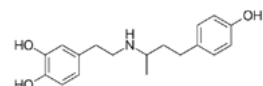


- **Phosphodiesterase 3 inhibitor**
 - PD3 breaks down cAMP in myocytes
 - Inhibition $\rightarrow$ $\uparrow$ cAMP $\rightarrow$ contraction
 - Vascular smooth muscle $\uparrow$ cAMP ($\beta 2$) $\rightarrow$ dilation
- $\uparrow$ Inotropy
- $\uparrow$ Vasodilation
- Hypotension

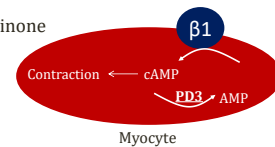


Inotropes

Dobutamine



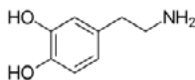
- **Mostly beta-1 agonist**
 - Increases heart rate and contractility
- Weak beta-2 agonist
 - Vasodilation
- $\uparrow$ Inotropy
- $\uparrow$ Vasodilation
- Hypotension
- Similar effects to milrinone



Inotropes

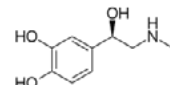
Dopamine

- Does not cross blood brain barrier (no CNS effects)
- Peripheral effects highly dependent on dose
- **Low dose:** dopamine agonist
 - Vasodilation in kidneys
- **Medium dose:** beta-1 agonist
 - Increased heart rate and contractility
- **High dose:** alpha agonist
 - Vasoconstriction



Inotropes

Epinephrine



- Also dose dependent effects
- **Low dose:** beta-1 and beta-2 agonist
 - Increased heart rate & contractility, vasodilation
- **High dose:** alpha agonist
 - Vasoconstriction

Inotrope Risks

- Numerous registries and clinical trials demonstrate increased mortality with routine use of inotropes
- **Dangerous drugs** used in very sick patients under monitored conditions

A Typical Acute Heart Failure Course

- ER presentation:
 - Dyspnea, edema, sleeping in chair
- Admitted to hospital
 - Nitro drip to relieve dyspnea
 - IV Furosemide to remove fluid
- Hospital Day 2
 - Weight down 4kg, feels better
 - Nitro drip stopped
 - Changed to oral furosemide
- Hospital Day 3: Discharge

A More Complex Heart Failure Course

- ER presentation:
 - Dyspnea, edema, sleeping in chair
 - Known LVEF 10%
- Admitted to hospital
 - Nitro drip to relieve dyspnea
 - IV Furosemide to remove fluid
- Hospital Day 2
 - Poor urine output, Cool extremities, Cr rises 1.1→1.4
 - Dobutamine drip started

A More Complex Heart Failure Course

- Hospital Day 3-5
 - Good urine output
 - Weight loss 4kg
 - Breathing improves
- Hospital Day 6
 - Dobutamine stopped
 - Furosemide drip stopped
- Hospital Day 7
 - Oral furosemide given
- Hospital Day 8: Discharge

Heart Failure Readmission

- Recurrence of HF after discharge common
 - Post-discharge follow-up VERY important
 - "Readmissions" a focus of public health policy
 - **High risk of readmission within 30 days**
 - Highest risk category among Medicare population

Acute Heart Failure

- Most patients require chronic, daily diuretic
 - Helps to maintain euvolemic status
 - Often oral furosemide or other loop diuretic
- Some patients require daily long acting nitrate
 - Often oral isosorbide mononitrate

Acute Heart Failure

- Rare patients: continued treatment for **low output**
 - Systolic heart failure only
 - Chronic, IV infusion inotrope (i.e. "home dobutamine")
 - Left ventricular assist device (LVAD)
 - Heart transplant

Digoxin

- Only available **oral** inotrope
- "Dig and diuretic" once the mainstay of HF treatment
- What changed?
 - Digoxin shown to have no mortality benefit
 - Digoxin not effective for diastolic heart failure
 - About 50% of all cases
 - Digoxin carries significant risk of side effects

Digoxin Mechanism

Two important cardiac effects

- #1: Inhibits Na-K-ATPase pump
 - More Na in cell → more Ca++ in cell
 - More Ca++ → more contractility
- #2 Suppresses AV node conduction (parasympathetic)
 - Can be used to slow heart rate in rapid atrial fibrillation

Digoxin

Benefits in Heart Failure

- Useful for **systolic HF** patients
 - Symptoms despite maximal therapy on other drugs
 - i.e. persistent dyspnea despite good volume status
- Can be administered for acute heart failure
- Can be administered long term to maintain CO

Digoxin

Benefits in Heart Failure

- Increased cardiac output
- Improved symptoms and quality of life
- No established mortality benefit

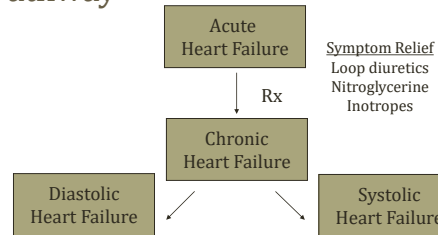
Chronic Heart Failure

Chronic Heart Failure

Jason Ryan, MD, MPH

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Heart Failure Treatment Pathway



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Chronic Heart Failure

- **LOTS** of therapies for systolic heart failure
 - Drugs: ACE-inhibitors, beta blockers, aldosterone antagonists
 - Defibrillators
 - Bi-ventricular pacemakers
- **NO** direct therapies for diastolic heart failure
 - Guidelines recommendations: treat HTN, diabetes, A. fib
 - Mainstay of therapy: monitor for symptoms, diuretics

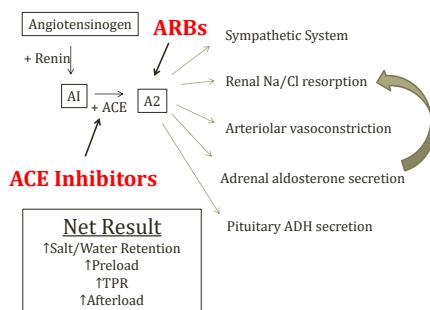
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Systolic Heart Failure

- Chronic over-activation of two physiologic systems
- **Renin-angiotensin-aldosterone system**
- **Sympathetic nervous system (β_1 stimulation)**
- Blockade $\rightarrow$ $\downarrow$ mortality and disease progression

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Renin-Angiotensin System



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RAAS Drugs

- **ACE Inhibitors**
 - Captopril, Enalapril, Lisinopril, Ramipril
 - Block conversion AI $\rightarrow$ AII
- **Angiotensin Receptor Blockers (ARBs)**
 - Candesartan, Irbesartan, Valsartan
 - Directly block AII receptor
- Both classes: $\downarrow$ mortality, $\downarrow$ hospitalizations
- Side effects
 - Hyperkalemia (aldosterone)
 - Renal failure (GFR)

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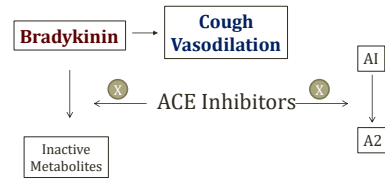
ACE Inhibitors

Unique Side Effects

- Due to increased **bradykinin**
- Dry Cough
 - Occurs in ~10% of patients
- Angioedema
 - Swelling of face, tongue
 - Can be life-threatening



Bradykinin



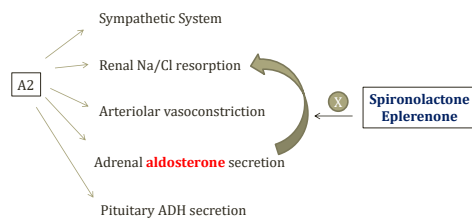
Beta Blockers

- Once contraindicated in systolic heart failure
 - Negative inotropes
- Not used in acute heart failure
 - May worsen cardiac output and symptoms

Beta Blockers

- Three agents beneficial in chronic systolic HF
 - Metoprolol (β_1)
 - Carvedilol ($\beta_1\beta_2\alpha_1$)
 - Bisoprolol (β_1)
- **↓ mortality, ↓ hospitalizations**

Aldosterone Antagonists



Spironolactone, Eplerenone

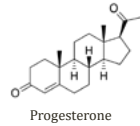
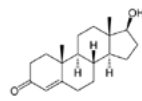
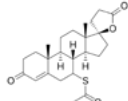
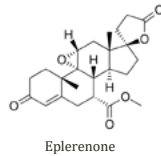
Potassium-sparing diuretics

- $\uparrow$ Na/ H_2O excretion (diuretics)
- "Spare" potassium
 - Unlike other diuretics, do not increase K^+ excretion
- HYPERkalemia is side effect
- Reduced mortality
- Reduced hospitalization rate

Spirolactone, Eplerenone

Potassium-sparing diuretics

- Similar structure to testosterone
 - Blocks testosterone effects
 - **Gynecomastia** in men
 - Eplerenone: No gynecomastia
- Derivative of progesterone
 - Activates progesterone receptors
 - **Amenorrhea** in women



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Neprilysin Inhibitors

Sacubitril

- Neprilysin: Degrades atrial/brain natriuretic peptide
- Inhibition → **↑ANP/BNP**
 - Antagonists to RAAS system
 - Vasodilatation
 - Natriuresis (sodium excretion)
 - Diuresis (water excretion)
 - Reduced sympathetic tone



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Neprilysin Inhibitors

Sacubitril

- Entresto: oral combination sacubitril/valsartan
 - Valsartan: angiotensin receptor blocker (ARB)
- **↓ mortality**
- **↓ hospitalizations**



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Neprilysin Inhibitors

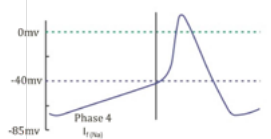
Side Effects

- Studied in combination with valsartan
- Many side effects similar to ARBs
- Hypotension
- Hyperkalemia
- Angioedema
 - Rare, feared adverse effect
 - Neprilysin also degrades bradykinin (like ACE)
 - Angioedema may occur
 - Cannot be given together with ACE inhibitors

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Ivabradine

- Selective sinus node inhibitor
- Elevated HR → worse prognosis
- Slows heart rate without ↓ contractility
- Inhibits SA pacemaker **"funny current" (I_f)**
- Used in patients on max-dose beta blocker with ↑HR
- Limited evidence of ↓ mortality and ↓ hospitalizations



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Chronic Systolic Heart Failure

Drug Therapy

- ACE inhibitors/ARB
- Beta Blockers
- Aldosterone antagonists
- Neprilysin inhibitors
- Ivabradine

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ICD

Implantable Cardiac Defibrillator

- Annual risk SCD >20% some studies
- Most due to ventricular tachycardia



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ICD

Implantable Cardiac Defibrillator

- Improve mortality in appropriate patients
- Implantation carries some risk:
 - Bleeding, infection
 - Inappropriate shocks

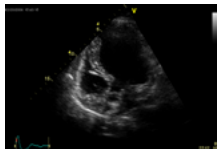


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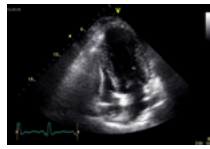
Biventricular Pacemakers

Cardiac Resynchronization Therapy (CRT), BiV pacer

Out of Synchrony

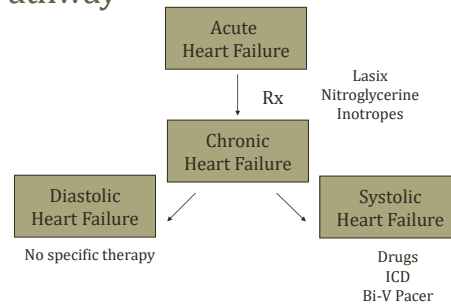


After Pacemaker



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Heart Failure Treatment Pathway



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Cardiac Embryology

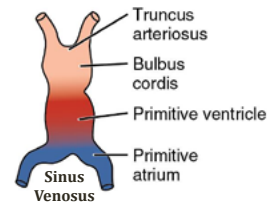
Cardiac Embryology

Jason Ryan, MD, MPH



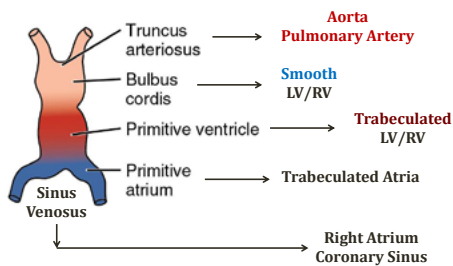
Primitive Heart

22 days

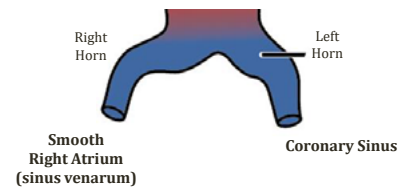


Primitive Heart

22 days

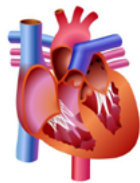


Sinus Venosus



Cardinal Veins

- Form SVC/IVC (not from heart tube)
- Connect to right atrium
- Superior vena cava
 - R common cardinal vein and R anterior cardinal vein
- Inferior vena cava
 - Posterior veins



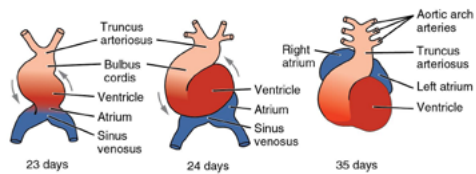
Adult Heart



Patrick J. Lynch, medical illustrator/Wikipedia



Primitive Heart



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OpenStax Colleg/Wikipedia

Cardiac Looping

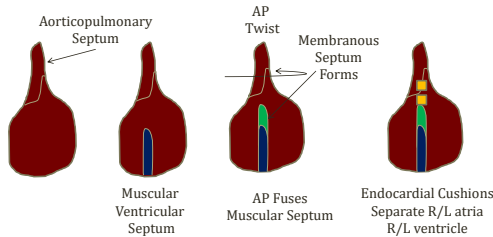
- Heart tube “loops” at about 4 weeks gestation
- Establishes left-right orientation in chest
- Requires **cilia** and **dynein**
- Dextrocardia (heart on the right side of body)
 - Seen in Kartagener syndrome
 - Part of primary ciliary dyskinesia



Nevit/Wikipedia

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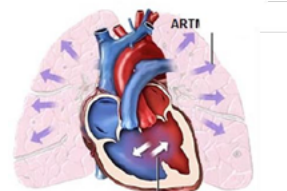
Ventricular Septum Formation



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Ventricular Septum Pathology

- Membranous VSD (most common type)
- Muscular VSD



Wikipedia/Public Domain

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Endocardial Cushions

- Contribute to several cardiac structures
 - Atrial septum
 - Ventricular septum
 - AV valves (mitral/tricuspid)
 - Semilunar valves (aortic/pulmonic)
- **Endocardial cushion defects**
 - Atrioventricular canal defects
 - Atrioventricular septal defects
 - ASD, VSD, Valvular malformations
 - Common in **Down syndrome**



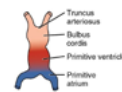
Endocardial Cushions Separate R/L atria R/L ventricle

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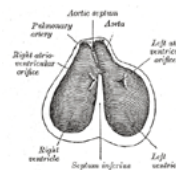
Aorticopulmonary septum

Spiral Septum

- Formed from **neural crest cells**
- Migrate to truncal and bulbar ridges
- Separates aorta and pulmonary arteries
- Fuses with interventricular septum



OpenStax Colleg/Wikipedia



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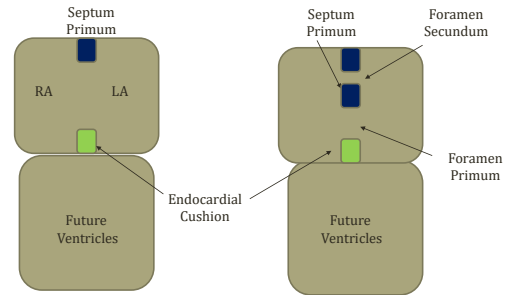
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Aorticopulmonary septum

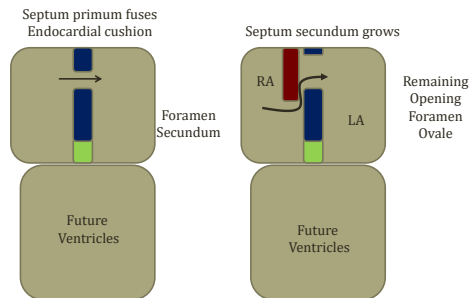
Spiral Septum

- Abnormal formation → congenital pathology
- Transposition of great vessels
 - Failure to spiral
- Tetralogy of Fallot
 - Skewed septum development
- Persistent truncus arteriosus
 - Partial/incomplete septum development

Atrial Septum



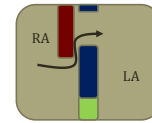
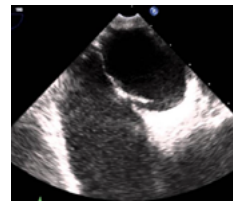
Atrial Septum



PFO

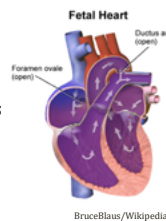
Patent Foramen Ovale

- Found in ~25% adults
- Failure of foramen ovale to close after birth
- Septum primum/secundum fail to fuse



Fetal Circulation

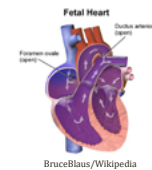
- **High resistance to flow** in lungs
- Oxygenated blood umbilical veins
 - About 80% saturated (30mmHg O₂)
- Travels directly to right atrium
 - Bypasses liver via ductus venosus
- Bypasses lungs via foramen ovale
- Some blood gets to RV (SVC)
 - Bypasses lungs via **ductus arteriosus**
 - Left pulmonary artery to aorta



BruceBlaus/Wikipedia

Changes at Birth

- **Pulmonary resistance falls**
- More blood to left atrium
- LA pressure > RA pressure
- Foramen ovale closes (fossa ovalis)
- Ductus arteriosus closes
 - In utero: ↓ O₂, ↑ prostaglandins maintain patency
 - Birth: ↑ O₂, ↓ prostaglandins (loss of placenta)



BruceBlaus/Wikipedia

	PVR	RA	LA
In Utero	↑	↑	↓
Birth	↓	↓	↑

Changes at Birth

- Placenta has **low resistance to flow**
- In utero: helps keep LA pressure low
- At birth: **increase in peripheral resistance**
- Rise in systemic blood pressure
- Rise in left ventricular pressure
- Contributes to rise in LA pressure



Shunts

Shunts

Jason Ryan, MD, MPH

Boards&Beyond

Shunts

RA	LA
RV	LV
PA	Ao

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Shunts

- Left side pressures >> Right side pressures
 - LA ~10mmHg >> RA ~6mmHg
 - LV ~120/10 >> RV ~24/6
 - Ao ~ 120/80 >> PA ~24/12
- Left to right connection → Left to right flow
 - VSD (LV→RV)
 - ASD (LA→RA)
 - PDA (Aorta → Left pulm artery)

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Shunts

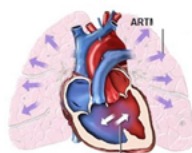
- At birth:
 - Left to right flow → volume overload of right heart
 - Blood flow to lungs unimpaired → no cyanosis
- YEARS later (untreated)
 - Pulmonary vessels become stiff/thick
 - Right ventricle hypertrophies
 - **Right sided pressures rise**
 - Shunt reverses (now R → L)
 - Cyanosis occurs (Eisenmenger syndrome)
- “Blue kids” not “blue babies”

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VSD

Ventricular Septal Defect

- Most common congenital anomaly
- Communication LV/RV
- **Harsh, holosystolic murmur**
 - Tricuspid area (LLSB)



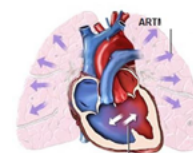
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VSD

Ventricular Septal Defect

- Characterized in many ways
 - Size
 - Location
 - Associated defects



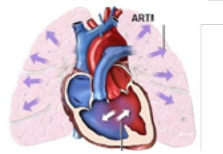
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Boards&Beyond

VSD

Ventricular Septal Defect

- Small VSD
 - Tiny hole → resists flow across defect ("restrictive")
 - Lots of turbulence → loud murmur
 - Small shunt (small volume of flow across defect)
- Large VSD
 - Large hole ("non-restrictive")
 - Significant shunting
 - Often closed surgically

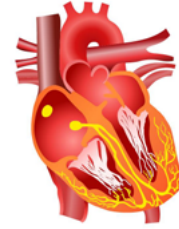


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ASD

Atrial Septal Defect

- Communication between left/right atrium
- Adds volume to RA/RV
- Delays closure of pulmonic valve
- **Wide, fixed splitting of S2**
- Increased flow across PV/TV
- **Systolic ejection murmur**
- Rarely a mid-diastolic murmur

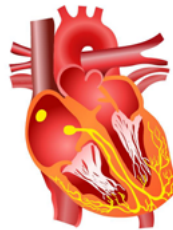


ASD

Atrial Septal Defect

- Oxygenated blood LA → RA
- ↑ O₂ saturation in RA, RV, PA
- "Shunt run"
 - Series of blood samples
 - SVC = 65%
 - IVC = 65%
 - RA = 75%
 - RV = 75%
 - PA = 75%

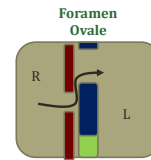
"Step up"



ASD

Atrial Septal Defect

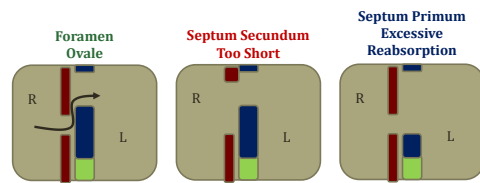
- **Secundum type** is most common
 - Defects at site of foramen ovale/ostium secundum
 - Poor growth of secundum septum
 - Or excessive absorption of primum septum
 - Located mid-septum
 - Often isolated defect



Septum secundum
Septum primum

ASD

Atrial Septal Defect



Septum secundum
Septum primum

ASD

Atrial Septal Defect

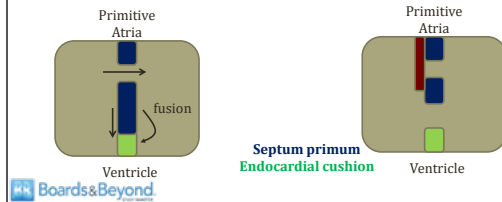


ASD

Atrial Septal Defect

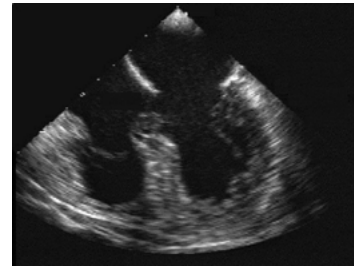
• Primum type

- Defect at site of ostium primum
- Failure of primum septum to fuse with endocardial cushions
- Located near AV valves; often occurs with other defects
- Seen in endocardial cushion defects (Down syndrome)



ASD

Atrial Septal Defect



PDA

Patent Ductus Arteriosus

- Ductus arteriosus shunts blood in utero
 - Left pulmonary artery → aorta
- Closes close after birth
 - "Functional" closure 18 to 24 hours (smooth muscle)
 - "Anatomic" occlusion over next few days/weeks
 - Becomes **ligamentum arteriosum**
- Patency maintained by prostaglandin E2
 - Major source in utero is placenta

PDA

Patent Ductus Arteriosus

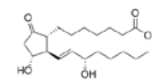
- Rarely remains patent (3 to 8 per 10,000 births)
- Associated with **congenital rubella syndrome**
 - ToRCHeS infection
 - Mother: Rash, fever, lymphadenopathy
 - Baby: Deafness, cataracts, cardiac disease
- PDA common
- Rare in developed countries (vaccination)
- Consider in infants whose mothers are immigrants

PDA

Patent Ductus Arteriosus

- **Continuous**, machine-like murmur
- **Widened pulse pressure**
 - Loss of volume in arterial tree through PDA
 - Low diastolic pressure → Increased pulse pressure
- **Differential cyanosis**
 - Occurs when shunt reverses R → L
 - Blue toes, normal fingers

Alprostadil



- **Prostaglandin E1**
- Maintains patency of ductus arteriosus
- Key effect: delivers blood to lungs
- Useful when poor RV → PA blood flow
 - Tetralogy of Fallot
 - Pulmonary atresia

Indomethacin

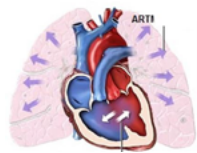
- **NSAID**
- Inhibits cyclooxygenase
- Decreases prostaglandin formation
- Can be used to close PDA

Qp:Qs

- Qp = Pulmonary blood flow
- Qs = Systemic blood flow
- Qp:Qs should be 1:1
- In shunts, Qp:Qs may be > 1:1
 - 1.5:1, 2:1, 3:1, etc.

Eisenmenger's Syndrome

- Uncorrected ASD/VSD/PDA
- Right heart chronically overloaded
 - RV Hypertrophy
 - Pulmonary hypertension
- Shunt reverses right → left
 - Cyanosis
 - Clubbing
 - Polycythemia (very high Hct)



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Fetal Alcohol Syndrome

- Caused by prenatal exposure to alcohol (teratogen)
- Characteristic facial features
- Impaired neurologic function
- **Congenital heart defects**
 - Atrial septal defect
 - Ventricular septal defect
 - Tetralogy of Fallot

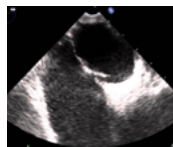


Teresa Kellerman/Wikipedia

PFO

Patent Foramen Ovale

- Found in ~25% adults
- Failure of foramen ovale to close after birth
- Can lead to **stroke** in patients with DVT/PE



Cyanotic Congenital Heart Disease

Cyanotic Congenital Heart Disease

Jason Ryan, MD, MPH

Boards&Beyond

Cyanosis

- **Central** cyanosis
 - Cardiac output normal
 - Blood is flowing
 - Not enough O₂
 - Lips
 - Nail beds
 - Conjunctivae
 - Warm extremities
- Peripheral cyanosis
 - Low blood flow
 - Severe heart failure
 - Cold extremities



WaltFletcher/Wikipedia

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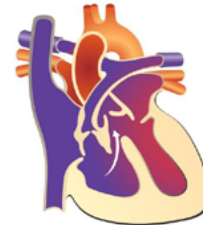
Blue Babies

- **Central cyanosis** early in life
- Blood not going through lungs after birth
 - Tetralogy of Fallot
 - Transposition of great vessels
 - Truncus arteriosus
 - Tricuspid atresia
 - Total anomalous pulmonary venous return

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Tetralogy of Fallot

- Constellation of four abnormalities
 - Ventricular septal defect (VSD)
 - Rightward deviation of aortic valve ("overriding aorta")
 - Subpulmonary stenosis
 - Right ventricular hypertrophy

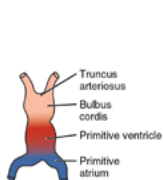


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Infundibulum

Conus Arteriosus

- "Funnel" leading to pulmonic valve
- Develops from **bulbus cordis**
- Smooth, muscular structure at RV outflow to PA



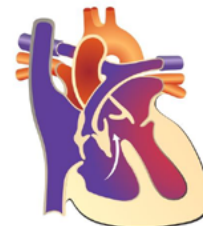
OpenStax College/Wikipedia

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Infundibulum

Conus Arteriosus

- **Septum displaced (moves toward RV) in TOF**
- Causes "overriding aorta"
 - 5-95% of aorta may lie over RV
- Causes VSD
 - Usually large ("non-restrictive")

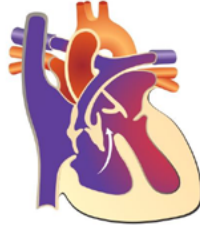


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Infundibulum

Conus Arteriosus

- **"Infundibular stenosis"**
 - Subpulmonary stenosis
 - RV outflow tract obstruction
- Abnormal pulmonary valve
 - Rarely main cause of obstruction
- Flow obstruction → RVH



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Tetralogy of Fallot

Physiology

- High **resistance to flow** RV → pulmonary artery
 - RV outflow pulmonic stenosis
- Diverts blood across VSD to left ventricle
- **Severity of flow obstruction determines symptoms**
- Severe obstruction: severe cyanosis
- Mild obstruction: less shunting ("pink" tets)

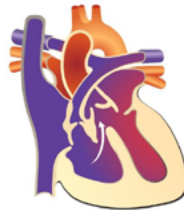


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Tetralogy of Fallot

Physiology

- Poor blood flow RV → lungs
- Left to right shunts beneficial
 - Bring back to pulmonary artery
 - Diverts blood to lungs
 - Improves oxygenation
- **Patent ductus arteriosus**
- Aortopulmonary collateral arteries
- Surgical shunt

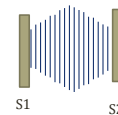


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Tetralogy of Fallot

Murmur

- Systolic ejection murmur
 - Crescendo-decrescendo
 - RV outflow and pulmonic stenosis
 - Heard best at left sternal border
- Single S2
 - S2 = closure of aortic and pulmonic valves
 - TOF: Diseased pulmonic valve → no sound
- VSD murmur (holosystolic) not typically heard
 - Large VSD → no murmur



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Tetralogy of Fallot

X-ray



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Tetralogy of Fallot

Other Features

- **Squatting** improves symptoms
 - Increased **afterload/TPR** (resists flow out of LV)
 - Pressure rises in the aorta/left ventricle
 - Less blood shunted RV → LV via VSD
 - More blood to lungs



Sherif Salama/Flickr

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Tetralogy of Fallot

Other Features

- **"Tet spells"**
 - Sudden cyanosis often when agitated
 - **Severe/complete RVOT obstruction**
 - O2, knees to chest, beta blockers (propranolol)

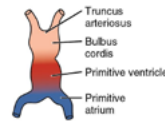


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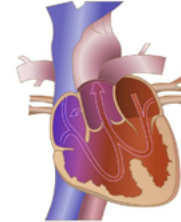
Truncus Arteriosus

- Common arterial trunk → Mixing of blood
- Failure of **neural crest cells** to drive formation of **aorticopulmonary septum**
- **Almost always has VSD**

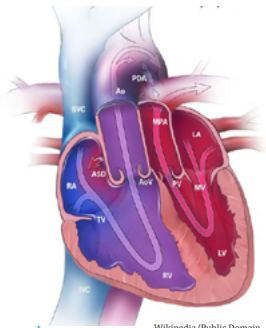


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Transposition of Great Vessels

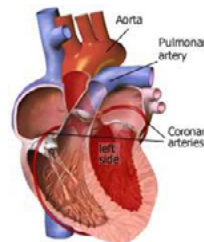


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Transposition of Great Vessels

- Normal heart:
 - **Aorta is posterior** and to right of pulmonary artery

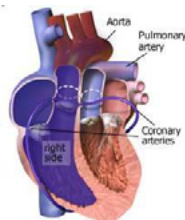


Kshaefer/WikiDoc

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Transposition of Great Vessels

- D-transposition (most common type):
 - **Aorta forms anterior** and rightward of pulmonary artery
 - Aorta arises from right ventricle
 - Pulmonary artery from left ventricle

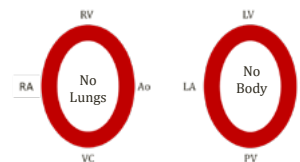


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Transposition of Great Vessels

- RV → Aorta → body → RA → RV
- LV → Pulmonary artery → LA → LV
- Two completely separate circuits
- NOT compatible with life unless shunt present
 - Usually PDA or VSD



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L-TGA

L-Transposition of the Great Arteries

- "Double switch": Aorta/PA and LV/RV
- "Congenitally corrected TGA"
- Venous blood → RA → LV → PA → Lungs
- Lungs → PV → LA → RV → Aorta
- Two circuits *not* separated
- Wrong connections (RV-Aorta, LV-PA)
- **Eventually right ventricle fails**

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Maternal Diabetes

- Infants at increased risk congenital anomalies
- Common congenital heart defects
 - **Transposition of great vessels**
 - Truncus arteriosus
 - Tricuspid atresia
 - VSD
 - PDA

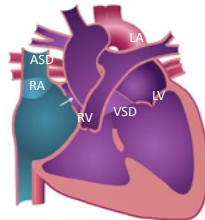


Øyvind Holmstad/Wikipedia

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Tricuspid Atresia

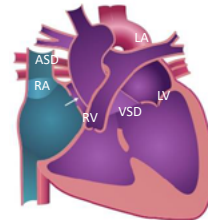
- Abnormal AV valves from endocardial cushions
- **No tricuspid valve**
- No blood RA → RV



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Tricuspid Atresia

- All cases have R→L shunt
 - Always seen with ASD
 - Allows blood flow to LA
- All cases have L→R shunt
 - Allows blood flow to lungs
 - LV → RV via VSD
 - Ao → PA via PDA

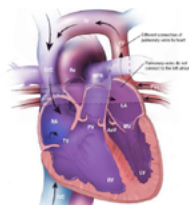


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TAPVR

Total Anomalous Pulmonary Venous Return

- Normal: pulmonary veins drain to left atrium
- TAPVR: pulmonary veins drain to venous system
 - Innominate (brachiocephalic) veins → SVC
 - Coronary sinus
 - Portal vein



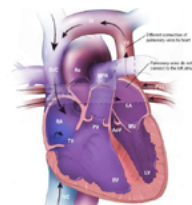
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TAPVR

Total Anomalous Pulmonary Venous Return

- RV → Lungs → Pulm Veins → RA → RV
- RA and RV dilate
- Must have a right to left shunt
 - **ASD** (most common)
 - **PDA**
- Mixed (low O2) blood to body

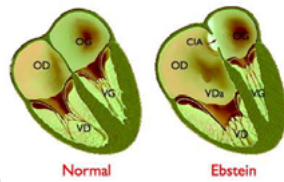


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Ebstein's Anomaly

- Apical displacement of TV → small RV
- "Atrialization" of RV tissue
- Severe tricuspid regurgitation
- Can lead to right heart failure

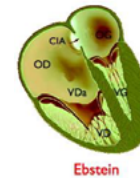
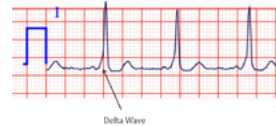


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Ebstein's Anomaly

- Right to left shunting and cyanosis if ASD
- High RA pressure
- Associated with **WPW**
 - Electrical bypass tract often present
 - Delta wave on EKG



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Maternal Lithium

- Teratogen
- Completely equilibrates across the placenta
- Teratogenic effects primarily involve heart
- **Ebstein's anomaly** most common



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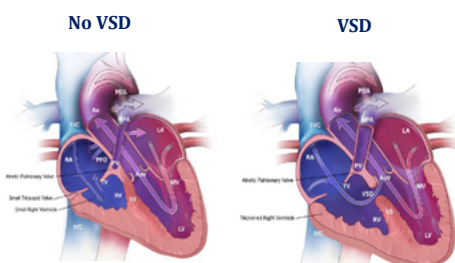
Øyvind Holmstad/Wikipedia

Pulmonary Atresia

- Failure of pulmonic valve orifice to develop
- No flow from RV to lungs
- In utero blood bypasses lungs (normal development)
- At birth: No blood flow to lungs through PV
 - PVR should fall but does not
- Often co-exists with VSD for outflow of RV
 - Similar to a severe form of Tetralogy of Fallot
- Survival depends on ductus arteriosus
- **Alprostadil** given to keep DA open

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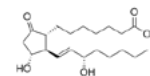
Pulmonary Atresia



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Alprostadil



- Prostaglandin E1
- Maintains patency of ductus arteriosus
- Key effect: **delivers blood to lungs**
- Useful when poor RV → PA blood flow
 - Tetralogy of Fallot
 - Pulmonary atresia

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Conotruncal Heart Defects

- Outflow tract anomalies
 - Trunk = Truncus arteriosus
 - Conus = Conus arteriosus
- Tetralogy of Fallot
- Truncus arteriosus
- Transposition of the great arteries
- **22q deletion syndromes**
 - DiGeorge syndrome (Thymic Aplasia)
 - Immunodeficiency, hypocalcemia
 - **Conotruncal anomalies**

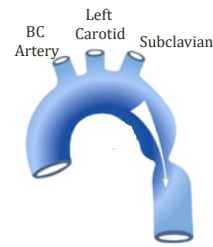
Coarctation of the Aorta

Coarctation of the Aorta

Jason Ryan, MD, MPH

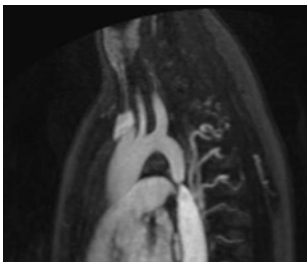
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Coarctation of the Aorta



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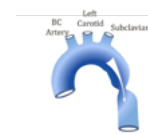
Coarctation of the Aorta



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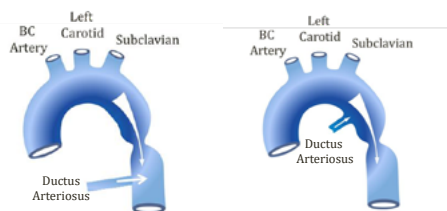
Coarctation of the Aorta

- Congenital disorder
- Usually involves thoracic aorta distal to subclavian
- Near insertion of ductus arteriosus
- "Juxtaductal" aorta
- Subtypes based on location of ductus arteriosus
- High **resistance to flow** in aorta



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Coarctation of the Aorta



Pre-ductal Coarctation

Post-ductal Coarctation

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Ductus Arteriosus

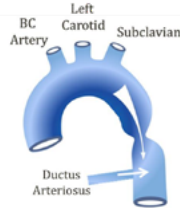
- Shunts blood in utero
- Left pulmonary artery → aorta
- Patency maintained by $\downarrow$ O₂ and $\uparrow$ prostaglandins
- At birth: $\uparrow$ O₂ and $\downarrow$ prostaglandins
- "Functional" closure 18 to 24 hours after birth
 - Smooth muscle constriction
- "Anatomic" occlusion over next few days/weeks
- Becomes **ligamentum arteriosum**

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Coarctation of the Aorta

Preductal or Infantile

- Ductus arteriosus supplies lower extremities
- Poor development of **collateral vessels**

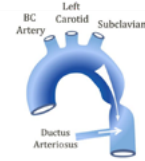


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Coarctation of the Aorta

Preductal or Infantile

- At birth ductus arteriosus open (not closed yet)
- Deoxygenated blood to lower extremity
- **Lower extremity cyanosis** may occur

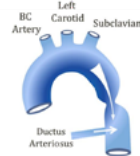


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Coarctation of the Aorta

Preductal or Infantile

- Ductus closure → symptoms may develop
- All flow through aorta with severe narrowing
- **Abrupt increase afterload**
- Rise in LVEDP
- Acute heart failure
- LV can dilate → fail → shock
- All caused by **closure of DA**



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Coarctation of the Aorta

Preductal or Infantile

- Key associations: Turner syndrome (45, XO)
- Short stature, webbed neck
- 5-10% have coarctation of the aorta



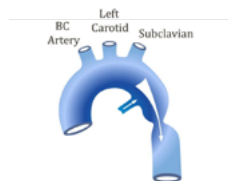
Johannes Nielsen/Wikipedia

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Coarctation of the Aorta

Postductal or Adult type

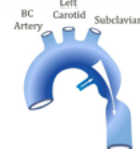
- Ductus arteriosus does not supply lower extremities
- Collaterals develop
- May go undetected until adulthood



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Coarctation of the Aorta

- Lower extremities → low blood pressure
 - ↑ Renin release
 - Salt/water retention
 - Vasoconstriction (Ang II)
 - Weak pulses ("brachio-femoral delay")
- Upper extremities and head → **high blood pressure**
- **Secondary hypertension**



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Coarctation of the Aorta

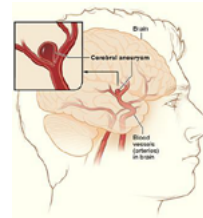
- Key association: **bicuspid aortic valve**
- Found in up to 60% of coarctation cases



Patrick J. Lynch, medical illustrator

Coarctation of the Aorta

- Key association: **intracranial aneurysms**
- Occur in about 10% of patients with coarctation



Wikipedia/Public Domain

Coarctation of the Aorta

Signs/Symptoms

- Only sign may be **hypertension** in arms
- Murmur over back between **scapula**
- Weak femoral pulses
- Pain with walking (claudication)

Coarctation of the Aorta

Signs/Symptoms

- **Rib notching**
 - High pressure above coarctation
 - Intercostals enlarge to carry blood around obstruction
 - Bulge into ribs
 - "Rib notching" seen on chest x-ray



WikiRadiography

Coarctation of the Aorta

Signs/Symptoms

- **3-sign**
 - Bulge before and after coarctation
 - "3 sign" on chest x-ray



WikiRadiography

Coarctation of the Aorta

Physiology

- **Autoregulation** maintains regional blood flow
 - Normal upper/lower perfusion despite high/low pressures
- **Upper extremities**
 - High blood pressure → high flow
 - Arterioles constrict to limit flow to normal level
 - Local effect – not mediated by sympathetic/parasympathetic
 - Resistance to flow is high ($Q = \Delta P / R$)
- **Lower extremities**
 - Low blood pressure
 - Arterioles dilate to increase flow to normal level ($Q = \Delta P / R$)
- Result is normal ("compensated") flow

Coarctation of the Aorta

Complications

- Heart failure
 - Pressure overload of left ventricle
- Aortic rupture/dissection
- **Endocarditis/endarteritis**
 - High-low pressure across narrowing
 - Endothelial injury
 - Low pressure distal to narrowing
 - Bacteria may attach more easily

Hypertension

Hypertension

Jason Ryan, MD, MPH



Hypertension

- Normal blood pressure: 120/80
- Stage 1 hypertension:
 - Systolic 130-139
 - Diastolic 80-89
- Need more than one measurement



Etiology

- Most (90%) is primary ("essential") HTN
 - Cause not clear
- Remainder (10%) secondary



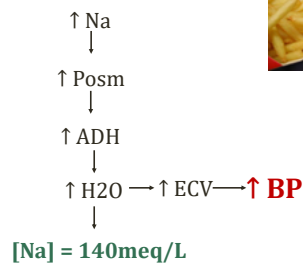
Hypertension

Risk Factors

- Family history
- African-American race
- High salt intake
- Alcohol
- Obesity
- Physical inactivity



Sodium Intake



Wikipedia/Public Domain



Hypertension

Associations

- Stroke
- Heart disease
 - MI
 - Heart failure
- Renal failure
- Aortic aneurysm
- Aortic dissection

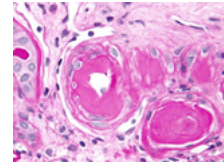


Hypertension Effects

- Atherosclerosis – lipid/fibrous plaques in vessels
- Arteriosclerosis – thickening of artery wall
 - Response to chronic hypertension

Hypertension Effects

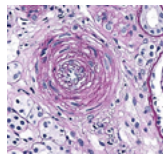
- **Hyaline** arteriosclerosis
 - Thickening of small arteries
 - Seen with aging
 - Also common with diabetes



Nephron/Wikipedia

Hypertension Effects

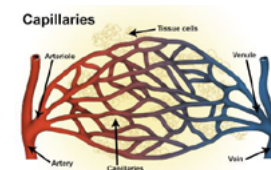
- **Hyperplastic** arteriosclerosis
 - Arteries look like "onion skin"
 - Occurs when hypertension is severe (usually DBP > 120)
 - "Malignant" hypertension
 - Retinal hemorrhages, exudates, or papilledema



Pacolarosa/Wikipedia

Arteriolar Rarefaction

- Loss of arterioles
- Arterioles close off and get resorbed



Wikipedia/Public Domain

Hypertension Effects

- Pulse pressure may increase
 - Example: Normal 120/80; HTN 170/100
 - Stiff arteries → ↓ compliance



Distensible
Vessel
120/80



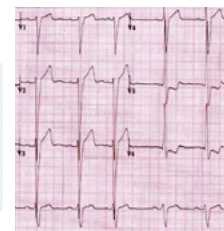
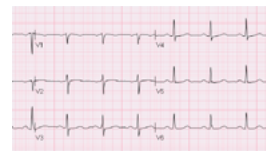
Stiff
Vessel
170/100

$$C = \Delta V / \Delta P$$

$$\Delta P = \Delta V / C$$

Hypertension Effects

- Afterload on heart is increased
- Left ventricle: concentric hypertrophy
 - Large voltage on EKG
 - Displaced apical impulse
 - S4



Hypertensive Urgency

- Severe hypertension without end-organ damage
- No agreed upon BP value
- Usually >180/120

Hypertensive Emergency

- Also no definite value
- BP usually >180/120
- Patient longstanding HTN, stops meds
- Neurologic impairment
 - Retinal hemorrhages, encephalopathy
- Renal impairment
 - Acute renal failure
 - Hematuria, proteinuria
- Cardiac ischemia

Hypertensive Emergency

- Associated with MAHA
- Endothelial injury → thrombus formation
- Improved with BP control



Database Center for Life Science (DBCLS)

Malignant Hypertension

- Historical term
- Most cases hypertension: "benign"
 - Modestly elevated blood pressure
 - Stable over years
- "Malignant hypertension"
 - Rare form, often fatal
 - Severe elevation of blood pressure (diastolic >120mmHg)
 - Rapidly progressive over 1 to 2 years
 - Renal failure, retinal hemorrhages, ischemia

Secondary Hypertension

Secondary Hypertension

Jason Ryan, MD, MPH

Boards&Beyond

Etiology

- Most (90%) is primary ("essential") HTN
 - Cause not clear
- Remainder (10%) secondary

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Blood Pressure

Determinants

- Cardiac output
 - Increased with renal salt/water retention
- Total peripheral resistance
 - Key vessels: arterioles
 - Increased by vasoconstrictors (i.e. catecholamines)
 - Increased by sympathetic nervous system

$$BP = CO \times TPR$$

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Chronic Kidney Disease

- Over 80% of patients have hypertension
- Multiple causes:
 - Sodium retention
 - Increased renin-angiotensin-aldosterone activity
 - Increased sympathetic nervous system activity



Anna Prodesiak/Wikipedia

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Obstructive Sleep Apnea

- Sleep-related breathing disorder
- Apnea during sleep
- Often associated with hypertension
- Treatment may reduce BP



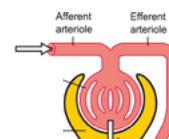
PruebasBMA / Wikipedia

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NSAIDs

Ibuprofen, naproxen, indomethacin, ketorolac, diclofenac

- Nonsteroidal anti-inflammatory drugs
- Inhibit cyclooxygenase in kidneys
- Decrease synthesis of prostaglandins
- PGE-2: Renal vasodilator



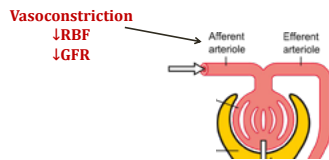
Madher088 / Wikipedia

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NSAIDs

Ibuprofen, naproxen, indomethacin, ketorolac, diclofenac

- ↓ Na/Water excretion
- May cause **hypertension**
- May exacerbate **heart failure**



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Oral Contraceptive Pills

OCPs

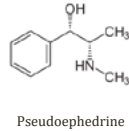
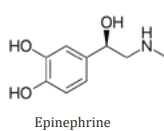
- Estrogen and progesterone analogs
- Cause mild increase in blood pressure



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Pseudoephedrine

- Nasal decongestant
- Alpha-1 agonist
- Vasoconstriction → ↓ nasal blood flow



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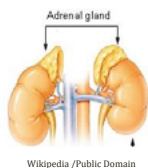
Cyclosporine & Tacrolimus

- Immunosuppressants
- Calcineurin inhibitors
- **Renal vasoconstriction** → salt/water retention
- **Diltiazem**: drug of choice
 - Impairs metabolism (↑ drug levels)
 - Treats HTN and allows lower dose cyclosporine to be used

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Primary Aldosteronism

- Excessive levels of aldosterone secretion
- Not due to increased activity of RAAS system
- Adrenal adenoma (Conn's syndrome)
- Bilateral idiopathic adrenal hyperplasia



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Primary Aldosteronism

- ↑ Na reabsorption distal nephron
- ↑ ECV → ↑ CO → Hypertension
- ↑ K excretion → hypokalemia

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Aldosterone Escape

- Excess aldosterone does not lead to volume overload
- Usually no pitting edema, rales, increased JVP
- Na/Fluid retention → hypertension
- Compensatory mechanisms activated
 - Increased ANP
- Increased sodium and free water excretion
- Result: diuresis → **normal volume status**

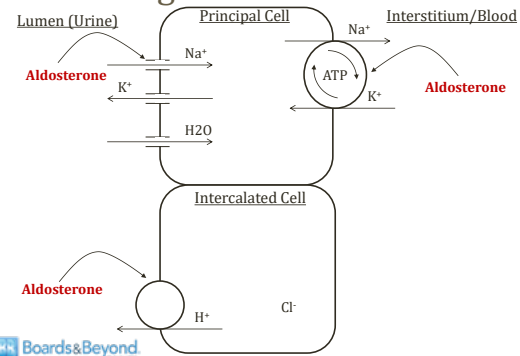
Primary Aldosteronism

- Clinical features
 - Resistant hypertension
 - Hypokalemia
 - Normal volume status on physical exam
- Diagnosis
 - **Renin-independent** aldosterone secretion
 - Low plasma renin activity
 - High aldosterone levels
- Drugs of choice: **Spirolactone/Eplerenone**
 - Aldosterone antagonists

Liddle Syndrome

- Genetic disorder
- **Increased activity of ENaC**
- Similar clinical syndrome to hyperaldosteronism
 - Hypertension
 - Hypokalemia
- **Aldosterone levels low**

Collecting Duct



Pheochromocytoma

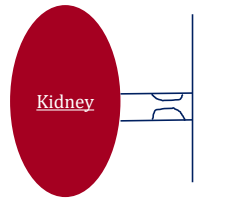
- **Catecholamine-secreting tumor**
 - Epinephrine, norepinephrine, dopamine
- Usually arises from adrenal gland
- Triad: Palpitations, headache, episodic sweating
 - PHEochromocytoma
- Most patients have hypertension
- Diagnosis: Catecholamines breakdown products
 - Metanephrines
 - Vanillylmandelic acid (VMA)

Cushing's Syndrome

- Excess **cortisol**
- Often from steroid administration
- Other causes
 - Cushing's Disease (pituitary oversecretes ACTH)
 - Tumors (i.e. small cell lung cancer secretes ACTH)
 - Adrenal tumor secretes cortisol
- Cortisol → hypertension
 - Increased vascular sensitivity to adrenergic agonists

Renal Artery Stenosis

- Vascular disease of renal arteries
- Decreased blood flow to kidneys
- Key exam finding: **renal bruit**



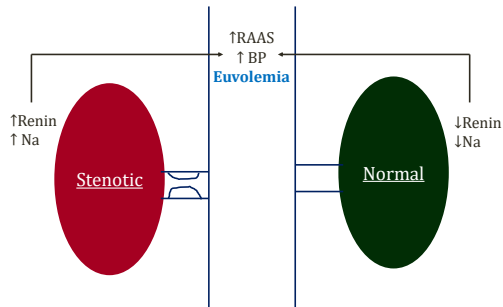
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Renal Artery Stenosis

- Increased renin, salt-water retention → HTN
- Often unilateral stenosis
- Normal kidney compensates
- Results: **No signs of volume overload**

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Renal Artery Stenosis



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Renal Artery Stenosis

Angiotensin II

- Normal GFR depends on **angiotensin II**
 - All → efferent arteriole vasoconstriction
 - Maintains GFR
- ACE inhibitors can precipitate renal failure

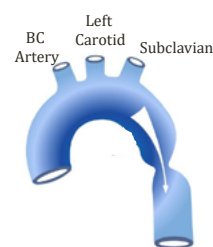
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Fibromuscular Dysplasia

- Vascular disease → obstruction to flow
- Common among **women**
- Often occurs in 40s-50s
- Non-atherosclerotic, non-inflammatory
- Often involves **medial layer** fibroplasia
- Stenosis and aneurysms of vessels ("string of beads")
- Most common in renal and carotid arteries
- Can lead to renal artery stenosis

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Coarctation of the Aorta

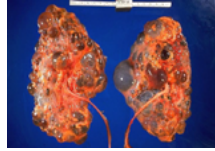


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ADPKD

Autosomal dominant polycystic kidney disease

- Genetic disorder
- Mutations of PKD1 or PKD2
- Presents in adulthood with HTN and renal cysts
- Increased RAAS activity



Wikipedia/Public Domain

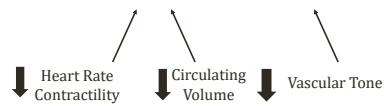
Hypertension Drugs

Antihypertensives

Jason Ryan, MD, MPH



$$BP = CO \times TPR$$



Beta Receptors

- β_1 receptors in heart, kidneys
 - Increase heart rate and contractility
 - Stimulate renin release
 - Blockade $\rightarrow$ $\downarrow$ CO, $\downarrow$ ECV $\rightarrow$ $\downarrow$ BP
- β_2 receptors
 - Dilate blood vessels (muscle, liver)
 - Bronchodilate
 - Blockade does not lead to lower blood pressure



Beta Blockers

β_1 -selective antagonists

- Atenolol, Metoprolol, Esmolol
- Used for hypertension
 - Blockade $\rightarrow$ $\downarrow$ CO, $\downarrow$ ECV $\rightarrow$ $\downarrow$ BP
- Metoprolol: Systolic heart failure
 - Blocks sympathetic stimulation of heart
 - Reduces mortality



Beta Blockers

$\beta_1\beta_2$ (nonselective) antagonists

- Propranolol, Timolol, Nadolol
- Can be used for hypertension
- Nadolol, Propranolol: Used in **portal hypertension**
 - Beta 1 blockade: $\downarrow$ CO, $\downarrow$ ECV $\rightarrow$ $\downarrow$ BP
 - Beta 2 blockade: $\downarrow$ portal blood flow
- Timolol: Used in **glaucoma**
 - Beta 1 and Beta 2 $\rightarrow$ aqueous humor production



Beta Blockers

$\beta_1\beta_2\alpha_1$

- Carvedilol, Labetalol
- Labetalol: Hypertensive Emergency
 - Rapid reduction in blood pressure
- Carvedilol: Systolic heart failure
 - Blocks sympathetic stimulation of heart
 - Reduces mortality



Beta Blockers

Partial Agonists

- Pindolol: $\beta_1\beta_2$ (nonselective)
- Acebutolol: β_1 -selective
- "Intrinsic sympathomimetic activity"
 - Beta agonist when sympathetic activity is low
 - Beta blocker when sympathetic activity is high
- Can cause **angina** through beta 1 activation
- Special pharmacologic properties

Beta Blockers

Side effects

- Fatigue, erectile dysfunction, depression
 - More common with older beta blockers (propranolol)
- Hyperlipidemia
 - Mild increase in triglycerides
 - Mild decrease in HDL
 - Effect varies with different beta blockers

Beta Blockers

Side effects

- Caution in **diabetes**
- Blockade of epinephrine effects
 - Epinephrine raises glucose levels
 - Blockade $\rightarrow$ hypoglycemia
- Blockade of hypoglycemia symptoms
 - $\downarrow$ glucose $\rightarrow$ sweating/tachycardia
 - Symptoms "masked" by beta blockers



Victory/Flier

Beta Blockers

Side effects

- Caution in **asthma/COPD**
 - β_2 receptors: bronchodilators
 - β_2 blockade may cause a flare
 - β_1 blockers ("cardioselective") often used
- Decompensated **heart failure**
 - β_1 blockers lower cardiac output $\rightarrow$ worsening of symptoms
 - Commonly used in compensated heart failure
 - Mortality benefit

Beta Blockers

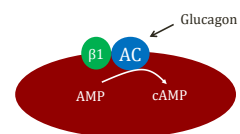
Overdose

- Depression of myocardial contractility $\rightarrow$ shock
- Bradycardia/AV block

Beta Blockers

Overdose

- Treatment: **Glucagon**
 - Activates adenylyl cyclase at different site from beta receptors
 - $\uparrow$ cAMP $\rightarrow$ $\uparrow$ intracellular Ca
 - Increased contraction and heart rate



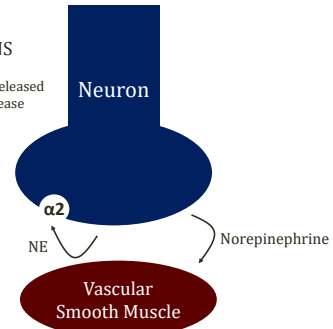
α1 Blockers

Tamsulosin, Alfuzosin, Doxazosin, Terazosin

- α1 receptors in periphery: vasoconstrict
- Blockade → vasodilation → ↓ TPR → ↓ BP
- Used in **benign prostatic hypertrophy**
 - Relax smooth muscle of bladder/prostate
 - Increase urine flow
- Common side effect: **Postural hypotension**
- Tamsulosin: "Uroselective"
 - Less hypotension effect

Alpha 2 Receptors

α2 receptors in CNS
Presynaptic receptor
Feedback to nerve when NE released
Activation leads to JNE release



Clonidine

α2 agonist

- Old, rarely used hypertension drug
- Key side effect: **Rebound hypertension**
 - Abrupt cessation of drug (usually at high dose)
 - Severe hypertension (SBP>200; DBP>120)
 - Symptoms of high BP and sympathetic over-activity
 - Nervousness, sweating, headache, chest pain
- Also causes **sedation**

Methyldopa

α2 agonist

- Drug of choice in **pregnancy**
- Also causes sedation
- Key side effect (rare): **Hemolytic anemia**



Calcium Channel Blockers

- Three major classes of calcium antagonists
 - dihydropyridines (nifedipine)
 - phenylalkylamines (verapamil)
 - benzothiazepines (diltiazem)
- **Vasodilators** and **negative chronotropes/inotropes**

Calcium Channel Blockers

- Vascular smooth muscle effects
 - Nifedipine>Diltiazem>Verapamil
- Heart rate/contractility effects
 - Verapamil>Diltiazem>Nifedipine

Calcium Channel Blockers

- Dihydropyridines (nifedipine) → vasodilators
 - Main effect: ↓TPR
- Non-dihydropyridines (Verapamil, diltiazem)
 - Similar to β1 blockers
 - Main effects: ↓HR; ↓ contractility

Calcium Channel Blockers

Dihydropyridines (nifedipine)

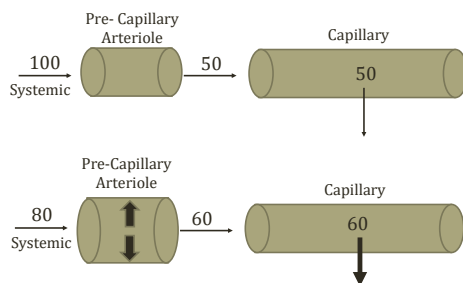
- Used for hypertension
- Flushing, headache, hypotension
 - Peripheral vasodilation
- Key side effect: edema
 - Increased **capillary** hydrostatic pressure
 - **Pre-capillary arteriolar vasodilation**



James Heilman, MD/Wikipedia

Calcium Channel Blockers

Dihydropyridines (nifedipine)



Calcium Channel Blockers

Verapamil, diltiazem

- Used for hypertension
- Also used in heart disease
 - Arrhythmias (atrial fibrillation)
 - Stable angina (lower oxygen demand)
- Potential side effect: Negative inotropes
 - Can precipitate heart failure

Calcium Channel Blockers

Other Side Effects

- Constipation
 - Most commonly with **verapamil**



Elya/Wikipedia

Calcium Channel Blockers

Other Side Effects

- Hyperprolactinemia
 - Seen with **verapamil**
 - Blocks calcium channels CNS → ↓ dopamine release
 - Causes hypogonadism
 - Men: ↓ libido, impotence
 - Pre-menopausal women: irregular menses, galactorrhea

Calcium Channel Blockers

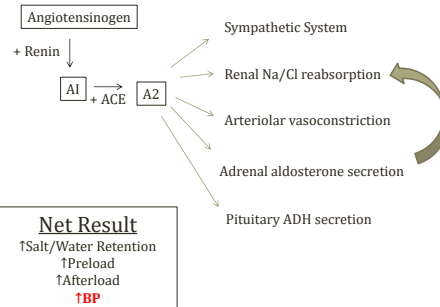
Other Side Effects

- Gingival hyperplasia
 - Seen in all types CCB
 - Also with phenytoin, cyclosporine

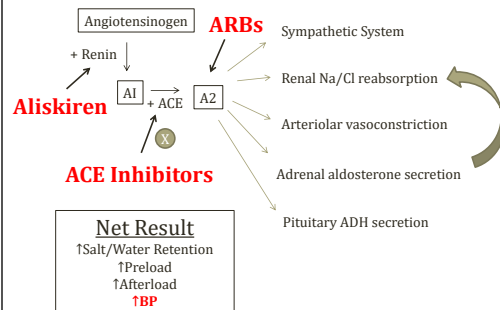


Lesion/Wikipedia

Angiotensin II



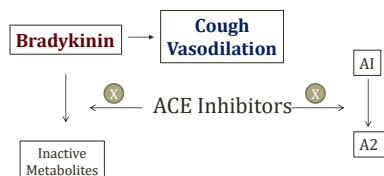
Angiotensin II Inhibition



ACE Inhibitors

- ACE Inhibitors
 - Captopril, Enalapril, Lisinopril, Ramipril
- Angiotensin Receptor Blockers (ARBs)
 - Candesartan, Irbesartan, Valsartan
- Side effects
 - Hyperkalemia (↓aldosterone)
 - Renal failure (↓GFR)

Bradykinin



ACE Inhibitors

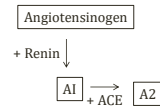
Unique Side Effects

- Due to increased bradykinin
- Dry Cough
 - Occurs in ~10% of patients
- Angioedema
 - Swelling of face, tongue
 - Can be life-threatening



Aliskiren

- Direct renin inhibitor
- Reduces angiotensin I levels (unique effect)



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Diuretics

- Loop diuretics
 - Furosemide, bumetanide, torsemide, ethacrynic acid
- Thiazide diuretics
 - **Hydrochlorothiazide**; chlorthalidone; metolazone
- Potassium sparing diuretics
 - Spironolactone, Eplerenone, Triamterene, Amiloride

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Hydralazine

- Direct arteriolar vasodilator
- Rarely used for hypertension
- Combined with nitrates for heart failure
- Safe in pregnancy
- Causes drug-induced lupus

Boards&Beyond

Drug-induced Lupus

- Syndrome similar to lupus
 - Often rash, arthritis, low blood cell counts
 - Milder than SLE
 - Usually no associated renal failure/CNS disease
- Key finding: **anti-histone antibodies**
- Three drugs
 - Hydralazine
 - Procainamide
 - Isoniazid

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Hypertensive Emergency

- Unique drugs used for therapy
 - Intravenous, rapid acting
- Lowering BP too fast can cause ischemia
 - Autoregulation of vascular beds → vasoconstriction

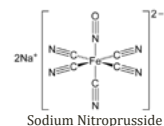


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KOMU News/Pike

Hypertensive Emergency

- **Nitroprusside**
 - Short acting drug
 - ↑ intracellular cGMP
 - ↑ nitric oxide release
 - Venous and arteriolar vasodilation
 - ↓ preload (VR); ↓ afterload
- **Cyanide toxicity** with prolonged use
 - Multiple cyanide groups per molecule
 - Inhibits electron transport
 - Toxic levels with prolonged infusions



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Hypertensive Emergency

- **Fenoldopam**
 - D1 agonist
 - Arteriolar vasodilation
 - Increased urinary sodium/water excretion
 - Maintains renal perfusion while vasodilating

Hypertensive Emergency

- Labetalol
 - β 1 and α 1 Blocker
- Esmolol
 - Rapid acting intravenous β 1 blocker
- Nicardipine, Clevidipine
 - Intravenous dihydropyridine calcium channel blocker

Orthostatic Hypotension

Postural Hypotension; Orthostasis

- $\downarrow$ blood pressure due to gravity with standing
- Compensation from **sympathetic nervous system**
 - Increased VR, CO, HR, TPR
 - Impaired with **low volume, low TPR, blunted ANS**
- Severe $\downarrow$ BP (>20 mmHg) = orthostatic hypotension
 - Dizziness, syncope
- Common etiologies:
 - Hypovolemia
 - Hypertensive medications

Orthostatic Hypotension

Postural Hypotension; Orthostasis

- Alpha-1 blockers
- ACE-inhibitors
 - Especially in patients on diuretics
 - Volume depletion $\rightarrow$ $\uparrow$ RAAS
 - "First dose hypotension"

Reflex Tachycardia

- Vasodilation $\rightarrow$ $\downarrow$ BP $\rightarrow$ $\uparrow$ SNS
- Reflex response: $\uparrow$ HR
- Can be caused by vasodilators
 - Hydralazine
 - Alpha-1 blockers
 - Dihydropyridine calcium channel blockers
 - Nitroglycerine
- May exacerbate chronic stable angina
- Drugs may be co-administered with β blocker

Choosing Drugs

- Diabetes
 - ACE inhibitors: Protective of kidneys
 - Beta blockers can lower glucose and mask hypoglycemia
 - HCTZ can increase glucose
- Systolic Heart Failure
 - ACEi, beta blockers, aldosterone blockers: mortality benefit
 - Calcium channel blockers $\rightarrow$ $\downarrow$ contractility

Choosing Drugs

- Hypertension in Pregnancy
 - Methyldopa
 - Beta blockers, nifedipine, hydralazine
 - Avoid: **ACE inhibitors, ARBs, direct renin inhibitors**
 - Associated with congenital malformations
- Significant renal failure or ↑K
 - Avoid: ACE-inhibitors, ARBs (↓GFR, ↓aldosterone)
 - Avoid: Potassium sparing diuretics (↑K)
 - Avoid: Other diuretics (↓ECV → ↓GFR)
 - Calcium blockers, beta blockers usually ok

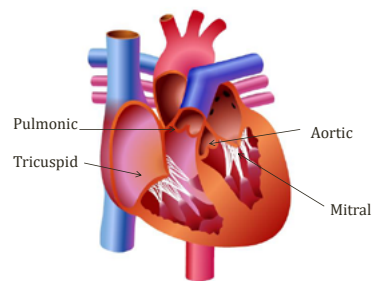
Valve Disease

Valve Disease

Jason Ryan, MD, MPH



Heart Valves



Valve Disease

- **Stenosis**
 - Stiffening/thickening of valve leaflets
 - Obstruction to forward blood flow
- **Regurgitation**
 - Malcoaptation of valve leaflets
 - Leakage of blood flow backwards across valve



Valve Lesions - Systole

- Occur when heart contracts/squeezes
- Aortic stenosis
- Mitral regurgitation
- Pulmonic stenosis
- Tricuspid regurgitation



Valve Lesions - Diastole

- Occur when heart relaxes/fills
- Aortic regurgitation
- Mitral stenosis
- Pulmonic regurgitation
- Tricuspid stenosis



Valve Disorders

Treatments

- Only **severe** valvular lesions treated
- Mostly **surgical diseases**
- Surgical repair
 - Often done for mitral valve prolapse → mitral regurgitation
- Valve replacement
 - Bioprosthetic (pig or cow)
 - Mechanical (requires life-long anticoagulation)
- Valvuloplasty (stenotic lesions)



Stenotic Valve Disorders

- Stiff valve
- “Gradient” across valve
- High pressure upstream
- Lower pressure downstream

Rheumatic Fever

- Occurs weeks after **streptococcal pharyngitis**
- Common in **children**
- Autoimmune: type II hypersensitivity reaction
- Antibodies to bacterial **M proteins** cross-react

Rheumatic Fever

- **Jones** criteria
 - Joints: Polyarthrititis (>5 joints)
 - ❤️: Carditis (valvulitis, myocarditis, pericarditis)
 - Nodules (subcutaneous)
 - Erythema marginatum (rash on trunk)
 - Sydenham chorea (jerking movement disorder)

Rheumatic Heart Disease

- Damage to heart valves by rheumatic fever
- **Mitral valve** most commonly involved
- Often presents years after acute rheumatic fever
- Many patients do not recall acute symptoms
- Common in **developing countries**
 - Limited access to medical care for pharyngitis
 - Often seen in **immigrants** to US

Carcinoid Heart Disease

- Caused by carcinoid tumors of intestines
- Secrete **serotonin**
- Fibrous deposits **tricuspid/pulmonic valves**
- Leads to stenosis and regurgitation
- Serotonin inactivated by lungs
- Left sided lesions rare

Aortic Stenosis

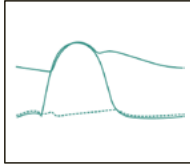
Pathophysiology

- Stiff aortic valve
- Systolic problem
- **Increased afterload**

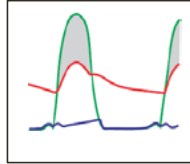
Aortic Stenosis

Hemodynamics

- **LV pressure systolic >> aortic pressure**
 - LVSP = 160mmHg (normal = 120)
 - SBP = 120mmHg (normal = 120)
 - Gradient = 40mmHg
- **↑ LVEDP** due to ↑ afterload



Normal



Aortic Stenosis

Aortic Stenosis

Clinical features

- Systolic crescendo-decrescendo murmur
- **Syncope**: failure to ↑CO due to ↑ afterload
- **Angina**: ↑ LVEDP → ↓ coronary blood flow
- **Left heart failure**: ↑ LVEDP

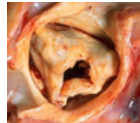
Aortic Stenosis

Causes

- **Senile aortic stenosis**
 - "Wear and tear"
 - Collagen breakdown
 - Calcium deposition
- **Bicuspid aortic valve**
- Rarely rheumatic heart disease



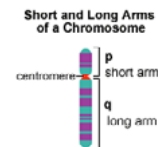
Patrick J. Lynch, medical illustrator



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Supravalvular Aortic Stenosis

- Narrowing of ascending aorta above aortic valve
- Seen in **Williams syndrome**
- Genetic deletion syndrome



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Mitral Stenosis

Pathophysiology

- Stiff mitral valve
- Diastolic problem
- **LA pressure >> LV diastolic pressure**
 - Left atrial pressure 20mmHg (normal = 10)
 - LVEDP 5mmHg (normal = 10)
 - Gradient = 15mmHg
- **Decreased preload**

Mitral Stenosis

Clinical features

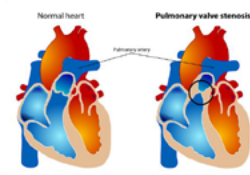
- Caused by **rheumatic fever**
- Most common symptom: dyspnea
 - ↑ LA pressure → pulmonary congestion
- Murmur: diastolic rumble with opening snap

Tricuspid Stenosis

- Very rare valve disorder
- **Diastolic murmur at left lower sternal border**
- Caused by rheumatic fever (with mitral disease)
 - Tricuspid regurgitation more common
- Carcinoid heart disease

Pulmonic Stenosis

- **Congenital defect** in children
 - Fused commissures with thickened leaflets
- Carcinoid heart disease



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Regurgitant Lesions

- Acute and chronic forms
- Acute regurgitation (often from endocarditis)
 - May cause shock
 - Activation of sympathetic nervous system
 - Increased contractility
 - Increased afterload
- Chronic regurgitation
 - No shock
 - Leads to chronic heart failure
 - Sympathetic activation only if severe heart failure

Aortic Regurgitation

Pathophysiology

- Blood leaks across aortic valve
- Diastolic problem
- **Increased preload, stroke volume**
- Increased afterload
 - More stroke volume → aorta → ↓ compliance (stiffening)
- Blowing diastolic murmur

Aortic Regurgitation

Causes

- **Dilated aortic root** → leaflets pull apart
 - Often from HTN or other aortic aneurysm
 - Rarely from tertiary syphilis (aortitis)
- **Bicuspid aortic valve**
 - Turner syndrome
 - Coarctation of the aorta
- Endocarditis
- Rheumatic heart disease
 - Almost always with mitral disease

Aortic Regurgitation

Clinical features

- Leaking blood back into LV causes **low diastolic BP**
 - 120/80 (normal) → 120/40
 - Low diastolic pressure
- **Wide pulse pressure**
 - High cardiac output with low diastolic pressure
- Wide pulse pressure symptoms
 - "Water hammer" pulses
 - Head bobbing
 - Many, many others (mostly historical)

Mitral Regurgitation

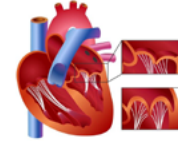
Pathophysiology

- Blood leaks across mitral valve
- Increased LA volume → Starling mechanism
- Increased left ventricular filling from LA
- **Increased preload, stroke volume**
- Reduced afterload

Mitral Regurgitation

Causes

- Primary MR caused by **mitral valve prolapse**
 - Also called degenerative or myxomatous
- Billowing of mitral valve leaflets above annulus
- Common cause of **mitral regurgitation**
- Causes a **systolic click**
 - Don't confuse with opening snap of mitral stenosis



Mitral Regurgitation

Secondary causes

- Ischemia → damage to papillary muscle
- Left ventricular dilation
 - Dilated cardiomyopathy
 - Leaflets pulled apart
 - **"Functional" MR**
- Hypertrophic cardiomyopathy

Mitral Regurgitation

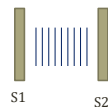
Causes

- Endocarditis
- Rheumatic heart disease
- Congenital
 - Cleft mitral valve
 - Endocardial cushion defect
 - Down syndrome

Mitral Regurgitation

Clinical Features

- Holosystolic murmur at apex



Afterload Reduction

Aortic and Mitral Regurgitation

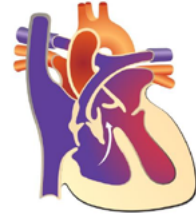
- In theory, ↓ afterload can improve forward flow
- For severe, acute regurgitation this helps
- For chronic disease, clinical trials with mixed results
- In general, these are surgical diseases
- Common test scenario "Best medical option?"

Tricuspid Regurgitation

- Small amount of TR normal ("physiologic TR")
- Holosystolic murmur at left sternal border
- Pathologic causes
 - Functional TR from **RV enlargement**
 - **Endocarditis** - classically IV drug users
 - Carcinoid
 - Ebstein's anomaly

Pulmonic Regurgitation

- Most common cause: repaired **Tetralogy of Fallot**
 - Repair of RVOT obstruction damages valve
- Endocarditis (rare)
- Rheumatic heart disease (rare)



Tetralogy of Fallot

Shock

Shock

Jason Ryan, MD, MPH

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Shock

- Life-threatening fall in blood pressure
- Poor tissue perfusion
- **Low cardiac output**
 - Loss of contractility
 - Low intravascular volume
- **Peripheral vasodilation**

$$BP = CO \times TPR$$

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Types of Shock

- Cardiogenic
 - Cardiac disorder → fall in cardiac output
- Hypovolemic
 - Fall in intravascular volume → fall in cardiac output
 - Hemorrhage
- Distributive
 - Peripheral vasodilation
 - Septic, anaphylactic
- Obstructive

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Types of Shock

- **Different treatments** for different types of shock
- Often can determine type from **history**
 - Myocardial infarction → cardiogenic shock
 - Massive bleeding → hypovolemic shock
- Shock of unclear etiology: Swan-Ganz catheter

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Swan-Ganz Catheter

Pulmonary artery catheter



Pulmonary Capillary Wedge Pressure
PCWP
"Wedge Pressure"
Equal to LA pressure

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Swan-Ganz Data

- RA Pressure (Normal ~ 5mmHg)
- RV Pressure (20/5)
- PA Pressure (20/10)
- PCWP Pressure (10)
- Mixed venous O2 sat
 - Oxygen concentration after all veins mix

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Fick Equation

$$\begin{aligned}\text{Oxygen Consumed} &= \text{O}_2 \text{ Out Lungs} - \text{O}_2 \text{ In Lungs} \\ &= \text{CO} (\text{Art O}_2 - \text{Ven O}_2)\end{aligned}$$

$$\text{Cardiac Output} = \frac{\text{O}_2 \text{ Consumption}}{(\text{Art O}_2 - \text{Ven O}_2)}$$

O₂ Consumption $\propto$ body size
Arterial O₂ Content = O₂ sat on finger probe
Venous O₂ Content = O₂ from Swan-Ganz

Swan-Ganz catheter gives cardiac output

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Flow Equation

- Used to determine **systemic vascular resistance**

$$\begin{aligned}\Delta P &= \text{CO} * \text{SVR} \\ \text{MAP} - \text{RAP} &= \text{CO} * \text{SVR}\end{aligned}$$

$$\text{SVR} = \frac{\text{MAP} - \text{RAP}}{\text{CO}}$$

Swan-Ganz Catheter gives SVR

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Swan-Ganz Data

- Direct
 - RA Pressure (Normal ~ 5mmHg)
 - RV Pressure (20/5)
 - PA Pressure (20/10)
 - PCWP Pressure (10)
 - Mixed venous O₂ sat
- Calculated
 - Cardiac output
 - Systemic Vascular Resistance

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Hemodynamic of Shock

- Four major classes of shock
- All have different hemodynamics from Swan
- Swan can be used to determine etiology of shock
 - Cardiogenic
 - Hypovolemic
 - Distributive
 - Obstructive

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Cardiogenic Shock

- Hallmark is **low cardiac output**
- High cardiac pressures**
- High SVR (sympathetic response)
- Classic cause: large myocardial infarction
- Also seen in advanced heart failure (depressed LVEF)

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Hypovolemic Shock

- Poor fluid intake
- High fever, insensible losses
- Hemorrhage
- Low cardiac output**
- Low cardiac pressures**
- High SVR (sympathetic response)

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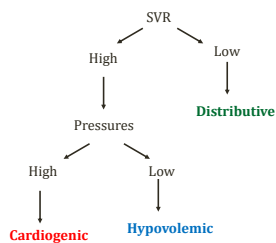
Distributive Shock

- **Hallmark is low SVR**
- Diffuse vasodilation and/or endothelial dysfunction
- **Sepsis (most common)**
- Anaphylaxis
- Neurogenic
- Cardiac output classically high (variable)
- Cardiac pressures variable

Type of Shock

	Cardiogenic	Hypovolemic	Distributive
Blood Pressure	↓	↓	↓
HR	↑	↑	↑
RA Pressure	↑	↓	↓/-
RV Pressure	↑	↓	↓/-
PCWP	↑	↓	↓/-
Cardiac Output	↓	↓	↑
SVR	↑	↑	↓

Major Shock Types



Physical Exam

- Cold skin → high SVR and low CO
 - Cardiogenic
 - Hypovolemic
- Warm skin → low SVR and high CO
 - Distributive
- Jugular venous pressure → high RA pressure
- Pulmonary rales → high LA pressure

Obstructive Shock

- Obstruction to blood flow from heart
- Low cardiac output despite normal contractility
- Tamponade
- Tension pneumothorax
- Massive pulmonary embolism
- Low cardiac output
- High SVR

Treatment of Shock

- Cardiogenic: inotropes
 - Milrinone, Dobutamine
- Hypovolemic: volume
 - Blood transfusions, IV fluids
- Distributive: vasopressors
 - Phenylephrine, epinephrine, norepinephrine
- Obstructive: resolve obstruction
 - Treat tamponade, embolism, tension pneumothorax

Swan in Valve Disease

RA (5)	15
RV (20/5)	45/15
PA (20/10)	45/30
PCWP (10)	30
LV (120/10)	120/5
Ao (120/80)	120/80

Mitral Stenosis

Swan in Valve Disease

RA (5)	5
RV (20/5)	20/5
PA (20/10)	20/10
PCWP (10)	10
LV (120/10)	150/10
Ao (120/80)	120/80

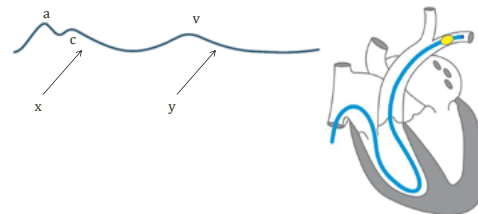
Aortic Stenosis

Swan in Valve Disease

RA (5)	15
RV (20/5)	45/15
PA (20/10)	45/30
PCWP (10)	30
LV (120/10)	120/30
Ao (120/80)	120/40

Aortic Regurgitation

Left Atrial Pressure



Giant V waves

- Seen in **mitral regurgitation** in **PCWP tracing**
- Similar to giant V waves in tricuspid regurgitation
 - Seen in venous pressure tracing



Pericardial Disease

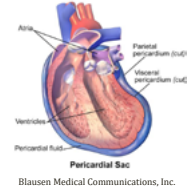
Pericardial Disease

Jason Ryan MD, MPH

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Pericardium

- Three layers
- Fibrous pericardium
- Serous pericardium
 - Parietal layer
 - Visceral layer
- Pericardial cavity between serous layers
- Innervated by **phrenic nerve**
- Pericarditis → referred pain to the shoulder

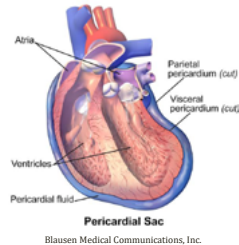


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Pericardial Diseases

- Pericarditis
- Tamponade
- Constrictive pericarditis



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Pericarditis

- Most common pericardial disorder
- Inflammation of the pericardium
- Immune-mediated (details not known)
- May recur after treatment

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Pericarditis

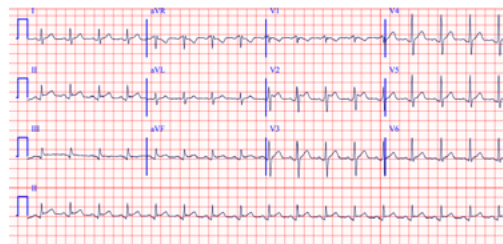
Clinical Features

- Chest pain
 - Sharp
 - Worse with deep breath (pleuritic)
 - Worse lying flat (supine)
 - Better sitting up/leaning forward
- Fever
- Leukocytosis
- Elevated ESR

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Pericarditis

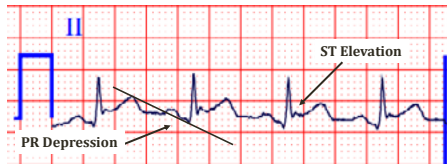
EKG Findings



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Pericarditis

EKG Findings



Pericarditis
Diffuse ST elevation
PR depression

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Pericarditis

EKG

- Technically, 4 stages of EKG changes
- Stage 1: diffuse ST elevations, PR depressions
- Stage 2 (~1 week later): Normal
- Stage 3: T wave inversions
- Stage 4: Normal

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Pericarditis

Physical Exam

- Pericardial friction rub
- **Scratchy** sound
- Systole and diastole



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Pericarditis

Etiology

- **Usually idiopathic**
- Viral
 - Classic cause is **Coxsackievirus**
 - Often follows viral upper respiratory infection (URI)
- Bacterial
 - Spread of pneumonia
 - Complication of surgery
 - Tuberculosis
- Fungal

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Pericarditis

Etiology

- Uremic (renal failure)
- Post-myocardial infarction
 - Fibrinous (days after MI)
 - Dressler's syndrome (weeks after MI)
- Autoimmune disease (RA, Lupus)

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Pericarditis

Treatment

- NSAIDs
- Steroids
- **Colchicine**
 - Inhibits WBCs via complex mechanism
 - Useful in gout and familial Mediterranean fever
 - Added to NSAIDs to lower risk of recurrence

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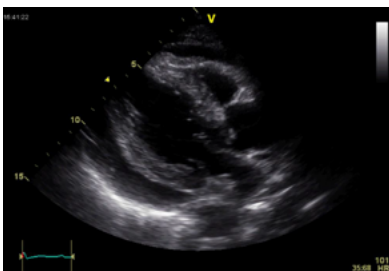
Myopericarditis

- Myocarditis = inflammation of myocardium
- Similar presentation to ischemia
 - Chest pain
 - EKG changes
 - Increased CK-MB, Troponin

Tamponade

- Accumulation of pericardial fluid
- **High pericardial pressure**
- Filling restriction of cardiac chambers
- Amount of fluid variable
 - Acute accumulation (bleeding): small amount of fluid
 - Chronic accumulation (cancer): large amount of fluid

Tamponade



Water Bottle Sign



Tamponade

Causes

- **Cancer metastases** to pericardium
- Uremia
- Pericarditis
- Trauma
- Treatment: Drainage of effusion

Tamponade

Clinical features

- Distant heart sounds
- Dyspnea
- Elevated jugular venous pressure

Tamponade

Clinical features

- Beck's Triad
 - Distant heart sounds
 - Elevated JVP
 - Hypotension
- Seen in rapidly-developing traumatic effusions
- Severe impairment LV function → **low cardiac output**
- Slower effusions: Pericardium stretches/dilates

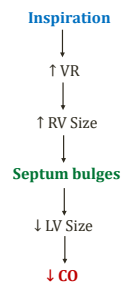
Pulsus Paradoxus

- Classic finding in tamponade
- Systolic BP always falls slightly on inspiration
- Exaggerated fall ($>10\text{mmHg}$) = pulsus paradoxus
- Severe fall = pulse disappears



Public Domain

Pulsus Paradoxus



Pulsus Paradoxus

- Also seen in **asthma and COPD**
- Inspiration: ↓ left sided flow
- Caused by pulmonary pressure fluctuation
- Exaggerated in lung disease
 - Normal lungs: 0 to -5mmHg
 - Lung disease: Change up to 40mmHg
- Large drop in left sided flow → pulsus paradoxus

Pulsus Paradoxus

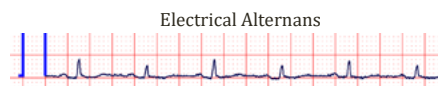
Measurement Technique

- Raise cuff pressure until no sounds heard
- **NORMAL respirations**
- Slowly lower cuff pressure
- First point (P1): intermittent sounds
- Second point (P2): constant sounds
- Pulsus = P1 - P2

Tamponade

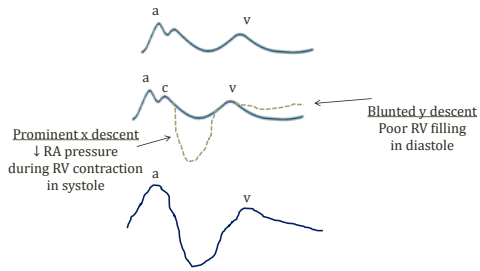
EKG

- Sinus tachycardia
- Low voltage – EKG sees less electricity due to effusion



Tamponade

Prominent x descent, Blunted y descent



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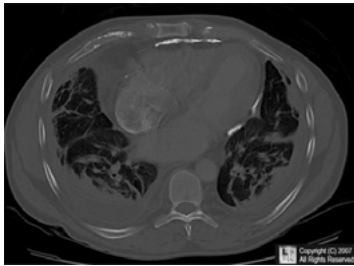
Equalization of Pressures

- Occurs when cardiac chambers cannot relax
- Pressure in RA, RV, LA, LV falls but then abruptly stops
- Seen in tamponade and pericardial constriction

Parameter	Normal	Tamponade
RA mean	5	20
RV Pressure	20/5	44/20
PCPW Pressure	10	20

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Constrictive Pericarditis



www.learningradiology.com, courtesy of Dr. William Herring, MD, FACP. Used with permission.

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Constrictive Pericarditis

- Fibrous, calcified scar in pericardium
- Loss of elasticity: stiff, thickened, sticky
- Can result from many pericardial disease processes
 - Pericarditis
 - Radiation to chest
 - Heart surgery

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Constrictive Pericarditis

Clinical Features

- Dyspnea
- Prominent **right heart failure**
 - Markedly elevated jugular venous pressure
 - Lower extremity edema
 - Liver congestion
 - May lead to cirrhosis ("nutmeg liver")



David Monniaux/Wikipedia

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Constrictive Pericarditis

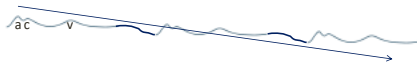
Other Features

- Pulsus paradoxus uncommon (~20%)
- High RA, RVEDP, PCPW pressures
- Equalization of pressures
- **Pericardial knock**



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Kussmaul's Sign



- Inspiration → ↑ VR → slight fall in mean JVP
- **Kussmaul's sign = ↑ JVP with inspiration**
 - Ventricle cannot accept ↑VR
 - Constrictive pericarditis
 - Restrictive cardiomyopathy
 - RV myocardial infarction
- Not seen in tamponade



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Pulsus and Kussmaul's

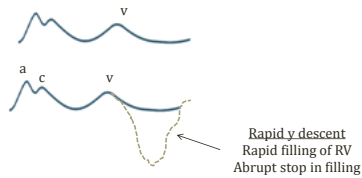
- Pulsus paradoxus: classic sign of tamponade
 - **Pulsus in tamPonade**
- Kussmaul's sign: classic sign of constriction
 - Also seen in restrictive heart disease
 - **Kussmaul's in Konstriction/Restriction**

	Tamponade	Constriction	Restrictive
Pulsus	Yes	No	No
Kussmaul's	No	Yes	Yes

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Constrictive Pericarditis

Rapid/prominent y descent



Myocardium adherent to pericardium
In diastole: **rapid relaxation** and suction of RA volume

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Venous Pressure Findings

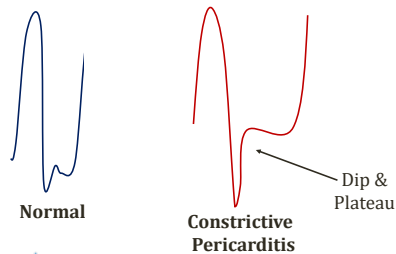
	Tamponade	Constriction
x descent	Rapid	--
y descent	Absent	Rapid

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Dip and Plateau

Square Root Sign

Right Ventricular Pressure
Rapid filling, abrupt stop



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Constriction and Restriction

- Constrictive pericarditis/Restrictive heart disease
- **Many common features**
- Prominent right heart failure
- Kussmaul's sign
- Rapid y descent
- Dip and plateau

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Aortic Dissection

Aortic Dissection

Jason Ryan, MD, MPH

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Aortic Dissection

CT Angiogram

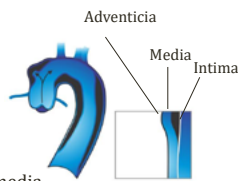


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* Images courtesy Dr. James Hellman and Wikipedia; used with permission

Aortic Dissection

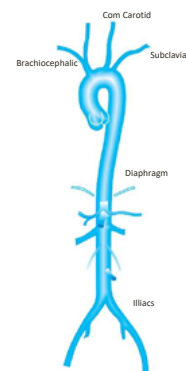
- Three layers to aorta
 - Intima
 - Media
 - Adventicia
- Dissection → **tear in intima**
- Blood "dissects" intima and media



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Propagation

- Blood enters dissection plane
- Spreads proximal, distal
- Can disrupt flow to vessels



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Types

- Type A
 - Involves ascending aorta and/or arch
 - Treated **surgically**
- Type B
 - Descending aorta
 - Can be treated **medically**
 - Control hypertension/symptoms
 - Surgical mortality high



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Symptoms

- "Tearing" chest pain radiating to back

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Other symptoms

- Propagation to aortic root
 - Aortic regurgitation
 - Pericardial effusion/tamponade
 - Myocardial ischemia (obstruction RCA origin)
- Propagation to aortic arch
 - Stroke (carotids)
 - Horner's syndrome
 - Vocal cord paralysis



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Recurrent Laryngeal Nerve

- Branch of vagus nerve
- Supplies larynx and voice box
- Compression:
 - Aortic dissection
 - Massive left atrial enlargement



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Other findings

- **Blood pressure differential** between arms
- **Widened mediastinum** on chest x-ray



JHeuser / Wikipedia

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Diagnosis

- Suggested by history, exam, chest x-ray
- Definitive diagnosis
 - **CT scan**
 - MRI
 - Transesophageal echocardiogram (TEE)
- D-dimer
 - Sensitive but not specific
 - Normal value makes aortic dissection unlikely

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Risk Factors

General Principles

- **Medial layer** of aorta
 - Tensile strength and elasticity
 - Key proteins: **collagen and elastin**
 - Weakness of medial layer → dissection/aneurysms
 - Common aneurysm feature: medial damage/destruction
- **Vasa vasorum**
 - Network of small vessels primarily in adventitial layer
 - Supplies blood to medial layer in thick vessels (i.e. aorta)
 - Thickening (HTN) → weakening of medial layer

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Risk Factors

General Principles

- Requires **tension** on wall
 - Common in proximal aorta (near aortic valve)
 - High tension from blood moving out of heart
 - Worsened by hypertension
- Requires **weakness of media layer**
 - Also caused by hypertension
 - Seen in collagen disorders (genetic)

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Risk Factors

General Principles

- **Cystic medial necrosis**
 - Development of cysts and necrosis in medial layer
- Occurs to mild degree with aging
- More rapid with:
 - Bicuspid aortic valve
 - Marfan syndrome
- Common in **ascending thoracic aneurysms**

Risk Factors

Aortic Dissection

- Aortic damage
 - **HTN - #1 risk factor**
 - Atherosclerosis
 - Thoracic aneurysm
- Abnormal collagen
 - Marfan Syndrome
 - Ehlers-Danlos
- Others
 - Bicuspid aortic valve
 - Turner Syndrome (bicuspid, coarctation)
 - Tertiary syphilis: Aortitis

Aortic Aneurysms

- Dilation/bulge of aorta
- More than 1.5x normal
- Involves all three layers
- Thoracic (TAAs)
- Abdominal (AAAs)



Thoracic Aortic Aneurysms

- Important risk factor for dissection
- Usually occur in proximal/ascending aorta
- Usually seen in **association with another disorder**
 - Marfan, Turner, Bicuspid aortic valve, **Syphilis**
- Family history of aneurysm important
- May be associated with **atherosclerosis**
 - More common in descending aorta
 - Occur in association with atherosclerosis risk factors
 - HTN, smoking, high cholesterol



Thoracic Aortic Aneurysms

Symptoms

- Most are asymptomatic
- Can cause **aortic regurgitation**
- Surgery if size >5.0cm



Abdominal Aortic Aneurysms

- More common than thoracic aneurysms
- Classically taught as a disease of **atherosclerosis**
- **Infrarenal aorta** most affected by atherosclerosis
- Also most common site of AAA
- Current research suggests many factors
 - Genetic, environmental, hemodynamic, immunologic



Abdominal Aortic Aneurysms

Risk Factors

- **Smoking**: strongest association with AAA
- Males: 10x more common men vs. women
- Age
 - Rare before 55
 - As high as 5% in men >65
- HTN, hyperlipidemia



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Abdominal Aortic Aneurysms

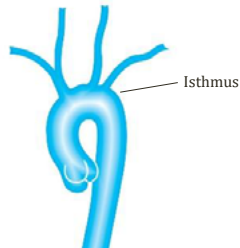
- Most are asymptomatic
- Some detected on physical exam
- **Pulsatile mass** from xiphoid to umbilicus
- Natural history is enlargement → rupture
- Followed with **ultrasound** or CT scan
- Surgery if >5.0cm



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Aortic Rupture

- Usually from trauma
- Most common site is isthmus



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Cardiac Tumors

Cardiac Tumors

Jason Ryan, MD, MPH

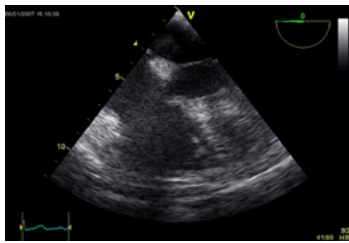
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Cardiac Tumors

- Myxoma
 - Most common 1° cardiac tumor
- Rhabdomyomas
 - Most common 1° cardiac tumor children
- Metastatic tumors
 - Most common cardiac tumor overall

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Myxoma



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Myxoma

- Common in the **left atrium (80%)**
 - Usually attached to atrial septum
 - Often at the border of fossa ovalis
- Benign (do not metastasize)

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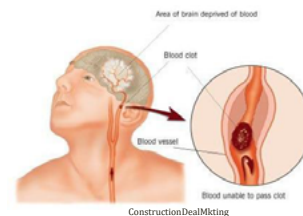
Myxoma

- Mesenchymal cells (undifferentiated cells)
- Endothelial cells
- Thrombus/clot
- Mucopolysaccharides

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Myxoma

- Often cause systemic symptoms
 - "B symptoms"
 - Fevers, chills, sweats
- Can embolize → stroke



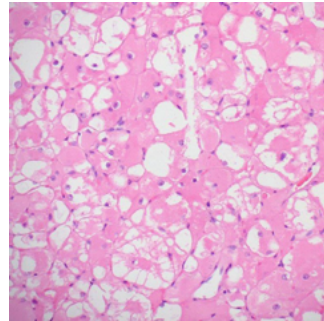
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Myxoma

- May disrupt **mitral valve function**
 - Regurgitation
 - Heart failure
- Can sit in mitral valve
 - "Ball in valve"
 - Mitral stenosis symptoms
 - Syncope or sudden death
- Auscultation: Diastolic "tumor plop"

Cardiac Rhabdomyomas

- Tumors of muscle cells
- Benign (do not metastasize)
- Usually children (most <1year)
- Sometimes detected prenatal
- Tumor **embedded in ventricular wall**
- Most regress spontaneously
- Rare symptoms from obstruction of blood flow



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Cardiac Rhabdomyomas

- Associated with **tuberous sclerosis (90%)**
- Autosomal dominant genetic syndrome
- Mutation in TSC1 or TSC2 gene
- TSC1: Hamartin
- TSC2: Tuberin
- Mutations → widespread tumor formation

Tuberous Sclerosis

- Involves MULTIPLE organ systems
- Numerous hamartomas and other neoplasms
- **Seizures** – most common presenting feature
- "Ash leaf spots": Pale, hypopigmented skin lesions
- Facial skin spots (angiofibromas)
- Mental retardation

Tuberous Sclerosis



Herbert L. Fred, MD and Hendrik A. van Dijk

Tuberous Sclerosis



Mohd Hanafi

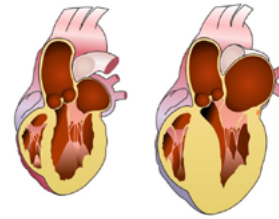
Hypertrophic Cardiomyopathy

Hypertrophic Cardiomyopathy

Jason Ryan, MD, MPH

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Hypertrophic Cardiomyopathy



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Hypertrophic Cardiomyopathy

Names

- Hypertrophic cardiomyopathy (HCM)
- Hypertrophic obstructive cardiomyopathy (HOCM)
- Idiopathic hypertrophic subaortic stenosis (IHSS)

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HCM

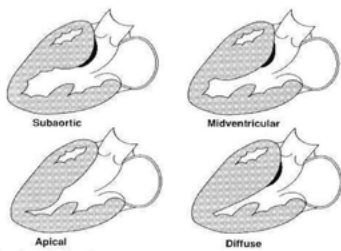
- Genetic disorder caused by gene mutations
- About 50% cases familial (50% sporadic)
- **Autosomal dominant**
- Variable expression
 - Significant variation in severity of symptoms
 - Many variations in location/severity of hypertrophy



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Morphologic Variants



Zorkun/Wikipedia

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HCM

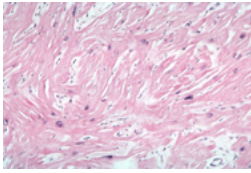
- Often **single-point missense mutations**
 - Point mutation → altered amino acid in protein
 - 15+ genes with 1500+ mutations identified
- Often involve genes for **cardiac sarcomere proteins**
 - Beta-myosin heavy chain (40% cases)
 - Myosin binding protein (40% cases)

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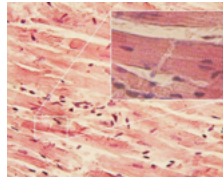
HCM

Histology

- Myocyte disarray (excessive branching)
- Hypertrophy
- Interstitial fibrosis



Zorkun/Wikipedia



Myokard/Wikipedia

HCM

Clinical Features

- Many patients asymptomatic
- **Heart failure**
 - Diastolic dysfunction
 - Impaired emptying due to LVOT obstruction
- **Chest pain (angina)**
 - Increased O₂ demand

HCM

Clinical Features

- **Sudden cardiac death**
 - Abnormal myocytes → ventricular arrhythmias
 - Most common cause SCD in young patients
- **Syncope**
 - Arrhythmias may lead to syncope
 - Thickened myocardium → LVOT obstruction
- **Mitral regurgitation**

HCM

Clinical Features

- Problem #1: Arrhythmia problem
 - Thick myocardium vulnerable to arrhythmias
 - Most serious is ventricular tachycardia → sudden death
 - Exercise (catecholamines) increase risk SCD
- **Sudden death in athletes**
 - Defibrillators for high risk patients
 - Avoidance of exercise

HCM

Clinical Features

- Problem #2: Outflow obstruction problem
 - Thickened myocardium obstructs blood leaving LV
 - Same physics and symptoms as aortic stenosis
 - Heart failure, chest pain, exercise-induced syncope
 - Treated with surgery
 - Beta blockers (↓ contractility)
 - Ca blockers (verapamil)



HCM

Clinical Features

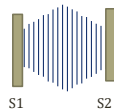
- #3: Mitral valve problem
 - High velocity in LVOT tugs mitral valve chords and leaflets
 - Causes systolic anterior motion (SAM) of mitral valve
 - Over time this leads to **mitral regurgitation**



HCM

Clinical Features

- **Systolic ejection murmur**
- Caused by outflow tract obstruction
- Sounds just like AS unless you do maneuvers
- Lots of associated abnormal heart sounds
 - S4
 - Holosystolic murmur of MR
 - Paradoxical split S2



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HCM

Maneuvers

- For any HCM maneuver, think about size of LV
- $\uparrow$ LV size $\rightarrow$ $\downarrow$ murmur
- $\downarrow$ LV size $\rightarrow$ $\uparrow$ murmur



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HCM

Maneuvers

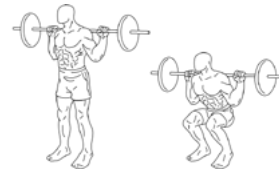
- Valsalva
 - Patient bears down as if having a bowel movement
 - Or blows out against closed glottis
 - Increase thoracic pressure $\rightarrow$ compression of veins $\rightarrow$ $\downarrow$ VR
 - Less VR $\rightarrow$ Less preload $\rightarrow$ Smaller LV cavity
 - Obstructing septum moves further into the outflow tract
 - Murmur **INCREASES** in intensity

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HCM

Maneuvers

- Squatting
 - Forces blood volume stored in legs to return to heart
 - Preload rises $\rightarrow$ size of LV increases $\rightarrow$ less obstruction
 - Murmur **DECREASES** in intensity



Wikipedia

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HCM

Other maneuvers

- Raising the legs
 - Increases venous return
 - More VR $\rightarrow$ More preload $\rightarrow$ Bigger LV cavity
 - This moves the obstructing septum out of the way
 - Murmur **DECREASES** in intensity
- Standing
 - Opposite mechanism of leg raise
 - Murmur **INCREASES** in intensity

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Aortic Stenosis

- Both HCM and AS cause a systolic ejection murmur
- Less effect of maneuvers on aortic stenosis
 - Fixed obstruction
- Opposite effects of maneuvers in aortic stenosis
 - Less preload $\rightarrow$ less flow $\rightarrow$ quieter AS murmur

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HCM

Maneuver Summary

- Valsalva → INCREASE
- Standing → INCREASE
- Squatting → DECREASE
- Leg Raise → DECREASE

HCM

Associations

- **Maternal diabetes**
 - Infants: transient hypertrophic cardiomyopathy
 - Usually thickening of interventricular septum
 - May have small LV chamber → obstruction in newborn
 - Resolves by a few months of age



Øyvind Holmstad/Wikipedia

HCM

Associations

- **Friedreich Ataxia**
 - Autosomal recessive CNS disease
 - Trinucleotide repeat disorder
 - Spinocerebellar symptoms
 - Often have concentric left ventricular hypertrophy
 - Also septal hypertrophy



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Cardiac Hypertrophy

Other Causes

- Hypertension
- Valve disease
- Athlete's heart



Pixabay

Cardiac Hypertrophy

Rare Pathologic Causes

- **Fabry Disease**
 - Lysosomal storage disease
 - Deficiency of α -galactosidase A
 - Neuropathy, skin lesions, lack of sweat
 - Left ventricular hypertrophy

Cardiac Hypertrophy

Rare Pathologic Causes

- **Pompe Disease**
 - Glycogen storage disease
 - Acid alpha-glucosidase deficiency
 - Enlarged muscles, hypotonia
 - Cardiac enlargement

Endocarditis

Endocarditis

Jason Ryan, MD, MPH

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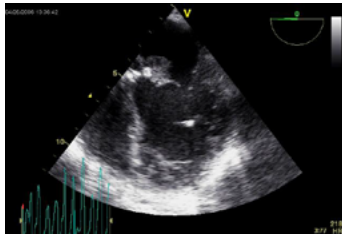
Endocarditis

- Inflammation of endocardium of heart
- Usually involves **cardiac valves**
- Often causes **new regurgitation murmur**
- Consequence of bacteremia

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Endocarditis

Echocardiogram



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General Symptoms

- Fever
- Chills
- Sweats
- Petechiae
 - Small vessel inflammation
 - Leakage of blood

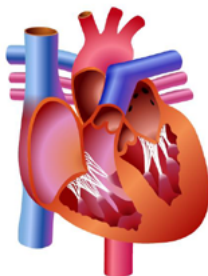


James Hellman, MD

Boards&Beyond

Regurgitant Valve Disease

- Aortic regurgitation
- Mitral regurgitation
- Tricuspid regurgitation



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Embolic Symptoms

- Brain (stroke)
- Spinal cord (paralysis)
- Eye (blindness)
- Legs (ischemia)
- Splenic or renal infarction
- Pulmonary embolism (tricuspid)
- Coronary artery (acute myocardial infarction)

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Endocarditis Stigmata

- **Physical exam findings** in endocarditis
- Caused by **septic emboli and immune complexes**
- Very rare in modern era

Endocarditis Stigmata

- Roth spots
 - **Retinal lesions**
 - Red with pale center
- Osler nodes
 - **Painful** bumps on pads of fingers and toes
- Janeway lesions
 - **Nontender** red macules on palms and soles
- Splinter hemorrhages
 - Reddish-brown lines under fingernails

Diagnosis

- Major Duke Criteria
 - Positive blood cultures
 - Vegetation on echocardiogram
- Minor Criteria
 - Fever
 - Risk factors
 - Roth spots, Osler nodes, Janeway lesions, splinters
- 2 major, 1 major 3 minor, or 5 minor

Microbiology

- Staphylococcus aureus
- Viridans streptococcus
- Streptococcus Bovis
- Enterococcus
- Staphylococcus epidermidis
- Culture negative endocarditis
- Libman-Sacks



Joydeep/Wikipedia

Staph Aureus

- Gram positive cocci
- Catalase positive
- Coagulase positive
- May infect **tricuspid valve in IV drug users**



Iqbal Osman/Pfizer

Staph Aureus

- Causes **acute** endocarditis
- Rapid, severe infection
- Symptoms occur over days
- Can occur in patients with **normal heart valves**
 - No pre-disposing valvular heart condition

Viridans Streptococcus

- Group of gram positive cocci
 - S. mitis, S. mutans, S. sanguinis
- Catalase negative
- Mouth flora
- Endocarditis may occur after **dental procedure**



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Viridans Streptococcus

- Low virulence bacteria
- Often affect **damaged valves**
 - Bacteria synthesize dextran
 - Dextran adheres to fibrin
 - Fibrin found with endothelial damage
- Classic predisposing condition: **mitral valve prolapse**

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Viridans Streptococcus

- Causes **subacute** endocarditis
- Less severe symptoms
- Symptoms occur over days to weeks

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Streptococcus Bovis

- Gram positive cocci
- Lancefield group D
- Normal gut bacteria
- Associated with **colon cancer**
 - All subtypes associated with cancer
 - Strongest association: **S. gallolyticus** (S. bovis type 1)

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Enterococcus Endocarditis

- Gram positive cocci
- Lancefield group D
- Normal gut bacteria
- Usually a subacute endocarditis course
- Commonly occurs in **older men**
- **Associated with manipulation of GI/GU tract**
 - Abdominal surgery
 - Urinary catheter
 - TURP for treatment of BPH

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Prosthetic Valve Endocarditis

- Occurs with mechanical or biologic valves
- Rarely cured with antibiotics
- Usually requires repeat valve surgery
- Similar bacteria to native valve endocarditis
- **Staphylococcus epidermidis**
 - Rarely cause endocarditis except in prosthetic valves



Stif Komar/Wikipedia

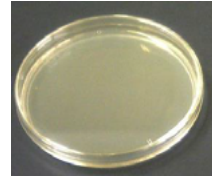
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Staphylococcus Epidermidis

- Catalase positive
- Coagulase negative (unlike *S. Aureus*)
- Most common coagulase negative staphylococcus
- **Normal skin flora**
- Low virulence
- Commonly cause infection of **prosthetic material**
 - Cardiac valves
 - Intravascular catheters
 - Prosthetic joints

Culture Negative Endocarditis

- Evidence of endocarditis with sterile blood cultures
- Caused by rare bacteria difficult to culture
- *Coxiella burnetii*
- *Bartonella*



Y tambe/Wikipedia

Coxiella Burnetii

- Zoonotic bacteria (transferred from animals)
- Obligate intracellular bacteria
- Found in **farm animals**
- Cattle, sheep and goats
 - Abortions in farm animals: *Coxiella* **placenta** infection
- Humans **inhale aerosolized bacteria** from animals
- **Causes Q fever**

Coxiella Burnetii

- Acute Q fever
 - Flu-like illness
 - May present as pneumonia
 - **More than half of cases: no symptoms**
- Chronic Q fever
 - **Most common manifestation is endocarditis**

Bartonella

- *Bartonella quintana*
 - Small, gram-negative rod
 - Transmitted by **lice**
 - Patients with poor hygiene
- *Bartonella henselae*
 - Found in **cats**
 - Causes cat scratch fever



BruceBlaus/Wikipedia



Inge Wallumrød/Pexels.com

NBTE

Non-bacterial, thrombotic endocarditis

- Libman-Sacks Endocarditis or Marantic endocarditis
- Lesions on valves that look like endocarditis
- Found on **both sides of valve**
- Mitral valve most common
- Formed by thrombus, immune complexes
- Seen in hypercoagulable states
 - **Advanced malignancy**
 - Systemic lupus erythematosus

NBTE

Non-bacterial, thrombotic endocarditis

- Often **asymptomatic** identified at autopsy
- Rarely cause regurgitation or murmurs
- Thrombus easily dislodges → **embolization**
- Most patients asymptomatic until embolism occurs
- May embolize to spleen, kidney, skin, extremities
- May cause stroke
- Can cause myocardial infarction

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Bacterial Endocarditis

Treatment

- Several weeks appropriate antibiotics
- Broad spectrum antibiotics initially
- Drug therapy changes when bacteria identified
- Valve surgery sometimes required
 - Large vegetation
 - Severe valve disease → heart failure

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Bacterial Endocarditis

Complications

- May form abscess beneath valve annulus
- Persistent fever, bacteremia often indicates abscess
- Aortic valve abscess can lead to heart block
 - AV node dysfunction



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Prophylaxis

- Primary prevention for bacterial endocarditis
- **Done before high-risk medical procedures**
- Antibiotics given to some high-risk patients
- New guidelines restrict to highest risk circumstances



Scotth23/Pixabay

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Prophylaxis

Conditions	Procedures
Prosthetic valves Prior endocarditis Cyanotic congenital heart disease Heart transplants	Dental work Respiratory procedures Skin surgery

Amoxicillin
Clindamycin

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